NeuroVLM: A generative vision-language framework for human neuroimaging

Point-by-point response to the reviewers

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Revised manuscript with changes marked **41**

Letter to the editor

We thank the editor and reviewers for their thoughtful comments. The revisions and response address every comment. The main changes are:

- Independent voxel-space evaluation of text-to-brain maps against simple baselines and existing, published models (1.1).
- Retrieval measured by recall@k, mean reciprocal rank, and median rank, also against hard negatives (1.1).
- A comparison of generated and retrieved captions (3.3).
- A data-flow diagram of every training stage (1.2).
- NeuroVault results reported as an upper bound, a lower bound, and a multi-target evaluation, with map types handled separately (1.3).
- A leakage audit beyond DOI matching (2.4).
- Out-of-distribution and failure-case analyses (1.4 and 2.12).
- Corrected numbers, names, and claims throughout.

We also updated the model and its evaluation:

- We added a latent-space loss when training the text-to-brain head, which makes its maps more accurate (1.1).
- Generated maps are calibrated to the scale of the training maps, with one factor per model fitted on the validation set.
- An MLP projector replaces the Qformer in both brain-to-text stages (2.1).
- The language model still generates every title, steered toward the canonical network and region names by masking tokens that cannot continue one (2.12).
- Decoding uses Latin-script tokens only (2.12).
- The synthetic training maps had included held-out papers. We rebuilt them from training papers only and retrained the models that use them (2.4).
- Every evaluation leaves out the duplicated papers found by the leakage audit (2.4).
- All paper figures were redrawn from the final model.

Comments are numbered in order within each reviewer, minor comments and code availability included. For example, 1.5 is the first minor comment of Reviewer 1. Each comment is quoted in a shaded box, followed by our response and the changes to the manuscript. Comment numbers link to those comments. Figures and tables in this letter are numbered R1, R2, and so on.

Line and page numbers in the changes to the manuscript refer to the revised manuscript with changes marked, which shows additions in blue and deletions in red.

Reviewer 1

Reviewer's comment

*This is an ambitious and timely manuscript introducing **NeuroVLM**, a unified framework for contrastive text–brain retrieval, text-to-brain generation, and brain-to-text generation. The integration of these capabilities within a modular architecture is appealing, and the use of a relatively large corpus of 30,826 publication-level text–neuroimage pairs is a notable strength. The public release of the code, pretrained models, and datasets should also be valuable to the community. The results provide encouraging evidence that the model learns meaningful associations between scientific text and coarse spatial brain representations. However, I think several central aspects of the evaluation and data construction require further clarification or analysis before the stronger claims regarding map generation and neuroimage interpretation are fully supported.*

Response: We added the requested analyses and clarifications (1.1 to 1.10).

1.1: More independent evaluation of the generative outputs

Reviewer's comment

1. More independent evaluation of the generative outputs is needed

The current generative evaluation is performed partly within NeuroVLM's own learned representation space. Generated maps and text are embedded using the model's contrastive projection heads and then ranked against candidate items in that same space. This is a useful measure of internal consistency, but it does not independently establish that a generated map is anatomically accurate or that a generated description faithfully reflects the source image.

The matched-versus-null results are statistically strong, although some absolute differences are relatively modest—for example, Dice is 0.47 versus 0.40, BERTScore is 0.56 versus 0.52, and semantic cosine is 0.47 versus 0.40. These findings support above-null alignment, but additional validation would help establish neuroscientific specificity.

I encourage the authors to add some combination of:

- *Evaluation using independent image and text representations not used during NeuroVLM training.*
- *Blinded expert ratings of anatomical accuracy, semantic accuracy, specificity, and unsupported claims.*
- *Hard negatives drawn from the same cognitive domain or from studies with similar spatial patterns.*
- *Comparison between generated outputs and their nearest training examples.*
- *Strong simple baselines, including nearest-neighbor retrieval, mean or domain-specific maps, linear regression from text embeddings to spatial coefficients, and atlas-based labeling.*

It would also be useful to report conventional retrieval measures such as recall@1, recall@5, recall@10, mean reciprocal rank, and median rank, in addition to the area under the full recall@k curve.

Response: We agree. Fig. R1 and Table R1 score text-to-brain maps in voxel space, against each test paper's reconstructed map. NeuroVLM leads, narrowly ahead of linear regression and ahead of Text2Brain, mean and domain maps, atlas labeling, Neurosynth, NeuroQuery, and the nearest training paper (Dice 0.53 against at most 0.49). Neurosynth, NeuroQuery, and Text2Brain are used as published, although their corpora each hold nearly half the test papers.

Independent evaluation. Both directions are also scored outside NeuroVLM's representation space. Maps are compared in voxel space, and NeuroVLM also leads against the raw coordinate maps. MiniLM cosine and BERTScore, whose encoders were not used in training, compare generated captions with the paper's text (3.3).

Matched and shuffled pairs. The submission's BERTScore null came from an earlier generation run. With the null computed from the same captions, BERTScore separates matched from shuffled pairs only weakly (0.553 against 0.548). BERTScore mostly rewards the style of the reference, an LLM summary. Even random training texts in that style score higher (Table R11).

Modest differences. The narrow gaps are real. Captions describe the map rather than the study, so they share little content with the paper's text. Unrescaled BERTScore stays near 0.58 even for two unrelated papers. The matched caption scores higher for 57% of maps by BERTScore, and more often by MiniLM. The paired test over all test maps resolves this small shift (Fig. R2).

Expert ratings. We did not run blinded expert ratings, because they are time consuming and hard to scale. Instead, we used existing expert labels and automatic checks. For anatomical and semantic accuracy, region titles are checked against anatomical atlases and network titles against expert atlas labels (3.6). Hard negatives test specificity. Unsupported claims are counted as study details, such as a sample size, that a map cannot support (3.3).

Hard negatives. Table R2 scores retrieval among all test papers and among hard negatives, the test papers of the query's own cognitive domain. Hard negatives are only slightly harder than random papers of the same number (recall@10 0.24 against 0.28).

Nearest training examples. Generated maps match each test paper's map better than the map of the nearest training paper does (Table R1). By MiniLM cosine, generated captions are as close to the paper's text as the nearest training caption. However, 57% of these retrieved captions report details of another study, against 2% of generated captions (3.3).

Simple baselines. Fig. R1 and Table R1 include each suggested baseline. The linear regression maps the SPECTER2 text embedding to the autoencoder latent, the spatial coefficients of each map. We fit it as ridge regression, which is linear regression with a small penalty against overfitting, chosen on the validation set. NeuroVLM's final head learns the same mapping through a hidden layer, with a loss that also scores the decoded map.

Retrieval measures. Table R2 reports recall@1, recall@5, recall@10, mean reciprocal rank, and median rank in both directions. Fig. R1 and Table R2 are new in the Supplementary Information.

Table R1. Text-to-brain maps on the PubMed test set. Mean Dice and Pearson r between each method's map and the test paper's reconstructed map ($n = 2{,}971$), as in Fig. R1. The nearest training paper is the one whose text is closest to the query.

Method	Dice	Pearson r
NeuroVLM	0.531	0.322
Simple baselines		
Linear regression (ridge)	0.526	0.311
Domain mean map	0.448	0.245
Atlas-based labeling	0.431	0.175
Global mean map	0.420	0.251
Nearest training paper	0.395	0.114
Published models		
Neurosynth	0.418	0.175
NeuroQuery	0.411	0.179
Text2Brain	0.485	0.293
Shuffled null	0.408	0.122

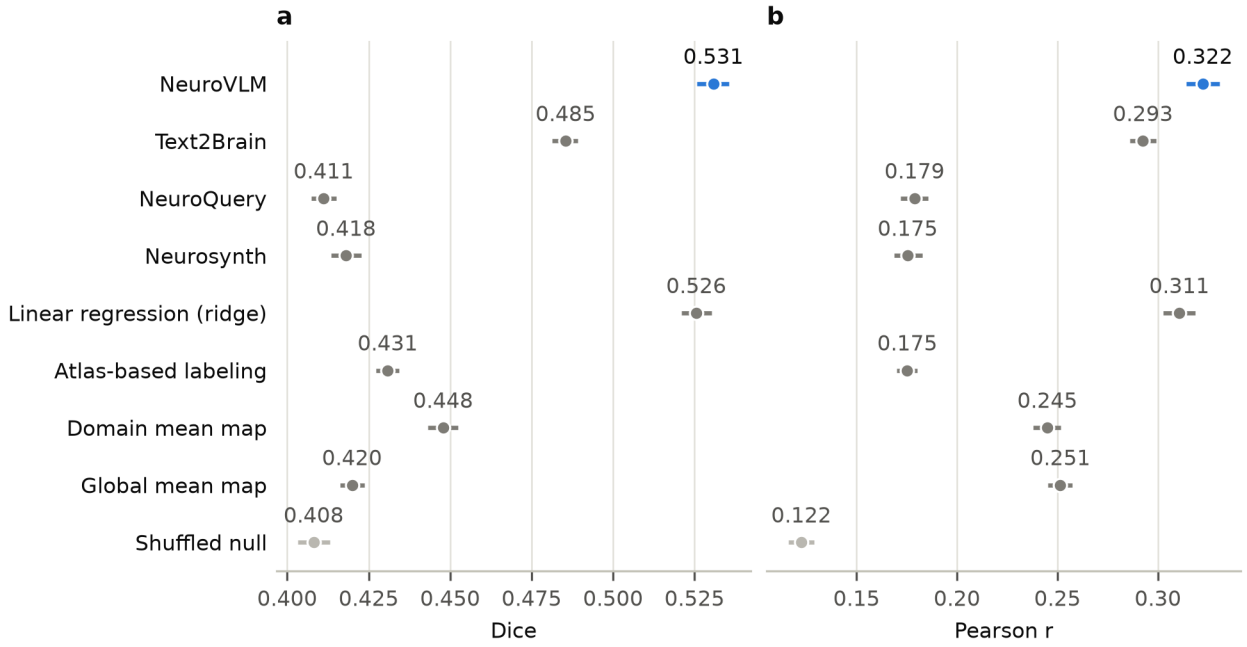


Fig. R1. Text-to-brain baselines. Text-to-brain maps from NeuroVLM and other methods, scored against each test paper's reconstructed map ($n = 2{,}971$). Panels show (a) Dice and (b) voxelwise Pearson r, with means and 95% bootstrap intervals. Dice uses each map's top 30% of voxels. Neurosynth, NeuroQuery, and Text2Brain were used as published. Their corpora include 1,386 (Neurosynth) or 1,295 (NeuroQuery and Text2Brain) test papers, which can only favor them. Text2Brain's training split is unreleased, so instead of retraining we scored the 1,157 papers in neither corpus, and NeuroVLM still beats all three.

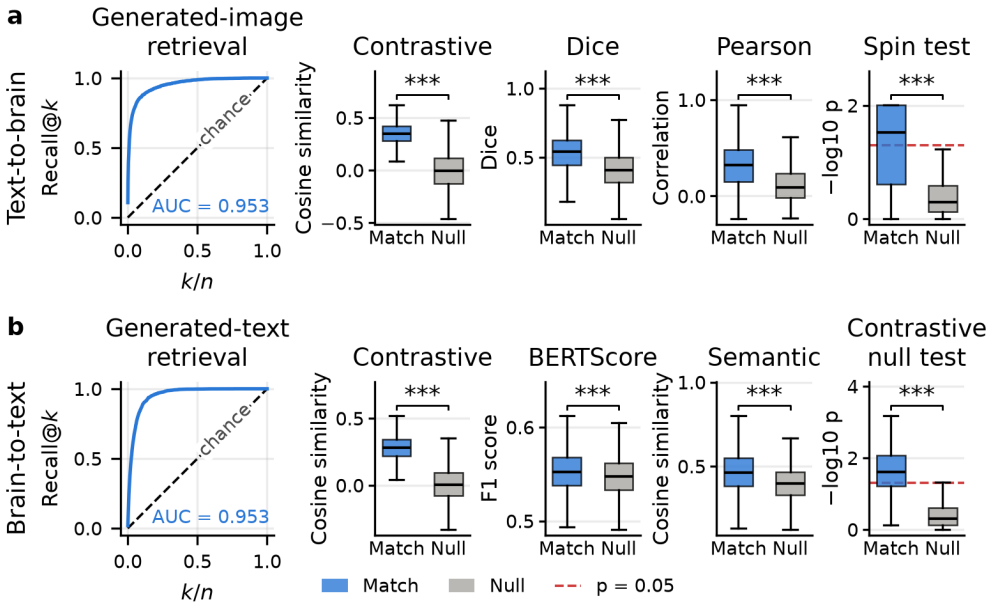


Fig. R2. New Fig. 4. Text-to-brain (a) and brain-to-text (b) generation on the PubMed test set ($n = 2{,}971$). Boxes compare matched (Match, blue) and shuffled pairs (Null, gray). Stars mark $p < 0.05$, 0.01, and 0.001.

Table R2. Contrastive retrieval on the PubMed test set. Recall@k, mean reciprocal rank (MRR), and median rank of the final model. Hard negatives are test papers from the query's own cognitive domain, and random negatives match their number.

	Recall@1	Recall@5	Recall@10	MRR	Median rank
All candidates (2,971 test papers)					
Text → brain	0.018	0.053	0.088	0.045	230
Brain → text	0.013	0.052	0.088	0.041	217
Chance	<0.001	0.002	0.003	0.003	1,486
Hard negatives (papers of the query's own domain, mean 585)					
Text → brain	0.054	0.160	0.242	0.118	44
Brain → text	0.059	0.176	0.263	0.126	40
Chance	0.003	0.017	0.034	0.019	250
Random negatives (random papers, as many as the hard negatives)					
Text → brain	0.069	0.196	0.283	0.142	35
Brain → text	0.071	0.201	0.289	0.144	35
Chance	0.003	0.017	0.034	0.019	250

Changes to the manuscript: Methods, Text-to-Brain, l. 365–385 (final head), and Performance Metrics, l. 398–404 and l. 415–426. Results, Generative Models: PubMed, l. 596–607, and Fig. 4 caption (p. 15). Supplementary Information, Projection Head, l. 1332–1339 (training of the final head), and Generative Performance, l. 1448–1450 (BERTScore null), with Table S4 (p. 32). New Supplementary Section Text-to-Brain Baselines and Retrieval, l. 1843–1859, with Fig. S6 (p. 42) and Table S8 (p. 43).

1.2: Publication-level supervision and dataset separation

Reviewer's comment

2. The publication-level supervision and separation of datasets across training stages need clarification

The primary training pairs associate publication titles and abstracts with maps reconstructed from reported coordinate tables. Because a single neuroimaging publication may include multiple experiments, contrasts, populations, and positive or negative effects, the correspondence between an abstract and one aggregated map may be substantially many-to-many.

Please clarify:

1. *Whether coordinates were combined across all experiments and contrasts within each publication.*
2. *How positive and negative effects were handled.*
3. *How multiple coordinate tables associated with one DOI were treated.*
4. *The final numbers of publications, experiments, contrasts, and coordinate tables.*
5. *Whether a contrast-level evaluation can be provided for a subset of the data.*

The complete train–test separation across model components also needs to be documented more clearly. In particular, Qwen3-0.6B was adapted using 1.2 million neuroscience articles. Please confirm that held-out PubMed and NeuroVault publications were excluded from this language-model corpus. Similarly, the manuscript should explain how test concepts and documents were excluded when constructing the synthetic fine-tuning data and canonical prototype bank. The synthetic stage uses generated maps and descriptions of common networks, regions, and cognitive concepts, creating another possible route through which test information could enter the system.

A data-flow diagram showing the datasets used for autoencoder training, contrastive training, language-model adaptation, Qformer training, synthetic fine-tuning, validation, and final testing would substantially improve transparency.

Response: 1. Experiments and contrasts. Yes, each paper's map combines the coordinates of all its experiments and contrasts. The released coordinates carry no experiment or contrast labels. Each map matches the paper-level text it is paired with.

2. Positive and negative effects. Positive and negative effects are pooled, because the released coordinates carry no sign labels. Each map therefore shows where effects lie, not their sign. The Methods state this.

3. Multiple coordinate tables. All coordinate tables of a paper are pooled into one map. Among open-access papers with a coordinate table, 42% report two or more.

4. Final counts. Table R3 gives the final counts, 30,826 publications with 857,298 peaks. Coordinate tables can be counted only in papers with open-access full text, and the table gives that count. Experiments and contrasts are unlabeled, so they cannot be counted.

5. Contrast-level evaluation. A contrast-level evaluation is possible only on NeuroVault, whose maps are released per contrast (1.3). With every map as its own target, the model beats chance in both directions (AUC 0.62 and 0.63).

Separation of datasets. Fig. R3 is the requested data-flow diagram. No validation or test paper trains the autoencoder, projection heads, or projectors. The language model's training script removes the validation, test, and NeuroVault evaluation papers from its corpus. Generated captions share only short phrases with their abstracts (median two words).

Synthetic data and prototype bank. The synthetic maps use training papers only (2.4). The prototype bank holds network, region, and function texts, not test documents. It names the networks that label the evaluation atlas maps, but no atlas network map trains any component.

Table R3. Final counts of the PubMed data. Peaks are coordinate rows, duplicates included. Tables are counted in the open-access full texts. Papers with more than 100 peaks or an empty map are in no split.

Quantity	Released data
Publications / peaks	30,826 / 857,298 (median 17 peaks per paper)
Coordinate tables	20,541 in the 11,943 open-access papers with a coordinate table (42% have two or more)
Experiments, contrasts	not labeled (only paper identifiers and coordinates are released)
Train / validation / test papers	23,894 / 2,987 / 2,987, and 958 in no split (2,971 test papers after the leakage audit)

Changes to the manuscript: Methods, Datasets, l. 137–144 (pooling per paper and splits), and Training, l. 169–181 (synthetic maps from training papers only, exclusions, and the evaluation papers removed from the language-model corpus). Supplementary Information, Coordinate Tables, l. 1044–1047. Discussion, l. 877–884. New Supplementary Section Data Flow and Training Details, l. 1745–1753, with Fig. S5 (p. 39) and Table S6 (p. 40).

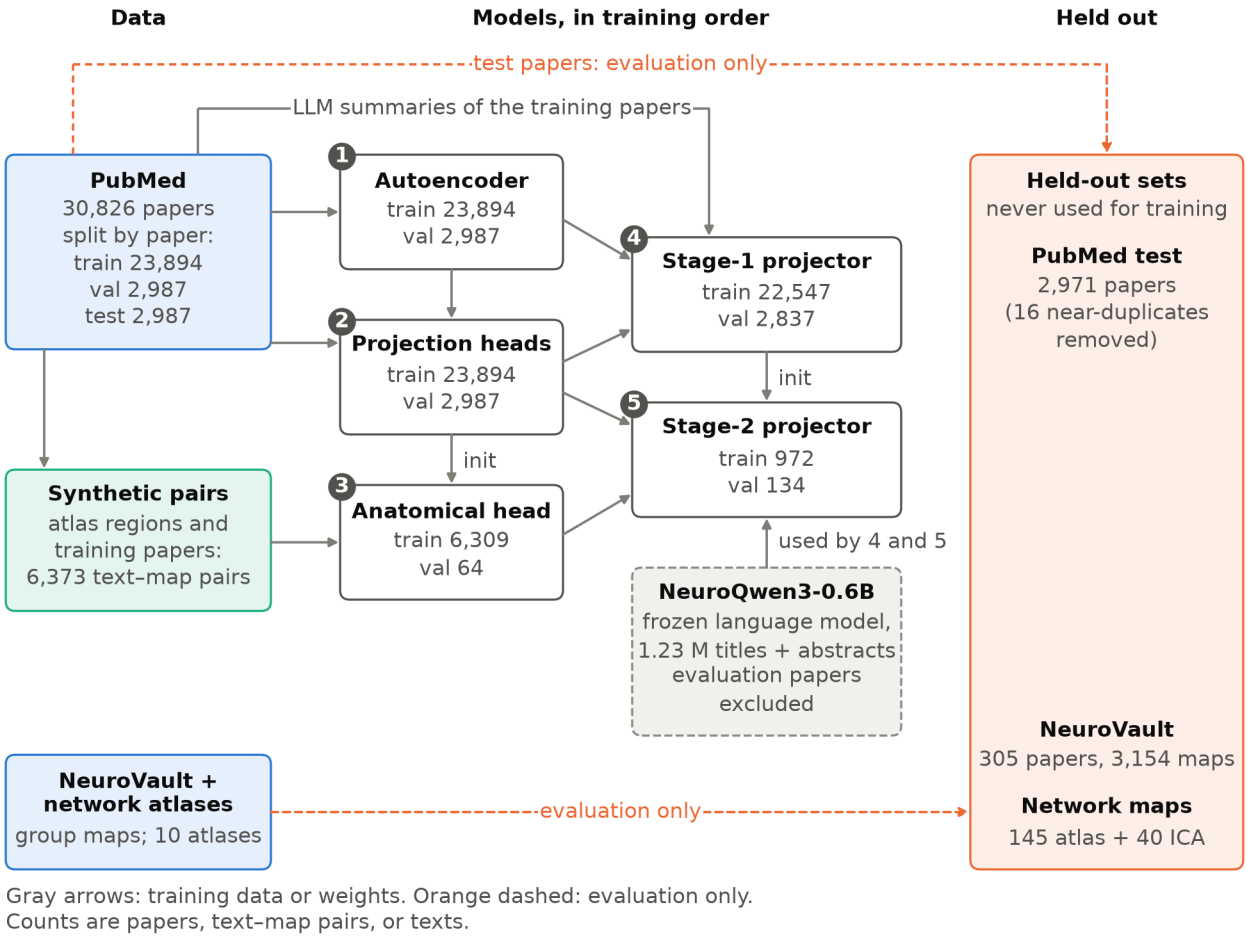


Fig. R3. Data flow of the final model. Boxes 1–5 are the trained components in training order, with their training and validation sizes. Gray arrows carry training data or weights, and orange dashed arrows mark sets used only for evaluation.

1.3: NeuroVault evaluation

Reviewer's comment

3. The NeuroVault evaluation would benefit from target-independent selection and map-type-specific preprocessing

The supplementary methods state that when several NeuroVault images were associated with the same title and abstract, the contrastive model selected the most similar image. The manuscript appropriately acknowledges that this inflates performance and provides an upper bound.

I suggest retaining this as an exploratory upper-bound analysis while adding a target-independent evaluation. For example, the authors could use contrast-specific NeuroVault metadata, evaluate all eligible images, apply a predefined image-selection rule, or use a multiple-instance retrieval metric.

The preprocessing of the NeuroVault images also requires reconsideration. The dataset contains t-, z-, F-, p-, and beta maps, as well as ROI masks and parcellations, but all maps are processed with a 99th-percentile threshold, cluster filtering, and binarization. These map types have substantially different meanings. In particular, the largest values in a conventional p-value map generally indicate weaker evidence, signed t-, z-, and beta maps may contain meaningful negative effects, and parcellations represent categorical labels rather than statistical intensity.

Please clarify whether map-specific transformations were applied before thresholding. If not, I recommend separate preprocessing and reporting for signed statistical maps, p-maps, masks, and parcellations. This would make the NeuroVault generalization results easier to interpret and ensure that spatial information is treated appropriately for each map type.

Response: We keep the model-selected map as an upper bound and add a lower bound, each paper's worst-matching map (Fig. R4). We add two target-independent scores. Multi-target retrieval counts any of a paper's maps as a hit, and the per-map score makes each map its own target. The model beats chance under every rule. Per-map text-to-brain and brain-to-text retrieval reach an area under the recall curve (AUC) of 0.62 and 0.63.

Map types. The submission thresholded every map alike, after removing masks, parcellations, anatomical maps, and variance maps. P-maps are inverted to $1 - p$ and signed maps keep their positive tail, as before. Using both tails, or adding negative tails as extra maps, reduces evaluation performance. Table R4 reports each map type separately, and Fig. 5b uses the same handling.

Maps labeled Other. Other is a type that NeuroVault uploaders choose, not a missing one. We typed these maps from their file names, descriptions, and values. Most are t, z, or F maps, effect estimates, or components (Table R4). Mean or variance images and maps we could not type are excluded, like the variance maps above. This moves every AUC by at most 0.004.

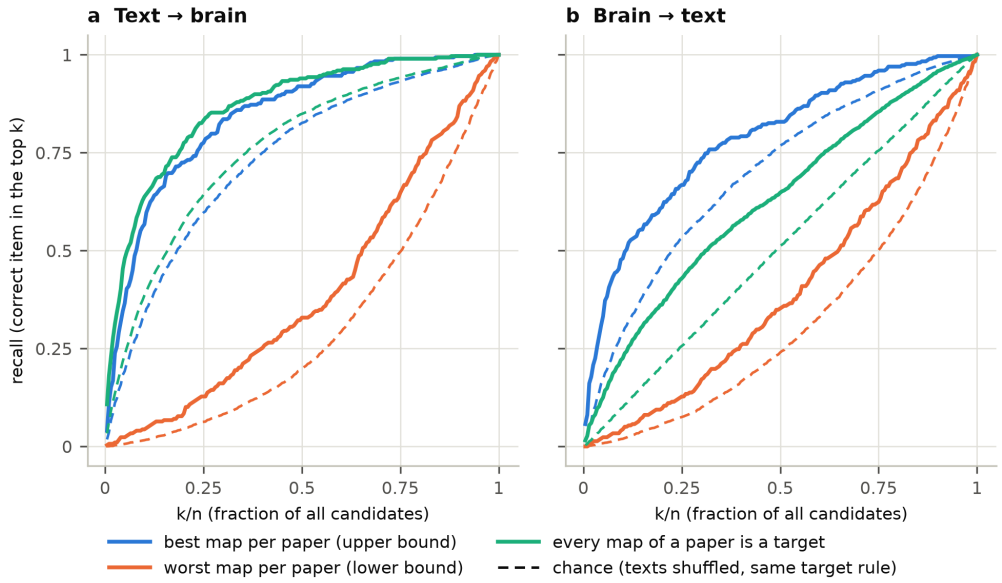


Fig. R4. NeuroVault retrieval. Recall for (a) text to brain and (b) brain to text, with each paper's best, worst, or every map as target. Dashed lines show chance from shuffled texts. The best map, or any of a paper's maps counting as a hit in (a), lifts chance above the diagonal, and the worst map lowers it.

Table R4. NeuroVault results by map type. AUC with every map as its own target (chance 0.50). Other marks maps their uploaders labeled Other, typed from file names, descriptions, and values. Masks, parcellations, anatomical and variance maps, mean images, and untyped maps were excluded.

Map type	Maps	Papers	Handling	AUC, text → brain	AUC, brain → text
T	1,761	189	positive tail	0.64	0.64
Z	811	75	positive tail	0.57	0.58
beta, univariate	43	5	positive tail	0.61	0.77
F	40	17	positive tail	0.64	0.55
beta, multivariate	14	5	positive tail	0.84	0.83
p	3	2	inverted (1 − p)	0.64	0.60
1 − p	2	1	used as is	0.54	0.44
Other, t, z, or F	204	23	positive tail	0.63	0.70
Other, effect estimate	103	15	positive tail	0.69	0.62
Other, component or gradient	93	8	positive tail	0.49	0.51

Changes to the manuscript: Fig. 5b caption (p. 16). Results, Contrastive Model: NeuroVault, l. 754–767. Supplementary Information, NeuroVault preprocessing, l. 1177–1192 (map types and exclusions) and l. 1204–1210 (bounds). New Supplementary Section NeuroVault Bounds and Map Types, l. 1954–1963, with Fig. S7 and Table S9 (p. 44).

1.4: Scope of the generative claims

Reviewer's comment

4. The scope of the generative claims should more closely reflect the available training targets

The authors appropriately acknowledge that the training images are proxy maps reconstructed from reported peak coordinates. Such maps omit subthreshold structure, cluster shape, effect size, uncertainty, contrast definitions, and other study-specific information. The additional smoothing, DiFuMo reconstruction, resampling, and binarization further emphasize broad spatial support rather than complete statistical information.

Given these limitations, the generated outputs may be more accurately described as ***coarse spatial-support predictions*** *or* ***spatial priors*** *rather than complete statistical maps. This framing would still support useful applications in literature navigation, network labeling, and exploratory hypothesis generation, while avoiding the implication that effect size, sign, uncertainty, or a study-specific statistical contrast has been reconstructed.*

Similarly, the brain-to-text outputs appear designed to provide concise, high-level network or functional descriptions rather than detailed interpretations of individual experiments. The manuscript should distinguish broad network labeling from study-specific interpretation and mechanistic inference.

The authors already note that the model can generate plausible maps for essentially any text input and should not be viewed as an oracle. A small quantitative analysis of failure cases, out-of-distribution inputs, and abstention or confidence calibration would strengthen this important caution.

Response: We agree and adopt this framing. The Abstract, Results, and Discussion call text-to-brain outputs coarse spatial priors, without effect size, sign, or uncertainty. They call brain-to-text outputs broad network-level labels, not interpretations of a specific study or mechanism. Failure cases are not rare. In 16% of test-set captions, the similarity to the paper's text is negative (2.12).

Out-of-distribution inputs. Fig. R5 flags a map when it correlates poorly with its autoencoder reconstruction or its latent lies far from the training latents. It flags a text that lies far from every training abstract. Thresholds set on the validation split flag about 5% of held-out test inputs. They

flag every shuffled, white-noise, and rescaled map, every off-topic text, and 61 to 80% of rotated or spun maps.

Abstention and confidence. Flagged inputs can be withheld, but the scores are warnings, not calibrated probabilities. Within the training distribution, the peak of a calibrated text-to-brain map works as a confidence score, since it tracks the map's retrieval rank (Spearman 0.63). It does not detect off-topic text, whose maps mostly peak within the range of test abstracts.

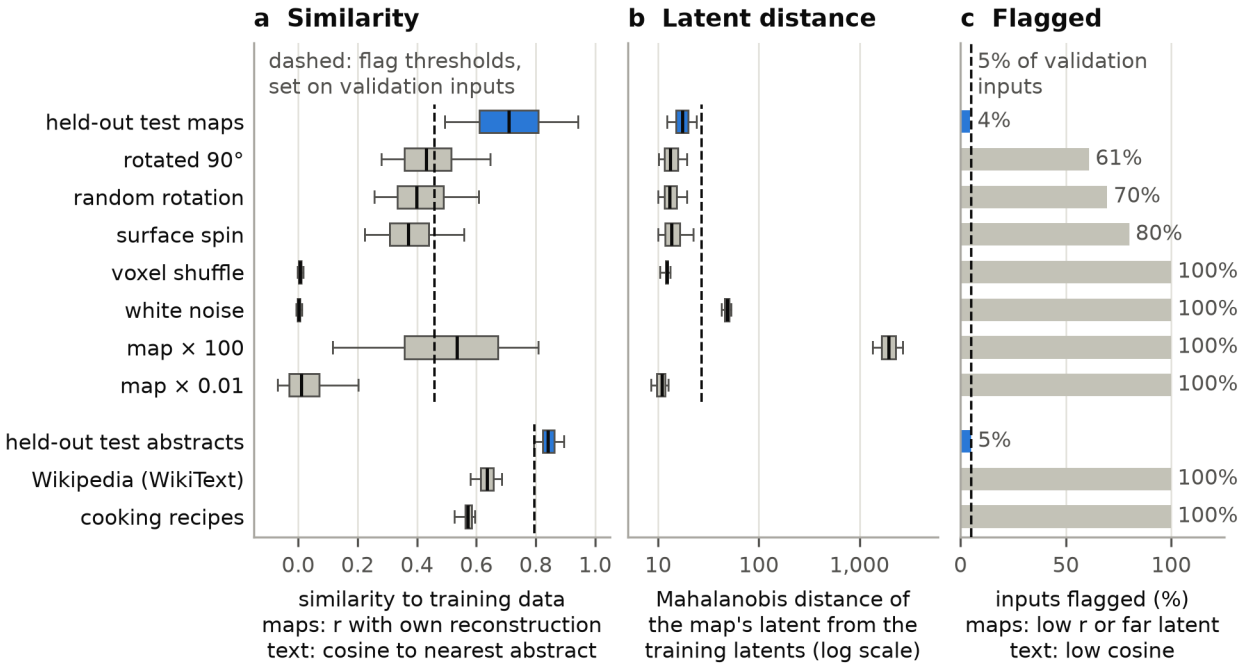


Fig. R5. Out-of-distribution warnings. Panels show (a) each input's similarity to the training data, (b) the distance of a map's latent from the training latents, and (c) the share of inputs flagged. Dashed lines mark thresholds set on the validation split. Blue marks held-out test inputs.

Changes to the manuscript: Abstract, l. 28–39. Introduction, l. 112–117. Methods, Text Decoder, l. 321–327 (out-of-distribution warnings), and Text-to-Brain, l. 374–385. Results, Generative Models, l. 432–434 (spatial priors, network-level labels). Discussion, limitations, l. 858–862 and l. 898–909. New Supplementary Section Out-of-Distribution Inputs, l. 2027–2040, with Fig. S8 (p. 45).

1.5: Inconsistent dataset and metric values

Reviewer's comment

*1. **Please reconcile inconsistent dataset and metric values.** The main methods list 3,000 NeuroVault maps and 100 network images, whereas the supplement reports 4,553 NeuroVault maps and the results describe 300 network images. The autoencoder AUROC is also reported as 0.94 in the Figure 5 caption and 0.975 in the main text.*

Response: We recomputed every count from the released data and corrected the text and captions (Table R5). The supplement's 4,553 maps predate image quality control and the removal of papers already in PubMed. Neither network-image count can be reproduced, so the released counts replace both. Both AUROCs are correct but refer to different datasets, which the Fig. 5 caption names.

Table R5. Corrected counts and AUROCs. Counts are recomputed from the released data. The AUROC is the autoencoder's voxelwise reconstruction AUROC.

In the paper	Correct value
3,000 or 4,553 NeuroVault maps	3,183 released maps from 312 papers, 3,154 maps from 305 papers after the leakage audit, and 3,074 maps from 298 papers for retrieval
100 or 300 network images	185 network maps, made of 145 atlas maps (119 with a network label) and 40 ICA components
AUROC 0.94 and 0.975	both correct, 0.94 on NeuroVault and 0.975 on the PubMed test set (0.974 with the final model)

Changes to the manuscript: Methods, Datasets, l. 132–144. Results, Contrastive Model: Network Labeling, l. 735–747, and Autoencoder, l. 826–830. Figs. 4 and 5 captions (pp. 15–16). Supplementary Information, NeuroVault preprocessing, l. 1177–1192 and l. 1209–1211.

1.6: SPECTER and SPECTER2

Reviewer's comment

*2. **Please use SPECTER and SPECTER2 consistently.** Section 2.6 describes SPECTER, whereas the text-to-brain methods and supplement describe SPECTER2.*

Response: Every text-encoding step uses SPECTER2, the allenai/specter2_aug2023refresh model with its ad hoc query adapter. The original SPECTER is never used. We corrected every mention to SPECTER2.

Changes to the manuscript: Introduction, l. 107–111. Methods, Text Encoder, l. 220–227, and Text-to-Brain, l. 365–373. Supplementary Information, Text preprocessing, l. 1085–1088. Fig. S1 caption (p. 23).

1.7: Two methodological analogies

Reviewer's comment

*3. **Two methodological analogies should be revised.** Transformer language models still depend on a fixed tokenizer vocabulary, despite their ability to represent context and novel compositions. In addition, a separately trained projection head is not technically equivalent to LoRA, although both approaches can preserve the weights of a larger pretrained encoder.*

Response: We agree with both points. We deleted the LoRA analogy and rewrote the tokenizer sentence. It states that contextual encoders also use a fixed vocabulary and that their advantage is composing words in context. SPECTER2 beats a context-free word2vec encoder (3.4). Swapping one word moves the generated map as expected, but negation has no effect (3.5).

Changes to the manuscript: Introduction, l. 81–84 (tokenizer sentence). Methods, Text-to-Brain, l. 365–385 (LoRA sentence deleted, projection head named).

1.8: Edge-device claim

Reviewer's comment

*4. **The edge-device claim would benefit from empirical support.** Please report total memory requirements, inference latency, hardware, numerical precision, and any quantization used. The relatively small text encoder is attractive, but the complete brain-to-text pipeline also includes the Qformer and Qwen language model.*

Response: We timed both full pipelines, projector and language model included, on a consumer GPU and a desktop CPU at several precisions (Table R6). We used the released weights, whose projector and text-to-brain head match the final ones in size. On the CPU, int8 quantization runs brain-to-text fastest, with the same memory as bf16. No phone or embedded device was tested, so "edge devices" becomes "a consumer CPU" in the Methods.

Table R6. Cost of one query. Median latency at batch 1 and peak memory of weights and activations, not file size. Brain-to-text decodes 128 tokens greedily. The GPU is an RTX 5090 and the CPU an 8-thread Ryzen 7 9800X3D. Dynamic int8 quantization skips NeuroQwen3's token embedding, adds an int8 copy of its tied output layer, and skips SPECTER2's adapter-wrapped layers, so it saves no memory over bf16.

Pipeline, one query	Parameters	GPU, bf16	CPU, fp32	CPU, bf16	CPU, int8
text-to-brain	171 M	7 ms, 0.4 GB	135 ms, 0.6 GB	53 ms, 0.4 GB	120 ms, 0.5 GB
brain-to-text, 128 tokens	685 M	3.2 s, 1.3 GB	13.1 s, 2.7 GB	6.3 s, 1.4 GB	3.5 s, 1.4 GB

Changes to the manuscript: Methods, Text Encoder, l. 226–230 (edge-device sentence replaced by CPU costs). Discussion, l. 935–936.

1.9: Figure text size, full descriptions, affiliations

Reviewer's comment

5. Some text in Figures 2, 3, and S4 is difficult to read at normal manuscript scale. Providing the complete generated descriptions in a supplementary table would be helpful. Please also correct the remaining "Organization" placeholders in the affiliation block.

Response: We redrew Figs. 2, 3, and S4 at the journal's text width with larger text. Figs. R6, R7, and R8 show the new Figs. 2, 3, and S4. Supplementary Table S5 gives the complete texts of Fig. 2, and Figs. 3 and S4 print each generated text in full beside its map. We also corrected the affiliations.

Changes to the manuscript: Affiliations, l. 16–17. Figs. 2, 3, and S4, redrawn (p. 11, p. 12, and p. 34). Results, Generative Models: Networks, l. 442–448. New Supplementary Table S5 with the complete texts of Fig. 2, l. 1567–1570 (pp. 35–38). Figs. 4 and 5 captions (pp. 15–16).

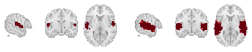
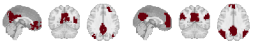
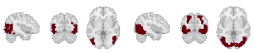
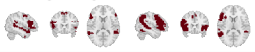
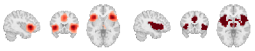
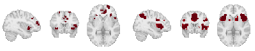
	Generated	Generated (brain-to-text)		
True	(text-to-brain)	Network	Anatomy	Cognition
Auditory		**Auditory Network.** The auditory network includes Heschl's gyrus, superior temporal gyrus, planum temporale, and inferior frontal cortex. It is commonly associated with the processing of sound, speech, and environmental sounds. The network is often described as a hierarchical system for auditory perception. Its coordinated activity is thought to support the integration of auditory information across the brain.	**Superior Temporal Gyrus.** The superior temporal gyrus is centered on the superior temporal sulcus. It is commonly associated with auditory processing, speech perception, and sound-related associations. Its anatomy supports representation of acoustic features such as frequency, timbre, and pitch.	**Early Auditory Processing.** Early auditory processing is the analysis of sound input at high frequencies. It involves primary auditory cortex in Heschl's gyrus, superior temporal gyrus, and planum temporale. This concept supports speech recognition, pitch detection, and low-frequency sound processing.
Default mode		**Default Mode Network.** The default mode network includes medial prefrontal cortex, posterior cingulate cortex, precuneus, angular gyrus, lateral parietal cortex, and medial temporal regions. It is commonly associated with internally oriented cognition, autobiographical memory, self-referential thought, and the integration of personal and social context.	**Precuneus.** The precuneus lies on the medial parietal cortex. It is commonly associated with autobiographical memory, self-related imagery, and mental context. This region is often described as supporting integration of personal and external information.	**Autobiographical Memory.** Autobiographical memory is commonly associated with the medial prefrontal cortex, posterior cingulate cortex, precuneus, angular gyrus, parahippocampal cortex, retrosplenial cortex, and lateral parietal cortex. This function supports remembering personal experiences, spatial context, and internally constructed events.
Visual		**Object Perception Network**	**Lateral Occipital Cortex**	**Visual Analysis**
Language		**Language Network**	**Middle Temporal Gyrus**	**Language Comprehension**
Somatomotor		**Somatomotor Network**	**Primary Somatosensory Cortex**	**Tactile Encoding**
Interoception		**Interoceptive Network**	**Anterior Insula**	**Interoceptive Processing**
Frontoparietal		**Frontoparietal Control Network**	**Inferior Frontal Gyrus**	**Working Memory**
Salience		**Salience Network**	**Anterior Cingulate Cortex**	**Executive Control**

Fig. R6. New Fig. 2. Auditory and default mode networks show the complete generated texts, and the other six their three generated titles. True is the atlas map. Maps show their top 5% of voxels in sagittal, coronal, and axial views. Supplementary Table S5 gives every text.

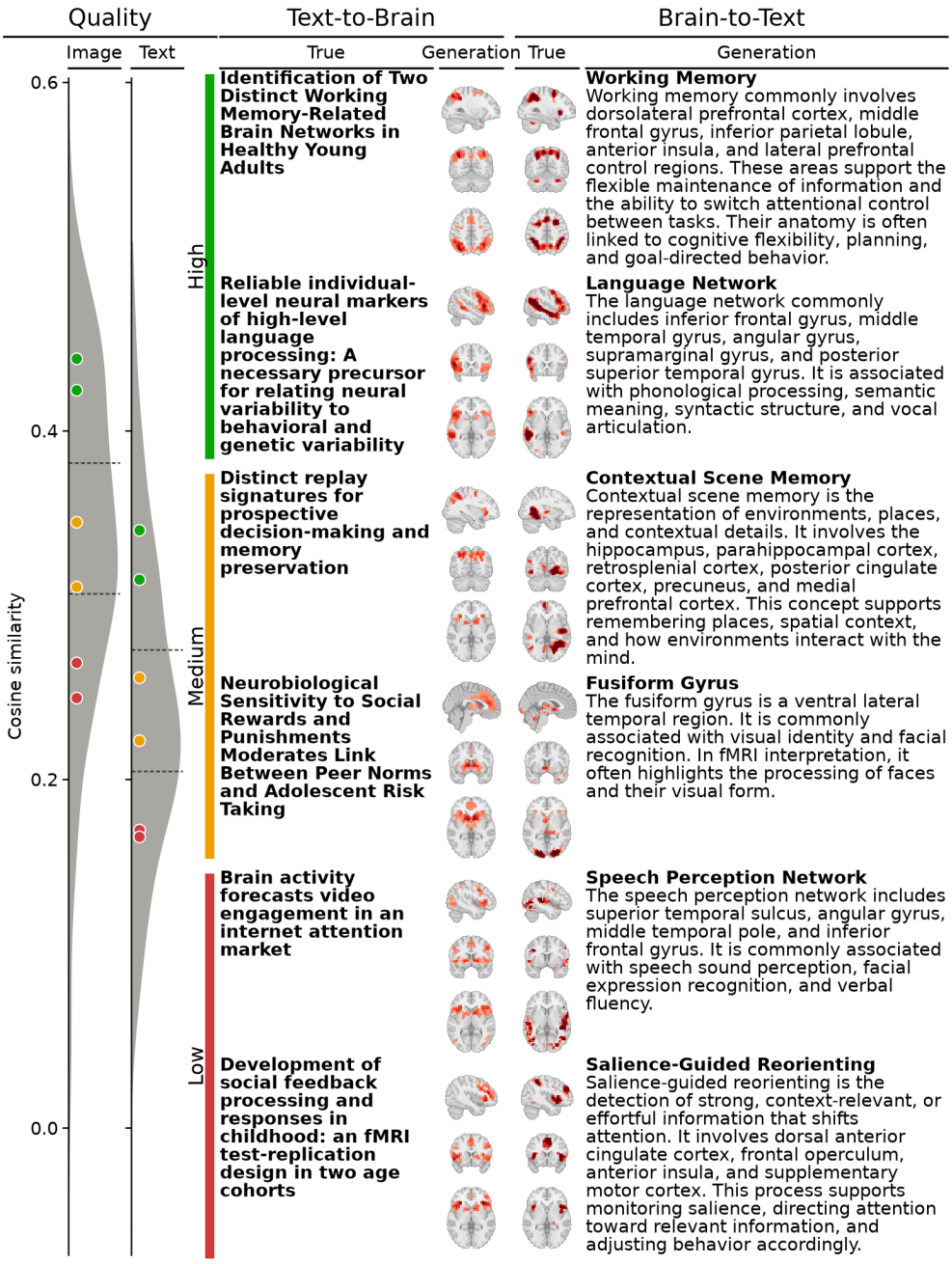


Fig. R7. New Fig. 3. NeuroVault papers from the high, medium, and low thirds of the shared-space cosines, chosen by a fixed rule (3.9). Low rows show a scattered true map (top) and a model miss on a clean map (bottom). Maps show their top 5% of voxels.



Fig. R8. New Fig. S4. Labeling of resting-state components from UK Biobank (top) and the Human Connectome Project (bottom). Bars give the five most similar labels from the contrastive models. The generated column is the [NETWORK] caption of the stage 2 MLP projector, its title in bold.

1.10: Overall assessment

Reviewer's comment

NeuroVLM is a creative and potentially useful framework, and the current results provide encouraging evidence that large collections of publication text and coordinate-derived maps can support cross-modal retrieval, broad functional labeling, and coarse generation. The manuscript would be considerably strengthened by independent validation of the generative outputs, clearer separation of datasets across all training stages, a revised NeuroVault evaluation, and more precise framing of the generated spatial representations. These revisions would help establish the appropriate scientific use cases and limitations of the framework.

Response: Comments 1.1, 1.2, 1.3, and 1.4 answer the four revisions named here, in the same order.

1.11: Code availability

Reviewer's comment

The repository is a useful community resource: it has an Apache-2.0 license, pip installation instructions, pretrained models and data, tutorials, tests, and CI across Python 3.10–3.13 on macOS and Ubuntu. The latest CI run passed 405 tests (one skipped), although reported code coverage was 47%.

The paper's results appear only partially reproducible. The repository includes data, training, and evaluation notebooks, but two preprocessing notebooks contain unresolved merge conflicts and are invalid; some analyses rely on manual steps, machine-specific paths, or external checkpoints; dependencies and Hugging Face artifacts are not fully pinned; and there is no figure-by-figure reproduction guide. Thus, the package appears usable for inference, but exact end-to-end reproduction would require additional fixes and documentation.

I reviewed the repository and existing CI results but did not install or run the code locally.

Response: We resolved the merge conflicts in both notebooks and added a figure-by-figure guide that names the script behind every paper figure. One command builds the pinned environment and fetches the Hugging Face inputs at pinned, checksummed revisions. It then retrains the model, redraws the figures, and checks each metric against the paper. Two runs from scratch gave byte-identical weights on the same GPU model and library versions.

Tests and coverage. Coverage is measured on the 57 files needed to train the models and draw the figures. The CNN and other experiments outside the paper are left out. Training takes hours, so continuous integration trains every model for one batch on CPU, which checks the whole pipeline end to end.

Remaining limits. Two inputs still come from outside the pinned set. They are the Automated Anatomical Labeling (AAL) atlas, whose download server fails, and the NiCLIP and NeuroConText outputs compared in Fig. S3.

Changes to the manuscript: New Data availability and Code availability sections after the Discussion, l. 954–986, replacing the links in Methods, Training, l. 185–190.

Reviewer 2

Reviewer's comment

The manuscript presents NeuroVLM, a unified framework for neuroimage–text contrastive learning, text-to-brain generation, and brain-to-text generation. The topic is meaningful, and the large-scale dataset provides potential value. However, the manuscript is not suitable for publication in its current form and several issues should be addressed.

Response: We addressed each issue (2.1 to 2.12).

2.1: Motivation of the Qformer

Reviewer's comment

1.The authors use a Q-Former for brain-to-text generation, but its motivation is unclear. Comparisons with simpler alternatives, such as an MLP projector or cross-attention module, should be provided.

Response: We agree. Our original Qformer attended over only two input tokens, the map latent and its semantic vector, so its attention had little to select. A multilayer perceptron (MLP) projector of the same size therefore replaces the Qformer in both stages. It raises stage 1 retrieval AUC from 0.82 to 0.90 (Fig. R9). A cross-attention module would also attend over only two tokens, so we did not add one.

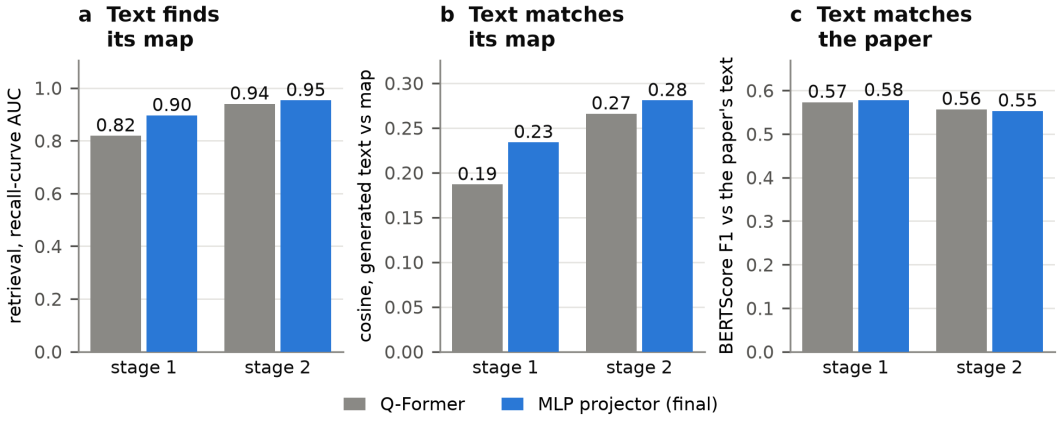


Fig. R9. Qformer and MLP projector. Gray bars show the original model with the Qformer and blue bars the final model with the MLP projector, on PubMed test maps. Panels show (a) retrieval AUC, (b) caption-to-map cosine, and (c) BERTScore F1 against the paper's text.

Changes to the manuscript: Methods, Text Decoder, l. 284–298, and Brain-to-Text, l. 341–362 (MLP projector). Results, Generative Models: Networks, l. 442–448 (projector named), and PubMed, l. 596–607. Supplementary Information, Brain-to-Text Projector, l. 1352–1374 and l. 1412–1428 (rewritten for the MLP projector), with Table S3 (p. 32). Figs. S1 and S4 captions (p. 23 and p. 34). New Supplementary Section Projector Comparison, l. 2073–2079, with Fig. S9 (p. 46).

2.2: Datasets and sample sizes per model

Reviewer's comment

2. The training data for the contrastive and generative models are not clearly described. The authors should specify the datasets and sample sizes used for each model and clarify whether the same image-text pairs were reused across different training stages.

Response: 1. Datasets and sample sizes. Fig. R3 and Table R7 give the data and sample size of each training stage (1.2).

2. Reuse across stages. Yes, the training papers are reused across stages. The autoencoder, the projection heads, and the stage 1 projector all learn from these papers. The anatomically centered head and the stage 2 projector learn from short concept texts, paired with maps built from training papers only (2.6). No validation or test paper trains these components. The language model's training script also removes them from its corpus (2.11).

Table R7. Training data per component. Training and validation sizes of each trained component of the final model, as in Fig. R3.

Training stage	Data	Train	Validation	Unit
Autoencoder	PubMed coordinate maps	23,894	2,987	papers
Text-to-brain and contrastive heads	SPECTER2 text and map latents	23,894	2,987	papers
Anatomically centered projection head	synthetic concept text–map pairs from training papers	6,309	64	pairs
Stage 1 projector (MLP)	map latents and LLM summaries	22,547	2,837	papers
Stage 2 projector (MLP)	[NETWORK], [REGION], [FUNCTION] texts (1,106 of 1,338 kept)	972	134	texts
NeuroQwen3-0.6B	PubMed titles and abstracts	1.22 M	12,317	articles

Changes to the manuscript: Methods, Training, l. 159–173. New Supplementary Section Data Flow and Training Details, l. 1745–1753, with Fig. S5 (p. 39) and Table S6 (p. 40), as for comment 1.2.

2.3: NeuroVault selection bias

Reviewer's comment

3. The NeuroVault evaluation has selection bias. When multiple images correspond to the same title and abstract, the contrastive model selects the most similar image. This procedure may substantially inflate the reported performance.

Response: We agree that choosing the most similar map inflates the score. That map is reported as an upper bound (1.3). A lower bound keeps each paper's worst-matching map, and a multi-target score counts any of the paper's maps (Fig. R4). Each rule has its own chance curve, from texts shuffled across papers. Under every rule, the model's AUC stays about 0.1 above chance.

Changes to the manuscript: Fig. 5b caption (p. 16). Results, Contrastive Model: NeuroVault, l. 754–767. Supplementary Information, NeuroVault preprocessing, l. 1177–1192 (map types and exclusions) and l. 1204–1210 (bounds). New Supplementary Section NeuroVault Bounds and Map Types, l. 1954–1963, with Fig. S7 and Table S9 (p. 44), as for comment 1.3.

2.4: Data leakage

Reviewer's comment

4.Potential data leakage across PubMed, Neurosynth Compose, and NeuroVault has not been sufficiently excluded. DOI-based filtering alone may not identify duplicate records, preprint and journal versions, or repeated coordinate tables.

Response: We audited leakage beyond DOI filtering. No DOI or PubMed identifier is shared across splits. The three checks below flag 16 test and 26 validation papers (Table R8). Every evaluation leaves them out. We also removed seven NeuroVault papers that duplicate a training paper.

1. Duplicate records. Each held-out paper was compared with the training and NeuroVault papers by fuzzy title matching and by the similarity of its title and abstract. Among the papers they flag is one paper stored as two records.

2. Preprint and journal versions. The same checks flag a preprint and its journal version, which carry different DOIs.

3. Repeated coordinate tables. The coordinate tables from PubMed and Neurosynth Compose were compared across all splits. Identical or near-identical tables mark a paper that repeats another paper's peaks.

Other training data. The synthetic maps of the anatomically centered head had pooled held-out papers, so we rebuilt them from training papers only. This fix and the removals each change AUC, Dice, and correlation by less than 0.01. The language model's training script also removes all evaluation papers from its corpus (2.11).

Table R8. Leakage audit. Held-out PubMed papers flagged by each check against other papers. Rows overlap, and the last row is their union.

Check against other papers	Test set (2,987)	Validation set (2,987)
Near-identical title	4	6
Near-identical title and abstract	3	6
Identical coordinate table	6	12
Near-identical coordinate table	12	21
Removed from every evaluation	16	26

Changes to the manuscript: Methods, Training, l. 169–181 (synthetic maps from training papers only, leakage audit, evaluation exclusions, and the evaluation papers removed from the language-model corpus). Methods, Datasets, l. 142–144 (NeuroVault papers that duplicate a training paper removed). Supplementary Information, NeuroVault preprocessing, l. 1177–1192.

2.5: Competitive baselines

Reviewer's comment

5.The manuscript lacks competitive baselines for the generative tasks. The proposed methods should be compared with existing approaches and simpler projection or retrieval-based baselines.

Response: Text to brain. Fig. R1 and Table R1 compare NeuroVLM's maps with existing approaches and simple baselines (1.1). NeuroVLM leads Neurosynth, NeuroQuery, mean and domain maps, and atlas labeling. It narrowly leads linear regression, the simple projection baseline, and

leads the nearest training paper's map, the retrieval baseline. Text2Brain, a published model, trails NeuroVLM (Dice 0.49 against 0.53), most on papers outside its training corpus.

Brain to text. We know of no published model that writes text from group-level brain maps. Existing decoders NiCLIP and NeuroConText rank fixed texts, and NeuroVLM is slightly ahead of both (Fig. S3). A simpler MLP projector beats the Qformer and replaces it (2.1). As a retrieval baseline, the nearest training caption is no closer to the paper's text than generated captions, and it carries details of other studies (3.3).

Changes to the manuscript: Methods, Text-to-Brain, l. 365–385 (final head), and Performance Metrics, l. 398–404 and l. 415–426. Results, Generative Models: PubMed, l. 596–607, and Fig. 4 caption (p. 15). New Supplementary Section Text-to-Brain Baselines and Retrieval, l. 1843–1859, with Fig. S6 (p. 42) and Table S8 (p. 43), as for comment 1.1.

2.6: Summarized and synthetic text

Reviewer's comment

6.The construction of the summarized and synthetic text data is unclear. The authors should describe the generation procedure and provide representative examples.

Response: One prompt generated the paper summaries and another generated the synthetic and curated texts. Both prompts are given below, and Table R9 gives a complete example of each kind of text.

Summaries. Llama-3.3-70B-Instruct summarized each paper with the first prompt as its system message, removing low-level study details. The user message was the paper's title and abstract, joined by a separator token. Decoding sampled with temperature 0.3 and top-p 0.9, so a rerun words each summary differently. Summaries re-created for test-set papers cover the same topics as the released ones. The stage 1 projector learns from these summaries.

Synthetic texts. Qwen3.5-27B wrote short entries with the second prompt. The seeds were region names from standard brain atlases, Cognitive Atlas concepts, and the network names of our network atlases. The user message was the seed alone, such as "Insula". Decoding used temperature 0.1 and top-p 0.9, with thinking mode off. Sampling each seed several times gave re-wordings of one concept. The anatomically centered head learns from these entries.

Curated texts. The stage 2 projector learns from 1,338 curated texts about networks, regions, and functions. Qwen3.5-27B wrote them in the same way. The original generation code was not kept. A script in the code repository repeats the process, and its texts match the released texts of the same seeds in content (MiniLM cosine 0.88 on average).

```
Model: meta-llama/Llama-3.3-70B-Instruct
Decoding: sampling, temperature 0.3, top-p 0.9, top-k 50, at most 256 new tokens
User message: title[SEP]abstract
System message:

You are writing a short, topic-focused neuroscience mini-review.

INPUT:
You will receive:
    title + " [SEP] " + abstract from an fMRI publication.

GOAL:
Write a single coherent paragraph (4-6 sentences) that:
- Describes the scientific topic implied by the paper (e.g., meditation and the default mode
  network, exercise and working memory), not the paper's methods.
- Explains the key cognitive processes (e.g., working memory, self-referential thought,
  emotion regulation) and the main brain systems or networks involved.
- May mention typical effects or relationships in general terms (e.g., ‘‘has been associated
  with changes in connectivity'' or ‘‘can influence executive control''), but should stay high-level.

ALLOWED:
- General statements about effects or relationships (e.g., ‘‘exercise has been associated
  with improved cognition'', ‘‘meditation can alter DMN connectivity'').
- Generic references to ‘‘studies'' or ‘‘research'' in the field.
- Qualitative descriptions of brain activity or connectivity (e.g., ‘‘increased activation
  in prefrontal regions'', ‘‘changes in functional connectivity'').

AVOID:
- Exact numerical details (sample sizes, p-values, coordinates, time points).
- Detailed task or protocol descriptions from the specific study.
- Directly quoting or closely paraphrasing long sentences from the abstract.
- Overly narrow focus on this single experiment; write as if summarizing the topic area.

STYLE:
- Neutral, scientific tone, like a short review or encyclopedia entry.
- One paragraph only. No bullet points or headings.
- Focus on giving a clear sense of ‘‘what this topic is about'' in the context of the field.
```

```
Write a short educational neuroscience entry for the provided fMRI concept.

Requirements:
- Explain the concept in general neuroscience terms.
- Always include relevant anatomical regions.
- Connect the anatomy to the cognitive or neural function.
- Use cautious, broad language; avoid overclaiming.
- Do not mention studies, patients, diseases, disorders, treatments, age groups, tasks, results,
  methods, or statistics.
- Do not use: critical for, responsible for, causal, proven, this study, activation, altered,
  reduced, increased.
- Do not invent highly specific subregions unless they are commonly associated with the concept.
- If the concept is broad, name the main commonly associated regions or networks.
- If the concept is anatomical, describe its broader network/function associations.
- Write 3-5 complete sentences.

Return valid JSON only:
{
  "title": "1-5 words",
  "description": "3-5 sentences"
}
```

Table R9. Examples of the training texts. Each row gives one complete text and the training stage that uses it. Llama-3.3-70B-Instruct wrote the summaries and Qwen3.5-27B the synthetic and curated texts.

Text (used by)	Complete example
Summary of a training paper (stage 1 projector)	The human parietal and opercular regions are involved in processing somatosensory information. Somatotopy refers to the organization of these regions according to the location of sensory inputs on the body. The parietal cortex, including areas such as the postcentral gyrus and intraparietal sulcus, contains multiple somatosensory areas with distinct somatotopic representations. Attentional modulation can enhance activity in these areas, particularly in higher-order regions.
Synthetic text "Insula" (anatomically centered head)	The insula is a deep cortical region situated within the lateral sulcus of the cerebral hemisphere. This area maintains connections with structures such as the anterior cingulate cortex and the amygdala. Processes occurring in this region are often linked to perceptions of internal bodily states and emotional awareness. It contributes to the integration of sensory information with subjective experience across various brain networks.
Curated [NETWORK] text "Salience Network" (stage 2 projector)	The salience network commonly includes anterior insula and dorsal anterior cingulate cortex. It is associated with detecting and prioritizing behaviorally relevant internal and external information. Its connections with limbic, autonomic, and frontoparietal systems support flexible shifts between internal monitoring and externally oriented control.

Changes to the manuscript: Methods, Brain-to-Text, l. 341–362. Supplementary Information, Text preprocessing, l. 1095–1174 (both prompts, their models, seeds, and settings, and the examples).

2.7: Dataset sizes

Reviewer's comment

1.The reported dataset sizes are inconsistent. The Methods section reports 3,000 NeuroVault images and 100 network images, whereas other sections report 4,553 and 300, respectively.

Response: As for comment 1.5, we recomputed every count from the released data and corrected the text and captions (Table R5). The supplement's 4,553 maps predate image quality control and the removal of papers already in PubMed. Neither network-image count can be reproduced, so the released counts replace both.

Changes to the manuscript: Methods, Datasets, l. 132–144. Results, Contrastive Model: Network Labeling, l. 735–747, and Autoencoder, l. 826–830. Figs. 4 and 5 captions (pp. 15–16). Supplementary Information, NeuroVault preprocessing, l. 1177–1192 and l. 1209–1211, as for comment 1.5.

2.8: Fig. 5 caption

Reviewer's comment

2.The caption of Fig. 5 reports both 0.78 and 0.85 for the NeuroVault AUC.

Response: The corrected caption names the curve behind each value. With each paper's best-matching map, the AUC is 0.85 from text to brain and 0.79 from brain to text (Fig. R10b). P-maps are inverted, so that high values mark strong effects, as for comment 1.3. Each curve has its own dashed chance curve, from texts shuffled across papers.

Lower bound. The lower bound and its chance curve lie below 0.50 by construction. The lower bound keeps the map that the model scores lowest for each paper. That map ranks below a typical candidate even when the texts are shuffled. The model still beats this chance level.

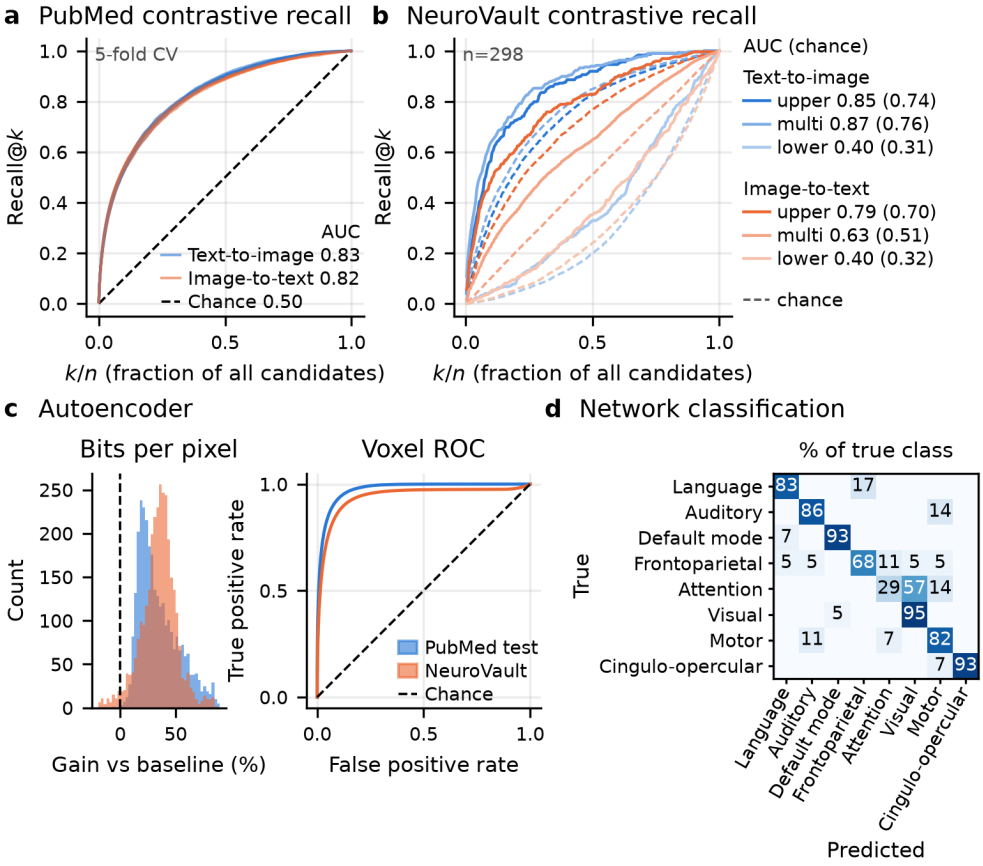


Fig. R10. New Fig. 5. Contrastive recall on (a) PubMed and (b) NeuroVault, (c) autoencoder reconstruction, and (d) network classification. In (b), each paper's target is its best map (upper bound), every map (multi-target), or its worst map (lower bound), and dashed lines show chance.

Changes to the manuscript: Fig. 5 caption (p. 16). Results, Contrastive Model: NeuroVault, l. 754–767, and PubMed, l. 771–780.

2.9: SPECTER naming

Reviewer's comment

3. The use of "SPECTER," "SPECTER2," and "Specter" is inconsistent. The exact model used in each component should be clarified.

Response: Every text-encoding step uses SPECTER2, the allenai/specter2_aug2023refresh model with its ad hoc query adapter (1.6). The original SPECTER is never used. We corrected every mention to SPECTER2. The language model is NeuroQwen3-0.6B, which is Qwen3-0.6B adapted to PubMed abstracts (2.11). Llama-3.3-70B-Instruct wrote the paper summaries, and Qwen3.5-27B wrote the synthetic and curated texts (2.6).

Changes to the manuscript: Introduction, l. 107–111. Methods, Text Encoder, l. 220–227, and Text-to-Brain, l. 365–373. Supplementary Information, Text preprocessing, l. 1085–1088. Fig. S1 caption (p. 23), as for comment 1.6.

2.10: LoRA analogy

Reviewer's comment

4.In Section 2.9.2, describing an MLP projection head as similar to a LoRA adapter is technically inaccurate.

Response: We agree and deleted the LoRA analogy (1.7). The head is a small network trained on frozen SPECTER2 embeddings. The revision calls it an anatomically centered projection head, not an adapter.

Changes to the manuscript: Methods, Text-to-Brain, l. 365–385 (LoRA sentence deleted, projection head named). Supplementary Information, Projection Head, l. 1272–1294, as for comment 1.7.

2.11: Training details

Reviewer's comment

5.The authors should report the model parameters, GPU hardware, training time, random seeds, Qwen fine-tuning strategy, and relevant training hyperparameters.

Response: Table R10, a new Supplementary Table, gives the parameters, training data, epochs, optimizer, learning rate, batch size, seed, and training time of every component. All components except the language model train on one RTX 5090 GPU, with fixed seeds and deterministic kernels. The whole pipeline, five-fold cross-validation included, trains from scratch in about 1.8 hours.

Qwen fine-tuning strategy. NeuroQwen3-0.6B is Qwen3-0.6B with all weights fine-tuned, without adapters, on PubMed titles and abstracts (Table R10). It trained for one epoch on one H100 GPU, and its training time was not recorded. Its training script removes the validation, test, and NeuroVault evaluation papers from this corpus. Comment 1.2 reports a memorization check.

Table R10. Training details. Rows give the trainable parameters, training data, epochs, optimizer settings, seed, and training time of each component of the final model. Times are for one RTX 5090 GPU, except for NeuroQwen3-0.6B.

Component	Parameters	Training data (train / validation)	Epochs (selection)	Optimizer, learning rate, batch	Seed	Time
Autoencoder	59.9 M	PubMed maps, 23,894 / 2,987	101 (lowest validation loss)	AdamW, 5e-5, 256	0	10 min
Contrastive heads, text and image	0.59 M and 0.30 M	text and map latents, 23,894 / 2,987	301 (last)	AdamW, 1e-5, 2,048	123	6 s
Text-to-brain head	0.59 M	text and map latents, 23,894 / 2,961	30 of at most 80 (best validation Dice)	AdamW, 1e-3, 512	0	22 s
Anatomically centered head	3.8 M	synthetic pairs, 6,309 / 64, and PubMed training pairs	200 of 2,000 (best PubMed validation recall)	AdamW, 2e-4, 2,048	123	103 s
Stage 1 projector (MLP)	28.3 M	maps and LLM summaries, 22,547 / 2,837	4 of at most 30 (lowest validation loss, early stopping)	AdamW, 3e-4 with cosine decay, 32	17	17 min
Stage 2 projector (MLP)	28.3 M	concept texts and maps, 972 / 134	30 (last)	AdamW, 1e-4 with cosine decay, 18	3 (best of 1–5)	5 min per seed
NeuroQwen3-0.6B	596 M	PubMed titles and abstracts, 1.22 M / 12,317 (385 M tokens)	1	AdamW, 5e-6 with cosine decay, 16 sequences of 1,024 tokens, bf16	12345	not recorded (one H100)

Changes to the manuscript: Methods, Training, l. 174–190 (exclusions, the evaluation papers removed from the language-model corpus, and training time). Methods, Text Decoder, l. 274–283 (NeuroQwen3-0.6B training). Supplementary Information, Autoencoder training, l. 1265–1268. New Supplementary Table S7 with the training details per component (p. 40).

2.12: Random results and failure cases

Reviewer's comment

6.Figs. 2 and 3 mainly show selected qualitative examples. Randomly selected results and representative failure cases should also be provided.

Response: We added random results and failure cases for every evaluation set. They include random test papers, the worst test-set cases in both directions, the weakest atlas networks, and NeuroVault papers that fail on their best-matching map. Failures are not rare, since 16% of test-set captions have a negative similarity to the paper's text. Fig. R11 shows the six worst brain-to-text cases. The other five sets are in the code repository.

Invented names. With unsteered title decoding, 23% of test-set [NETWORK] titles and 9% of [REGION] titles matched no canonical or atlas name, such as "Cerebro-Cervico-Brain". The language model still generates every title, but it is steered toward the canonical names. At the title position, tokens that cannot continue a canonical network or region name are masked, and the model's own probabilities choose the name. Nothing is classified or looked up.

Body text. The body text and [FUNCTION] titles are generated without this mask. They can still contain invented words or stray fragments, such as "useary" in Fig. R11, a limitation the Discussion

states.

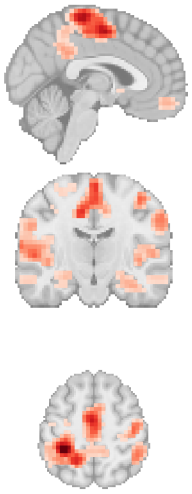
Non-Latin text. A few captions contained non-Latin characters from the language model's multilingual vocabulary. Decoding uses Latin-script tokens only, which removes them.

Changes to the manuscript: Figs. 2, 3, and S4 (p. 11, p. 12, and p. 34), with the Low rows of Fig. 3 as failure cases. Methods, Text Decoder, l. 299–306 (title decoding steered toward the canonical names, and Latin-script decoding). Supplementary Information, Brain-to-Text Projector, l. 1406–1413. Discussion, limitations, l. 899–905 (failure rate, typical failures, and invented words or stray fragments in the body text).

PubMed test set: the 6 worst brain-to-text cases (1-2 of 6)
The 6 test maps whose generated description is least similar to the paper's own text (title + summary; cosine in the shared text-brain space), out of all PubMed test maps. Maps: top 10% of voxels; texts complete.

1. PMID 27592810

true map



brain to text: cosine -0.713 (percentile 0.02 of 2,971)

Paper text (title + summary)

Retrieved emotional context influences hippocampal involvement during recognition of neutral memories.
During memory recognition, the emotional context in which a neutral memory is encoded can influence its retrieval and involvement of the hippocampus. The hippocampus, which plays a crucial role in memory formation and consolidation, interacts with the amygdala, a structure involved in emotional processing, to facilitate the encoding and retrieval of emotionally arousing experiences. When an emotionally arousing context is retrieved, the hippocampus, particularly an area adjacent to the amygdala, exhibits heightened activation and increased functional connectivity with the parahippocampal gyrus, a region involved in processing visual information. This interaction between the hippocampus and amygdala can lead to confident judgments of recognition for neutral memories, even after a delay. The retrieval of an emotionally arousing context can thus shape the memory of neutral events, influencing their recognition and recall. This highlights the dynamic nature of memory and its susceptibility to the emotional context in which it is retrieved.

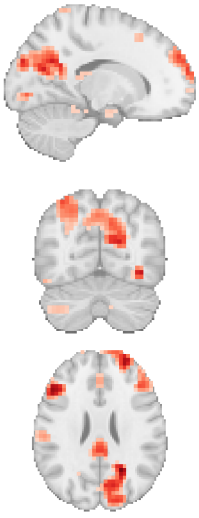
Generated description (complete)

[NETWORK] Sensorimotor Network The sensorimotor network includes precentral and postcentral gyri, as well as supplementary motor area. It is broadly tied to the execution of movement-related actions with high temporal precision. The network is often described as a tightly coupled system for bodily sensation and control. Its hubs are thought to support coordinated feedback from body parts during action planning or execution. **[REGION] Postcentral Gyrus** The postcentral gyrus is a primary somatosensory region in the brain. It contains cortical regions that support tactile and proprioceptive information processing related to body contact.
describe when it was used.
useary: When describing bodily sensorimotor representations, this region often serves as an intermediate hub for touch-related signals from fingertips or other skin surfaces of the hand or foot. **[FUNCTION] Body Sensory Mapping** Body sensory mapping is the cortical representation of body parts and their interactions with touch, pressure, proprioception, vibration, and temperature. It involves primary somatosensory cortex in regions such as parietal lobule 4B (PLC-4b), posterior insula, premotor cortex, thalamus, and supplementary motor area. This concept supports tactile navigation, spatial awareness, coordination between senses, and integration into self-motion perception.

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2. PMID 21715612

true map



brain to text: cosine -0.671 (percentile 0.05 of 2,971)

Paper text (title + summary)

Reciprocal anatomical relationship between primary sensory and prefrontal cortices in the human brain.
In the human brain, there is a reciprocal anatomical relationship between primary sensory cortices, such as those responsible for processing visual and auditory information, and the prefrontal cortex, a region involved in executive functions and decision-making. The gray matter volume of primary sensory cortices, including the visual and auditory cortices, is inversely correlated with the gray matter volume of the anterior prefrontal cortex. This means that as one area increases in size, the other decreases, forming a trade-off between the two regions. This relationship is observed along distinct anatomical pathways, with the dorsal and ventral parts of primary sensory cortices showing anticorrelation with the dorsolateral and ventromedial parts of the anterior prefrontal cortex, respectively. This anatomical trade-off suggests that primary sensory cortices and the anterior prefrontal cortex may be influenced by shared genetic or developmental factors, challenging the traditional view that brain area sizes simply scale with overall brain size.

Generated description (complete)

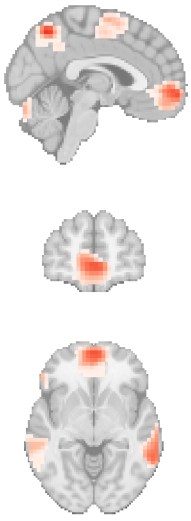
[NETWORK] Semantic Network The semantic network includes medial temporal cortex, hippocampus, anterior cingulate gyrus, and posterior parietal lobule. It is associated with internally-focused representations of the world—how things are connected or arranged in space. Its connections involve prefrontal cortex, amygdala, basal ganglia, retrosplenial cortex, and lateral occipital regions. This network helps organize internalized meaning into external experiences for processing.
[REGION] Medial Prefrontal Cortex Broadly tied to the processing of emotion and value, this region is often engaged during deliberation over personal preferences. It is associated with internal thought about mood or behavior. Its activity tends to accompany reflective evaluation rather than direct action. It is generally linked to self-awareness and long-term memory.
[FUNCTION] Memory Processing Memory processing commonly involves the hippocampus, amygdala, retrosplenial cortex, parahippocampal gyration, medial prefrontal cortex, and lateral temporal lobe. These regions support encoding of event-related information, retrieval of contextual memory components, integration with affective states, spatial context for episodic events, and semantic knowledge about objects or concepts.

Fig. R11. Worst brain-to-text cases. The six PubMed test maps whose caption is least similar to the paper's text, two per page. Each shows the map in sagittal, coronal, and axial views, the paper's title and summary beside the complete generated caption, and the caption's similarity with its percentile. (Page 1 of 3.)

PubMed test set: the 6 worst brain-to-text cases (3-4 of 6)

3. PMID 26485148

true map



brain to text: cosine -0.585 (percentile 0.08 of 2,971)

Paper text (title + summary)

Human-Robot Interaction: Does Robotic Guidance Force Affect Gait-Related Brain Dynamics during Robot-Assisted Treadmill Walking? During robot-assisted treadmill walking, the brain's sensorimotor cortex plays a crucial role in locomotor control. This region exhibits gait-related spectral modulations in the mu, beta, and low gamma bands, which are associated with specific phases of the gait cycle. When a robotic device provides assistance, such as the Lokomat gait orthosis, it can affect the brain's activity patterns, particularly in the right primary sensory cortex. Specifically, mu and beta rhythms are suppressed in this area during robot-assisted walking with high guidance force, indicating a reduced involvement of the sensorimotor area. In contrast, a lower level of guidance force allows for a more active participation of the sensorimotor cortex, which is essential for motor learning. This suggests that optimal training parameters for robot-assisted treadmill walking should minimize the use of high guidance force to maximize the engagement of the sensorimotor cortex.

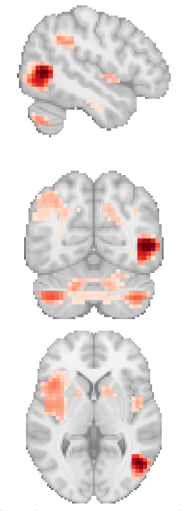
Generated description (complete)

[NETWORK] Language Network The language network includes inferior frontal cortex, middle temporal gyrus, posterior cingulate layer and lateral occipital cortex. It is broadly associated with comprehension of words, meaning, speech production, linguistic structure, semantics, and conceptual integration across domains. Its distributed organization supports the flexible mapping between semantic knowledge and phonological processing.
[REGION] Middle Frontal Gyrus The middle frontal gyrus is a large lateral region in the brain. It commonly involves tasks related to language and conceptual understanding. In fMRI, this region is often associated with controlled selection of information from memory or speech-related contexts.
[FUNCTION] Semantic Processing Sensory processing commonly includes anterior temporal cortex, middle frontal regions, and inferior parietal areas. These regions support meaning-based retrieval of words or concepts from stored knowledge. Semantic analysis often involves ventral visual word system, posterior medial prefrontal cortex for conceptual integration, and superior temporal gyrus for linguistic mapping.

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4. PMID 32595302

true map



brain to text: cosine -0.583 (percentile 0.12 of 2,971)

Paper text (title + summary)

Autism and Beauty: Neural Correlates of Aesthetic Experiences in Autism Spectrum Disorder In individuals with Autism Spectrum Disorder (ASD), the neural correlates of aesthetic experiences differ from those of typical individuals. When performing aesthetic judgments, individuals with ASD exhibit altered activity in key brain regions, including the temporoparietal area and insula, which are involved in processing and integrating sensory information. In contrast, the anterior region of the superior frontal gyrus, a region associated with executive functions and decision-making, shows reduced activity in individuals with ASD. Additionally, the amygdala, a structure involved in emotional processing, and the posterior region of the middle/inferior temporal gyrus, a region involved in visual processing, exhibit increased activity in individuals with ASD during aesthetic judgments of fractal images. These findings suggest that individuals with ASD may process and experience aesthetic stimuli in distinct ways compared to typical individuals.

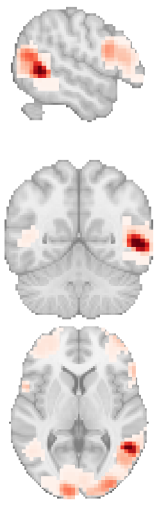
Generated description (complete)

[NETWORK] Cerebro-Cerebellar Network The cerebrally located network includes the lateral parietal cortex, prefrontal lobes, anterior cingulate gyrus, and posterior cingulum. It is commonly associated with coordination of action across time frames (e.g., timing), prediction based on current information, cognitive control over movement execution, regulation of attention span, and interoceptive awareness.
[REGION] Dentate Nucleus The dentate nucleus is a lateral region of the brain located near the cerebellum. It is associated with motor learning and maintenance, as well as cognitive functions related to action planning. Its anatomy links movement patterns with attentional control.
[FUNCTION] Predictive Coordination Predictive coordination is the integration of anticipatory and current action to optimize performance. It involves cerebellar vermis, dentate nucleus, thalamus, globus pallidum, ventral tegmental area, parietal cortex, prefrontal regions, motor cortex, sensory cortices, basal ganglia, and association cortex. This concept supports prediction-based decision-making across cognitive domains.

PubMed test set: the 6 worst brain-to-text cases (5-6 of 6)

5. PMID 31866810

true map



brain to text: cosine -0.577 (percentile 0.15 of 2,971)

Paper text (title + summary)

Chronic Tinnitus Exhibits Bidirectional Functional Dysconnectivity in Frontostriatal Circuit Chronic tinnitus is associated with abnormal functional coupling between the nucleus accumbens (NAc) and the prefrontal cortex, particularly the middle frontal gyrus (MFG), orbitofrontal cortex (OFC), and inferior frontal gyrus (IFG). This bidirectional dysconnectivity in the frontostriatal circuit may contribute to the evaluation and modulation of sensory signals. In individuals with chronic tinnitus, the NAc exhibits aberrant directional connectivity with the prefrontal cortex, which is linked to tinnitus characteristics, including distress and duration. Specifically, increased connectivity from the right NAc to the left MFG and from the right MFG to the left NAc is positively correlated with tinnitus handicap, while enhanced connectivity from the right NAc to the right OFC is associated with longer tinnitus duration. These findings suggest that the frontostriatal circuit plays a crucial role in the neural basis of tinnitus sensation and may provide insights into the neuropathological features of the condition.

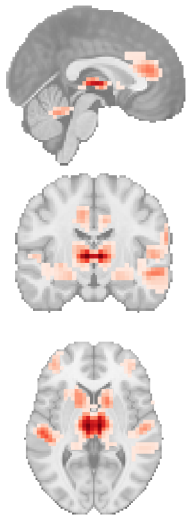
Generated description (complete)

[NETWORK] Ventral Attention Network The ventral attention network includes the frontal eye field, dorsal frontoparietal cortex, and lateral occipital regions. It is broadly associated with spatially guided attention. Its anatomy links to how visual information is processed in a coordinated manner for effective foveal vision during saccadic motion detection. The circuit supports both top-down directed attention and bottom-up involuntary responses to visually salient stimuli.
[REGION] Lateral Occipital Cortex The lateral occipital cortex is a region of ventral visual pathways. It supports analysis of shape and form, describing how objects appear in space. In general, it occupies the dorsal surface of inferior temporal sulcus.
[FUNCTION] Motion Perception Motion perception is the ability to detect and interpret movement in space. It involves motion-related visual cortex along with inferior frontal areas for downstream analysis of spatial relationships between objects or events. The network includes dominant occipital, intraparietal, superior temporal, dorsal frontoparietal, ventral premotor, and lateral prefrontal regions. Its anatomy supports both object-based and event-driven understanding of spatial information.

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6. PMID 29542095

true map



brain to text: cosine -0.576 (percentile 0.19 of 2,971)

Paper text (title + summary)

Tuning down the hedonic brain: Cognitive load reduces neural responses to high-calorie food pictures in the nucleus accumbens. When individuals are under cognitive load, neural responses to high-calorie food pictures in the brain's reward system, specifically the nucleus accumbens, are reduced. This region is involved in processing rewarding and pleasurable stimuli, such as food. Cognitive load, which can be induced by tasks requiring mental effort, appears to dampen the brain's response to high-calorie food images, potentially reducing their appeal. This effect may be related to the brain's ability to prioritize tasks and allocate resources, leading to a decrease in the neural processing of distracting stimuli like high-calorie food. The reduction in neural responses to high-calorie food pictures in the nucleus accumbens may contribute to a decrease in food cravings and consumption.

Generated description (complete)

[NETWORK] Auditory Network The auditory network includes primary and secondary complex acoustic networks, supported by the superior temporal gyrus. It is broadly tied to pitch, rhythm, timbre, speech perception, music processing, and sensorimotor sound experience. Its anatomy spans Heschl region, inferior frontal regions for feature association, parietal cortex for spatial organization of sounds, and posterior central sulcus areas involved in holistic perceptual systems.
[REGION] Brainstem This region is broadly associated with the regulation of arousal and the modulation of basic physiological processes. It is often linked to wakefulness, attentional control, and fundamental brain functions such as sleep-wake cycles. Its activity tends to correlate with general state transitions in the nervous system.
[FUNCTION] Early Auditory Processing This function supports the perception of basic auditory signals, such as sound waves and natural environmental sounds. It is associated with primary auditory cortex in temporal-limbic regions around the superior temporal gyrus and frontal-associated belt areas. This representation broadens the cortical basis for early acoustic signal processing.

Reviewer 3

Reviewer's comment

Hammonds et al. introduce a meta-analysis framework for aligning text-neuroimage pairs in a shared latent space. The framework can retrieve one modality from the other by comparing and ranking the latent vector embeddings. The framework can also generate one modality from the other by passing the latent vector embeddings through generative text and brain decoders. The authors tested the framework on multiple neuroimaging datasets. They found statistically significant retrieval and generation performance with sensible qualitative examples.

Response: No change was requested.

3.1: Novelty and what the model learns

Reviewer's comment

Augmenting existing meta-analysis tools with the flexibility provided by LLMs is an important and promising research direction. The manuscript appears technically sound and the NeuroVLM architecture seems appropriate for integrating the contrastive and generative models. However it is less clear how NeuroVLM compares to previous meta-analysis methods. The manuscript would benefit from a) clarifying the novelty of the framework by demonstrating the importance of generation and context and b) interpreting what the model learns.

Response: 1. Generation and context. Generated maps match each test paper's map better than the map of the nearest training paper (1.1). By MiniLM cosine, generated captions are as close to the paper's text as retrieved captions. Retrieved captions often report details of another study, and generated captions rarely do (3.3). Context also matters, since SPECTER2 beats a context-free word2vec encoder on every measure, by a modest margin (3.4).

2. What the model learns. Decoded latent dimensions match the Neurosynth maps of their top terms (3.5). Swapping one word moves the generated map toward the matching Neurosynth contrast, but negation has no effect. The cross-entropy loss matters most for retrieval.

Changes to the manuscript: Introduction, l. 99–105, Results, Generative Models: PubMed, l. 608–615, and Discussion, l. 863–868, which state the contribution of generation and context.

3.2: Comparison with existing methods

Reviewer's comment

My main concern is about the scope of the manuscript. NeuroVLM is clearly a useful tool but it is difficult to evaluate how much it improves on previous work. If the main contribution is higher performance then it would be helpful to include more direct comparisons. The only comparison between NeuroVLM and existing methods is in Figure S3 where it had comparable performance to NiCLIP and NeuroConText (albeit with fewer parameters). If the main contribution is instead more capabilities then it would be helpful to emphasize what NeuroVLM can do that previous methods cannot.

Response: 1. Direct comparisons. NeuroVLM gives the best text-to-brain maps, narrowly ahead of linear regression and ahead of Text2Brain, NeuroQuery, and Neurosynth (Fig. R1). Each method turns a test paper's title and abstract into a map, which is scored against that paper's map (Dice 0.53 against at most 0.49). In retrieval, NeuroVLM is slightly ahead of NiCLIP and NeuroConText on test papers that neither was trained on (Fig. R12).

2. New capabilities. NeuroVLM is, to our knowledge, the only model that writes text from group-level brain maps. It retrieves and generates in both directions through one shared space. Its generated maps and captions match or beat retrieved ones (3.3). The Introduction emphasizes these capabilities.

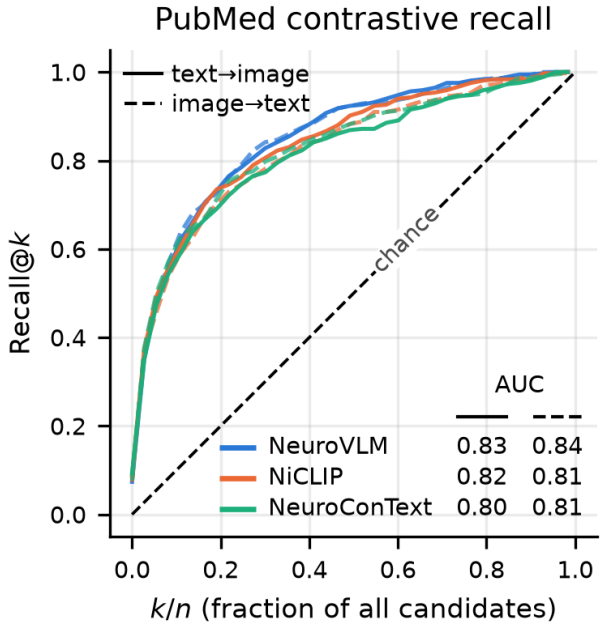


Fig. R12. New Fig. S3. Recall@k for NeuroVLM (blue), NiCLIP (orange), and NeuroConText (green) on the 366 PubMed test papers that none of the three models was trained on. Solid lines are text to image and dashed lines image to text.

Changes to the manuscript: Introduction, l. 99–105, on the new capabilities. Results, Generative Models: PubMed, l. 608–615, and Contrastive Model: PubMed, l. 771–780. Supplementary Information, Model Comparison, with Fig. S3 redrawn from the final model (p. 33). New Supplementary Section Text-to-Brain Baselines and Retrieval, l. 1843–1859, with Fig. S6 (p. 42).

3.3: Generation versus retrieval and granularity

Reviewer's comment

One capability that sets NeuroVLM apart from previous methods is that it can generate original text and neuroimages rather than only retrieving from a fixed set. However it would be helpful to show that the generated outputs are in fact more accurate and useful than retrieved outputs. For example if NeuroVLM uses an overfit decoder that collapses into a fixed set of outputs then it might be no better than existing retrieval methods. In fact the canonical projection matrix might explicitly drive the framework towards more retrieval-like outputs. Is there a way to quantify the granularity of NeuroVLM based on its output space?

Response: 1. Generated versus retrieved outputs. Generated maps are closer to each test paper's map than retrieved maps (1.1). By MiniLM cosine, generated captions are at least as close to the paper's text as the nearest training caption (Table R11). BERTScore favors training captions, even random ones, because they share the reference's summary style. Retrieved captions often carry another study's details, and generated captions rarely do (57% against 2%).

2. Collapse. The decoder does not collapse to a fixed set of outputs, since over 99% of test captions are unique (Fig. R13). The language model generates every caption. Network and region

titles are steered to canonical names by design, by masking tokens that cannot continue one (2.12). The body text is not masked.

3. Canonical projection. The canonical projection does pull captions toward its prototypes, as the reviewer suspected. With it, about one caption in ten nearly copies a curated training text, against almost none without it. We added this trade-off to the Methods.

4. Granularity. In the output space, test captions use nearly all canonical network and region names, and no title labels more than one test map in ten. The share of captions naming a study detail, such as a sample size, patient group, or task, falls from 36% at stage 1 to 2% at stage 2 (Fig. R13). Stage 2 captions describe the map rather than a single study.

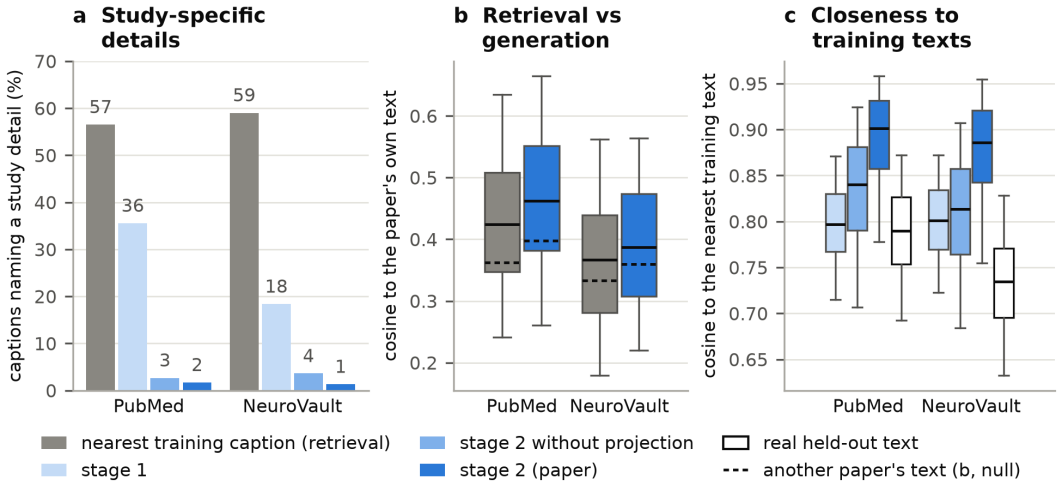


Fig. R13. Granularity of generated captions. Captions of PubMed test maps and NeuroVault papers, retrieved or generated at each stage. Panels show (a) the share naming a study detail and the MiniLM cosine (b) with the paper's own text and (c) with the nearest training text.

Table R11. Generated versus retrieved captions. Texts returned for PubMed test maps, scored against each paper's title and LLM summary by MiniLM cosine and BERTScore F1. Brackets give the MiniLM cosine against another paper's text.

Text returned for a test map	MiniLM cosine (null)	BERTScore F1
NeuroVLM, stage 2 caption (paper)	0.463 (0.397)	0.553
NeuroVLM, stage 1 caption	0.431 (0.358)	0.577
Retrieval, training caption nearest to the map	0.428 (0.362)	0.588
Random training text	0.363	0.580

Changes to the manuscript: Methods, Text Decoder, l. 316–320, on the canonical projection, which removes study details but adds near-copies. Discussion, l. 863–868. New Supplementary Section Caption Granularity, l. 2106–2115, with Fig. S10 (p. 47).

3.4: Context-free baseline

Reviewer's comment

Another capability that sets NeuroVLM apart is the context and flexibility provided by the LLM encoder. However it would be helpful to directly quantify how the LLM encoder performs relative to context-free baselines (such as mean Word2Vec embeddings) on the retrieval and text-to-neuroimage tasks.

Response: SPECTER2 outperforms a context-free baseline on both tasks, by a modest margin (Fig. R14). We replaced SPECTER2 with the mean word2vec embedding of each title and abstract and retrained only the projection heads. Mean retrieval AUC, the area under the recall curve, is 0.83 against 0.81. Text-to-brain Dice and voxel correlation are also higher with SPECTER2. We added this comparison to the Discussion and the Supplementary Information.

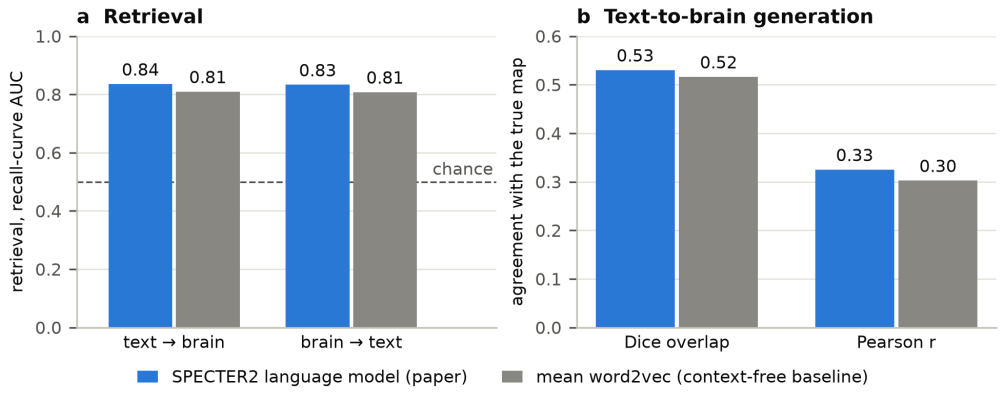


Fig. R14. Context-free encoder. PubMed test-set performance with SPECTER2 or with the mean word2vec embedding of each title and abstract, after retraining only the projection heads. Panels show (a) retrieval AUC in both directions and (b) text-to-brain Dice and voxel correlation.

Changes to the manuscript: Introduction, l. 109–111, and Discussion, l. 850–856. New Supplementary Section Context-Free Encoder, l. 2145–2151, with Fig. S11 (p. 48).

3.5: Interpretation, perturbations, and ablations

Reviewer's comment

It would be helpful to include more analyses like Figure 6 that interpret different components of the model. Interpreting the encoder weights could reveal the specific associations between text and neuroimage elements that NeuroVLM learned. Perturbing the inputs (such as adding negations) could show how NeuroVLM goes beyond linear additivity. Providing ablation results in the supplementary materials could compare the importance of different modeling decisions (such as the cross-entropy loss and canonical projection matrix).

Response: 1. Encoder weights. We interpret the latent dimensions that texts and maps share, rather than raw weights. For the clearest dimensions, the decoded map matches the Neurosynth map of the dimension's top term, such as listening in superior temporal cortex (Fig. R15, $r = 0.65$). Text and maps agree on what each dimension means. We added this figure to the Supplementary Information.

2. Perturbations. NeuroVLM responds to word swaps and to joined concepts (Fig. R16) but ignores negation. Swapping one word moves the map toward the matching Neurosynth contrast, and joining two concepts covers both. A sentence and its negation give nearly the same map ($r = 0.84$ to 1.00). We find no evidence of composition beyond addition. We added this limitation to the Discussion.

3. Ablations. The cross-entropy loss matters most for retrieval. Without it, retrieval AUC drops from 0.83 to 0.62, while the temperature and a one-directional loss barely matter (Table R12). The final text-to-brain head also beats heads trained with other losses. Comment 3.3 gives the effect of

the canonical projection on captions. We added these ablations to the Supplementary Information.

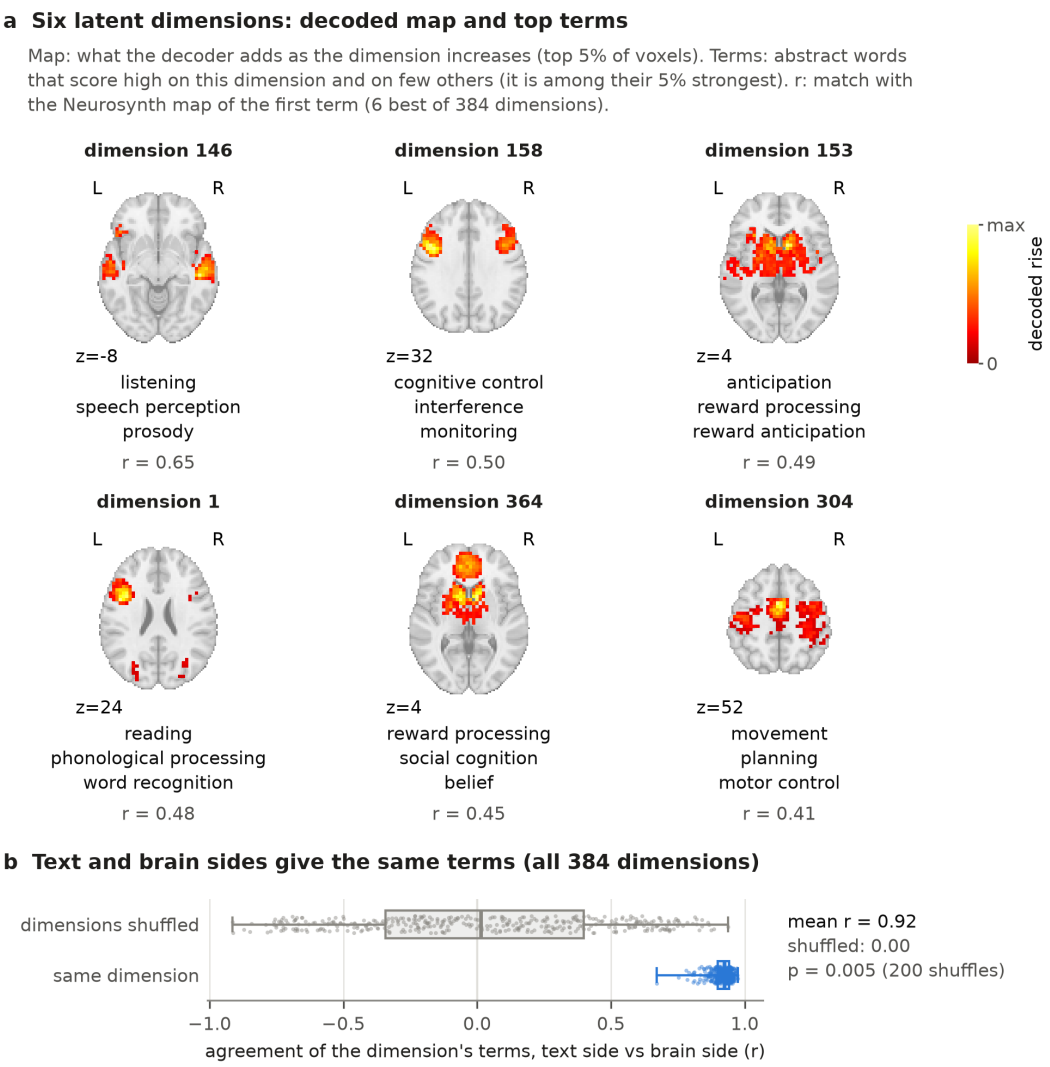


Fig. R15. Latent dimensions. (a) Decoded maps of six latent dimensions, each labeled with its three most associated Cognitive Atlas terms and its correlation r with the top term's Neurosynth map. (b) Agreement of each dimension's term associations between the text and brain sides, against shuffled dimensions.

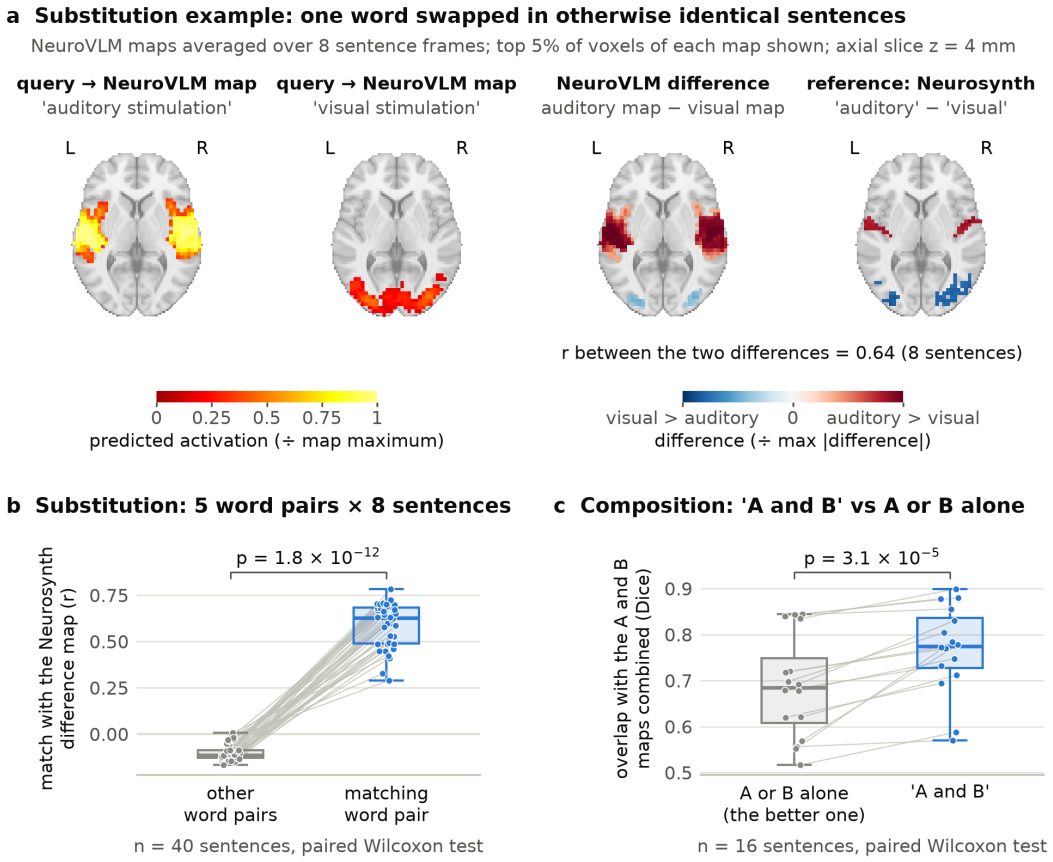


Fig. R16. Perturbation probes. (a) One word swapped, with NeuroVLM's difference map beside the matching Neurosynth contrast. (b) Match of such differences with Neurosynth for five word pairs, and (c) overlap of "A and B" maps with both concepts, against single-concept maps.

Table R12. Loss ablations. Heads retrained on the final model's latents with the loss shown and scored on the PubMed test set. Values are means of three seeds.

Loss	Test-set score
Contrastive heads, retrieval AUC	
Symmetric InfoNCE cross-entropy, $\tau = 0.07$ (final)	0.833
One direction only, text → brain / brain → text	0.834 / 0.827
$\tau = 0.03$ / $\tau = 0.20$ / learnable τ	0.829 / 0.827 / 0.833
No cross-entropy, cosine of matched pairs only	0.618
Text-to-brain head, Dice	
Latent MSE and map correlation, final head with its own training settings	0.530
Latent MSE only	0.512
Voxel-wise binary cross-entropy through the decoder	0.491
Latent cosine only	0.385

Changes to the manuscript: Introduction, l. 76–79, where the contrast with additive term models no longer implies non-linear composition. Discussion, limitations, l. 904–908, with a new sentence that NeuroVLM ignores negation. New Supplementary Section Latent Dimensions and Loss Ablations, l. 2186–2195, with Fig. S12 (p. 49) and Table S10 (p. 50).

3.6: Network, anatomy, and cognition outputs

Reviewer's comment

It is interesting that the model can generate separate brain-to-text outputs corresponding to the predicted network, anatomy, and cognition. How does quantitative performance compare between these outputs? What if the text-to-brain outputs were generated using only text about the network, anatomy, or cognition? These analyses would also shed light on the associations learned by NeuroVLM.

Response: We scored each generated title against its map. Every title is generated by the language model, steered toward the canonical vocabulary. The canonical projection steers the model's input, and at [NETWORK] and [REGION] titles, tokens that cannot continue a canonical name are masked. The model's own probabilities choose the name, so nothing is classified or looked up (2.12).

1. Comparison of the three outputs. [REGION] and [FUNCTION] titles are correct most often and [NETWORK] titles least (Fig. R17). A title is correct if it names the map's network, a region inside the map, or a function decoded from the map. On PubMed test maps, 63% of [REGION] titles are correct, against 24% by chance. Fig. R17 also scores atlas networks and ICA components.

2. Network, anatomy, or cognition text only. Maps generated from a network name, a region name, or a cognitive term alone all match their references better than a null baseline (Table R13). Region maps overlap their references least, because most regions are smaller than the top 10% of voxels that Dice compares.

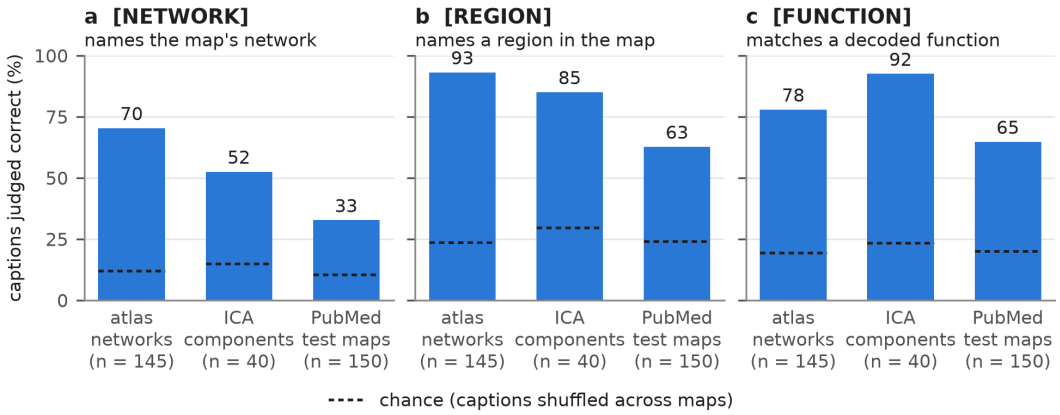


Fig. R17. Correctness of generated titles. Share of [NETWORK], [REGION], and [FUNCTION] titles judged correct for atlas networks, ICA components, and PubMed test maps. Dashed lines show chance, from titles shuffled across maps.

Table R13. Text-to-brain from one kind of text. Maps generated from bare network names, region names, or Cognitive Atlas terms, scored against the matching atlas map, region, or Neurosynth term map. Brackets give a null baseline, the same maps scored against the other items' references.

Input text	Reference	n	Dice (null)	Pearson r (null)
Network name	Yeo-7 and network atlas maps	126	0.29 (0.05)	0.29 (−0.01)
Region name	Harvard-Oxford and AAL regions	178	0.07 (0.01)	0.15 (0.00)
Cognitive Atlas term	Neurosynth map of the term	130	0.28 (0.13)	0.25 (0.02)

Changes to the manuscript: Methods, Text Decoder, l. 299–306, on title decoding steered toward the

canonical names. Discussion, limitations, l. 899–905, where [NETWORK] titles are the least reliable, especially for single-study maps.

3.7: Unsupported claims

Reviewer's comment

Some claims are not justified in the context of the study. For example "each component of NeuroVLM is modular and interchangeable with larger or smaller models depending on requirements" (line 101) and "our results underscore that the limiting factor for biological fidelity is often the structure of what the field reports and shares" (line 672). While intuitively true these statements would be better supported by including scaling and ablation analyses.

Response: 1. Modular and interchangeable components. The encoder swap of 3.4 supports a narrower claim, which we keep. Replacing SPECTER2 with word2vec required retraining only the small projection heads, and performance dropped modestly (Fig. R14). We softened the sentence to this claim and removed the reference to larger or smaller models, since we did not test scaling.

2. Limiting factor. The sentence compared coordinate-based supervision with full statistical maps, not model sizes. We rewrote it to say so and cited work on what coordinate-based meta-analysis loses.

Changes to the manuscript: Introduction, l. 109–111, modularity sentence. Discussion, l. 937–945, limiting-factor sentence, and l. 912–913, where the sentence on our strongest limitations is deleted.

3.8: Titles and abstracts only

Reviewer's comment

Could the authors justify why NeuroVLM was only trained on titles and abstracts? I imagine that other sources of information (such as table captions) might be useful for checking whether the activation coordinates are actually related to the claims in the titles and abstracts.

Response: Titles and abstracts exist for every paper, but table captions only for papers with open-access full text, fewer than half. Captions also say little about where coordinates lie, which limits their use for checking the abstract's claims (Table R14). A classifier predicts a table's main region from its caption at chance level (macro-F1 0.06 against 0.05), but does better from the abstract. We added these reasons to the Discussion.

Table R14. Uninformative table captions. Captions of coordinate tables from open-access papers, picked by hand from 60 random examples. The right column lists the Harvard-Oxford regions that hold most of each table's peaks.

Kind	Caption	Where its coordinates lie
boilerplate	Brain regions revealed by contrasts of interest.	anterior cingulate, frontal pole, insula
boilerplate	The main results of group analysis of block data.	posterior cingulate, amygdala, thalamus
boilerplate	Regional imaging relationships.	lateral occipital cortex, precuneus
condition code	Simple Effects contrast JUDGE –BODY.	anterior cingulate, lateral occipital, postcentral
condition code	Neural activations for the contrast AU > OC.	lateral occipital, paracingulate, posterior cingulate
statistics only	Functional results (p = .05 family-wise error (FWE) cluster corrected).	lateral occipital, pallidum
statistics only	Significant clusters exhibiting interactions of interest at $p < 0.001$ (uncorrected), n = 15.	insula, paracingulate, orbitofrontal
region list only	Significant clusters from amygdala connectivity analyses.	medial frontal cortex, insula

Changes to the manuscript: Discussion, l. 877–884, on table captions and on the pooling of a paper's tables as a limitation.

3.9: Percentiles of the qualitative examples

Reviewer's comment

It would be helpful to know where the qualitative examples in Figures 2 and 3 fall within the distribution of cosine similarity quality scores. Alternatively the authors could provide a percentile for each example.

Response: Table R15 gives each example's percentile within its set, from the 16th to the 99th. Fig. 3 also plots both similarity distributions and marks each example. Its rows follow a fixed rule. Each third of the distribution contributes the papers nearest its center with a clean generated text. High and Medium rows also require a title that names one domain and a true map with distinct clusters.

Table R15. Percentiles of the examples. Percentile of each example of Fig. 3 and Fig. 2 among all items of its set, by text-to-brain and brain-to-text cosine similarity. Fig. 3 ranks 305 NeuroVault papers and Fig. 2 ranks atlas networks.

Example	Text → brain percentile	Brain → text percentile
Fig. 3, High, working memory	86	79
Fig. 3, High, language	77	87
Fig. 3, Medium, memory	52	62
Fig. 3, Medium, reward	36	43
Fig. 3, Low, poor input map	16	18
Fig. 3, Low, model miss	22	19
Fig. 2, Visual	76	99
Fig. 2, Language	75	37
Fig. 2, Auditory	31	19
Fig. 2, Somatomotor	39	72
Fig. 2, Interoception	44	55
Fig. 2, Default mode	69	94
Fig. 2, Frontoparietal control	48	53
Fig. 2, Salience	33	46

Changes to the manuscript: Fig. 3, on one page, with rows chosen by a fixed rule and distribution strips that mark each example (p. 12). Fig. S4 caption, which gives the percentiles of its components (p. 34). Results, Generative Models: Networks, l. 445–447, with the percentiles of the Fig. 2 examples, and NeuroVault, l. 457–459, on the complete captions and the choice of rows.

3.10: Code availability

Reviewer's comment

I was able to install and run the code. The project is very well-documented and user-friendly. I believe that it provides a very usable resource for the community.

Response: No change was requested. The reproducibility changes of comment 1.11 also apply here.

NeuroVLM: A generative vision-language framework for human neuroimaging

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Abstract

Neuroimaging research has produced tens of thousands of articles that pair natural language and activation coordinate tables. Recent advances in vision-language models (VLMs) have provided methods to model text and images simultaneously. In this work, we present NeuroVLM, a model architecture for learning from 30,826 publications that pair text with brain coordinates. The architecture supports contrastive and generative objectives. The contrastive model ranks similarity between neuroimages and text. The generative models include text-to-neuroimage and neuroimage-to-text. These models are evaluated on network images from a variety of atlases, statistical maps from diverse publications, and images created from coordinate tables. These models generate coarse spatial priors from text and broad, network-level descriptions of neuroimages. They also label networks, find the publications most related to a neuroimage query, and find the neuroimages most related to a text query.

Keywords: neuroimaging, vision-language models, generative artificial intelligence, fMRI

1 Introduction

Over the past three decades, human functional magnetic resonance imaging (fMRI) neuroimaging research has changed the landscape of our understanding of how the brain gives rise to behavior and cognition. Tens of thousands of experiments have been performed, resulting in a massive, multimodal corpus of peer-reviewed publications that link natural language text to spatial brain coordinates. While powerful, expert decisions on how to interpret these results and which new experiments to run are constrained by one's own degree of familiarity with the literature. These constraints result in, for example, variations in how individual brain regions are named, or how collections of brain regions are grouped together and named as common networks [1]. Inconsistencies in nomenclature and interpretations in fMRI are exacerbated by small effect sizes [2–4] and large variability in data analysis approaches. Even when working on the same datasets, different research groups can come to different conclusions [5]. Furthermore, many fMRI studies are interpreted in isolation from other human brain maps, such as gene expression, neurotransmitter and modulator distribution, and electrophysiology [6], giving rise to a narrow view of the brain-behavior relationship [7].

Prior to the introduction of modern large language models (LLMs), text-mining of neuroscientific papers was demonstrated to be capable of generating plausible hypotheses by identifying gaps between text concepts mapped to a network graph [8]. In other domains, such as materials science, high-dimensional text embeddings have been shown to be capable of recapitulating concepts such as the periodic table of elements, and complex structure–property relationships in materials [9].

To overcome some of these limitations, a large, diverse training corpus is useful for integrating information across a broad neuroscientific context [7, 10]. This includes diverse datasets that pair text and brain activation maps, enabling models to translate between modalities: text-to-neuroimage and neuroimage-to-text. Past text-to-neuroimage models such as neurosynth [3] and neuroquery [11] use term-frequencies based on a predefined corpus [3, 11]. This limits handling of unseen phrases, synonymy, and context~~, and assumes near-independence of terms or linear additivity across terms~~. These methods ~~fail to generalize to out-of-corpus language and to account for non-linear interactions~~do not generalize to language outside their vocabulary.

Advances in language modeling architectures [12–14] move beyond term-frequencies and instead use transformers to model language. Transformer architectures are standard in LLMs~~, remove the dependence on~~ . They also use a fixed vocabulary ~~,~~ ~~and are context-sensitive~~of tokens, but they compose these tokens in context. Transformers are also used in vision-language models (VLMs), with the goal of aligning text-image latent vector pairs [15].

Recent works such as neurocontext [16] and niclip [17] use LLMs to encode text from neuroimaging publications. Contrastive loss [15] was used in these studies to learn a bi-modal embedding space, aligning text and neuroimage representations. These models provide similarity-based ranking and retrieval, given a corpus of publications and neuroimages. While powerful, contrastive models cannot generate brain images from input text, nor text outputs from 3D brain maps.

Here we present, NeuroVLM (Fig. 1a), a lightweight, artificial intelligence (AI) framework that trains both generative and contrastive models. This framework embeds text-neuroimage pairs to a shared latent space. This latent space can be trained with a contrastive objective to support ranking and retrieval, which allows us to label neuroimages, e.g., spatial ICA components, with text, and to select neuroimages most related to a text query. Decoding the latent space provides generative text-to-neuroimage and neuroimage-to-text models. To our knowledge, NeuroVLM is the only model that writes text from group-level brain maps. It retrieves and generates in both directions through one shared space.

Generation and context both matter. Generated maps are closer to each held-out paper's map than the map of the most similar training paper. Generated captions rarely carry the details of other studies, unlike retrieved captions. A contextual text encoder also beats a context-free word2vec baseline [18], by a modest margin.

To improve the accessibility of these models, we significantly reduced the computational requirements by using a much smaller text ~~encoders [35]~~ encoder, SPECTER2 [19], with 110 million parameters, rather than depending on LLMs with tens- to hundreds-of-billions of parameters. ~~Each component of NeuroVLM is modular and interchangeable with larger or smaller models depending on requirements~~The text encoder can be replaced by retraining only the small projection heads.

The resulting open-source models support a variety of applications. Use cases include generating ~~atlases or maps~~ coarse spatial priors from a text ~~corpus or~~ query, which could be used for first-pass hypothesis generation. NeuroVLM can also generate ~~natural-language interpretations~~ broad, network-level descriptions of human brain maps (Fig. 2) or automatically label functional networks (Fig. 5d). Finally, it supports cross-modal retrieval by finding the publications most related to a statistical map, or the statistical maps most related to a text query.

2 Methods

2.1 Datasets

Data were aggregated from PubMed central, neurosynth compose [20], the cognitive atlas [21], the network correspondence toolbox [1], and NeuroVault [22]. The networks toolbox maps [1], NeuroVault maps [22], and a subset of the PubMed articles were left out as test sets. The cognitive atlas is a dataset of text labels used to label activation maps post-training.

1. PubMed Central: 30,826 publications with titles, abstracts, and coordinate tables (857,298 peaks)
2. Cognitive Atlas [21]: 1,000 cognitive terms and definitions
3. Networks [1]: ~~100 labeled network images~~ 185 network maps, made of 145 atlas maps (119 with a network label) and 40 independent components
4. NeuroVault [22]: 3,~~000~~ 183 group-level ~~activation maps , titles, and abstracts~~ maps from 312 papers, with titles and abstracts

Each PubMed paper contributes one map, built from all of its peaks. The released coordinates carry no experiment, contrast, table, or sign labels. Each map therefore

pools the experiments, contrasts, and tables of a paper, and its positive and negative effects. It shows where a paper reports effects, not their sign. Papers were split into 23,894 training, 2,987 validation, and 2,987 test papers. The 958 papers with more than 100 peaks or an empty map are in no split. Seven NeuroVault papers (29 maps) duplicate a PubMed training paper and were removed, leaving 3,154 maps from 305 papers (Supplementary Fig. S5). All maps are group-level results of published studies. No participant-level data were used, so sex and gender were not analyzed.

2.2 Preprocessing

Coordinates were convolved in MNI space with a 9 mm activation Gaussian kernel, following [11]. The resulting activation maps were transformed using the dictionaries of functional modes (DiFuMo) atlas [23]. Reconstruction replaces voxel values with a weighted combination of DiFuMo components. Component coefficients were computed as the DiFuMo-weighted mean activation and were used to reconstruct voxel space by re-expanding the coefficients through the DiFuMo voxel maps. This returns a voxel-space map from a linear combination of DiFuMo components (~~see Sec.S2.1 for details~~Supplementary Fig. S2 and Section S2.1).

2.3 Training

Training ~~was performed with 30,826 text-neuroimage pairs, primarily from PubMed~~used the 23,894 training papers. Text included titles and abstracts of publications. Neuroimages are defined as sparsely clustered, MNI-space activation maps. PubMed ~~training~~ articles instead provide coordinate tables, which were smoothed in 3D, MNI space, similar to the approach used in ~~Neuroquery~~ NeuroQuery [11]. Continuous statistical maps are provided in the network and ~~neurovault datasets; these were used as additional test sets. DOIs used in the PubMed training set were excluded from all test sets. Software,including training scripts, are publicly available at:https://github.com/neurovlm/neurovlmPre-trained models and datasets are available on HuggingFace:https://huggingface.co/neurovlm~~ NeuroVault datasets. These were used only for evaluation. The autoencoder, the projection heads, and the stage 1 projector all learn from the training papers. The anatomically centered projection head and the stage 2 projector learn from short concept texts, paired with maps built from training papers only. No validation or test paper trains any component (Supplementary Fig. S5 and Table S6).

We audited leakage beyond DOI matching. No DOI or PubMed identifier is shared across splits. Each held-out paper was also compared with the training and NeuroVault papers by fuzzy title matching and by the similarity of its title and abstract. Coordinate tables were compared across all splits to find papers that repeat another paper's peaks. These checks flag 16 test and 26 validation papers, and every evaluation leaves them out. The PubMed test set therefore holds 2,971 papers. The language model's training script removes the validation, test, and NeuroVault evaluation papers from its corpus.

Every component except the language model trains on one NVIDIA RTX 5090 GPU, with fixed seeds and deterministic kernels. The whole pipeline, five-fold cross-validation included, trains from scratch in about 1.8 hours. Every input is read from Hugging Face at a pinned, checksummed revision (see Data availability). Supplementary Table S7 gives the parameters, data, epochs, optimizer settings, seed, and training time of each component.

2.4 Model Architecture

Building upon past work, we outline a general vision-language framework for neuroimaging applications: NeuroVLM (Fig. 1a). The encoders in this framework map from high-dimensional text and activation-map space to low-dimensional latent vectors (Fig. 1a, pink). After training, pairs of text and neuroimages encode to highly similar vectors. Decoders are able to generate text or activation maps from latent vector embeddings. These encoders and decoders collectively provide text-to-image and image-to-text generation. This framework is general and does not make assumptions on the choice of encoder or decoder architectures.

2.5 Autoencoder

An autoencoder was trained to compress 3D activation maps to 384-dimensional latent vectors (Fig. 1a, blue). Maps were down-sampled, masked, flattened, and binarized before passing through fully-connected linear layers with rectified linear unit (ReLU) activations. Exact model architecture, including layer sizes and activation functions, are provided in ~~Table S~~Supplementary Table S1. The downsampling and masking procedure was adapted from [11] and moves from a 1 million dimensional MNI space to 28,542 dimensional space. The encoder transforms from 28,542 to a 384 dimensional latent representation. A key component to successful training was the binarization of the activation map prior to training and the use of binary cross entropy loss. Any continuous map, e.g., a z-score or p-value map, can be binarized with an appropriate threshold and clustering method. Another key component in training was sparsity, reducing the influence of dense maps that were fully activated and lacked sparse clusters of activation.

2.6 Text Encoder

NeuroVLM uses ~~Specter [35]~~SPECTER2 [19], a small, efficient, and accessible transformer. ~~This small model allows deploying the model on edge devices with little computational requirements. Specter~~ Every text-encoding step uses the allenai/specter2_aug2023refresh model. SPECTER2 is based on SciBERT [24] and incorporates citation-based connectivity graphs. ~~Specter~~ It encodes title-abstract pairs to 768-dimensional latent vectors. Despite being trained on title-abstract pairs, ~~Specter~~ SPECTER2 has an ad hoc query adapter, suitable for short form user queries, which NeuroVLM uses for all text. The full model runs on a consumer CPU. On an 8-thread desktop CPU, a text-to-brain query takes 53 ms and 0.4 GB of memory in bf16. A 128-token brain-to-text caption takes 1.4 GB of memory, and 3.5 s with int8 quantization or 6.3 s in bf16.

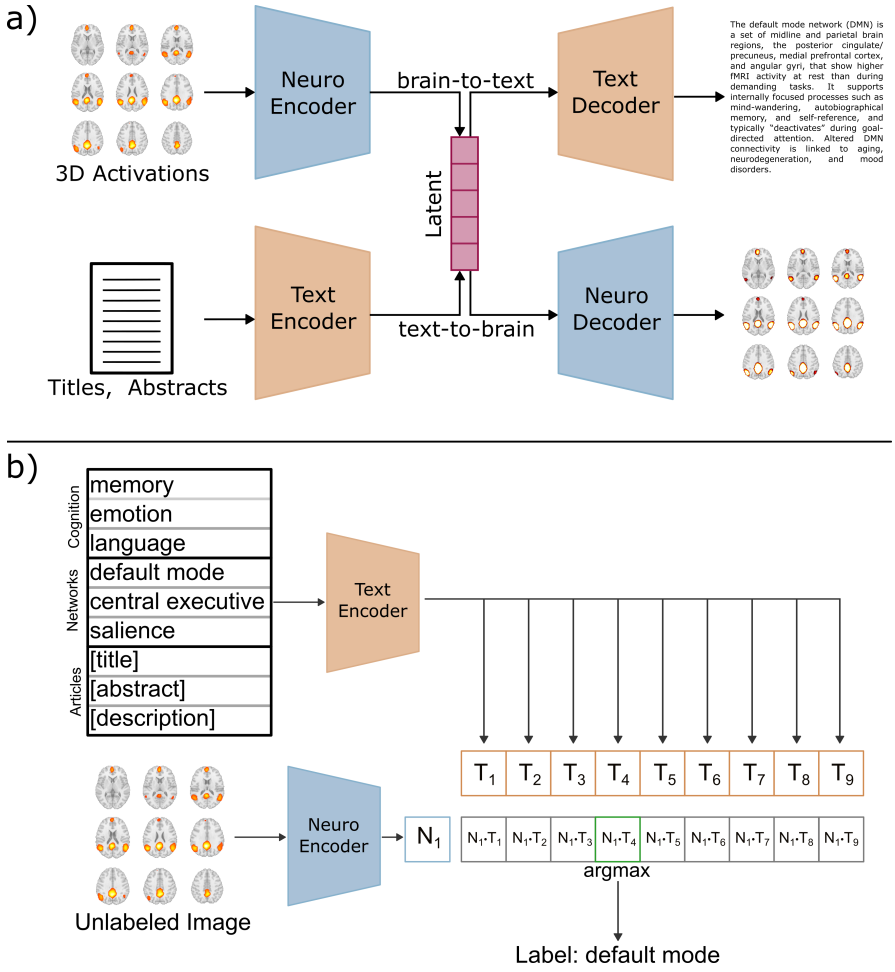


Fig. 1 Model architecture. a) NeuroVLM maps pairs of activation maps and text, from the same publication, to a shared latent vector (pink). Decoding the latent vector results in brain-to-text or text-to-brain generation. **b)** The shared latent vector is the basis for contrastive models, e.g., ranking and retrieving text most similar to a query brain map. Given an unlabeled, query activation map, the method ranks how similar a corpus of text is to the image. All encoded vectors, N_1 and $\{T_1, \ldots, T_9\}$ are unit normalized. The inner product between these vector pairs is cosine similarity. Taking argmax gives the most similar text. Any text corpus or image set may be used, suitable for a variety of use cases. The direction may also be flipped from what is presented: comparing a single text query to a set of activation maps.

2.7 Text Decoder

Brain-to-text generation requires a language model. ~~We trained~~ NeuroQwen3-0.6B is Qwen3-0.6B ~~on 1.2 million neuroscience articles and used this to generate descriptions of images using a query transformer (Qformer) [26]. The Qformer~~ [25] after continued

pretraining of all weights, without adapters, on 1,231,613 PubMed neuroimaging titles and abstracts published from 1990 to February 2026. It trained for one epoch on one H100 GPU and stays frozen in NeuroVLM. Its training script removes the validation, test, and NeuroVault evaluation papers from this corpus. A projector learns a mapping from the latent image space to the token embedding space of ~~LLM~~the language model. This training was supervised such that the decoded text was optimized to the true, paired text.

The ~~Qformer~~ projector receives two complementary inputs~~:~~, a raw 384-dimensional image latent from the image encoder and a 384-dimensional semantic latent ~~,~~from the contrastive model. ~~Each input is projected into a shared 512-dimensional hidden space, producing two memory tokens : one preserving the raw-image latent signal and one for semantic alignment information. A set of~~ A residual multilayer perceptron (MLP) maps the two normalized inputs to 32 ~~learned query embeddings then attends to these memory tokens through a 6-layer transformer decoder.The resulting query states are mapped into the 1024-dimensional embedding space of the language model , yielding 32~~ continuous prompt tokens ~~for downstream text generation~~in the 1024-dimensional embedding space of the language model (28.3 million parameters, Supplementary Table S3). A query transformer (Qformer) [26] of the same size attends over only these two input tokens and was less accurate (Supplementary Fig. S9). The projector trains in two stages. The stage 1 projector learns from PubMed training maps and summaries of their papers. The stage 2 projector starts from stage 1 and learns from short network, region, and function texts (see Brain-to-Text below).

Captions are decoded with the vocabulary restricted to Latin-script tokens. This removes the non-Latin characters that the multilingual language model otherwise writes into a few captions. Steering tokens select the kind of caption (Supplementary Section S5). The language model generates every title, steered toward the canonical names. After a `[NETWORK]` or `[REGION]` token, tokens that cannot continue a canonical network or region name are masked until the title ends. The model's own probabilities choose the name, so no title is an invented name. The body text and `[FUNCTION]` titles are decoded without this mask.

At inference, the images may come from a variety of distributions, including binary masks and t-, z-, F-, and p-maps. The one requirement of the brain-to-text model is to transform these maps to the range: $[0, 1]$. In addition to a variety of map distributions, thresholding and clustering depends on a variety of parameter choices. To help improve generalization of brain-to-text generation for out-of-distribution maps, relative to the training maps, we introduce a canonical, or stereotyped, projection matrix. This canonical projection matrix is constructed from clean examples of networks, regions, and cognitive concepts. Vector projection produces a denoised semantic conditioning vector constrained to lie near interpretable prototypes. The canonical projection adds a layer of interpretability~~and can warn if the input image lies too far from known images.See supplementary section S~~. It also pulls captions toward its prototypes. It lowers the share of test captions that name study details from 3% to 2%, but it raises the share that nearly copy a curated training text from 0.6% to 9.9% (Supplementary Fig. S10). See Supplementary Section S5 for details.

Warning scores flag inputs that lie far from the training data. A map is flagged when its Pearson correlation with its autoencoder reconstruction is low or when its

latent lies far from the training latents (Mahalanobis distance). A text is flagged when its cosine similarity to the nearest training abstract is low. With thresholds set on the validation split, they flag about 5% of held-out inputs, every shuffled, white-noise, and rescaled map, every off-topic text, and 61 to 80% of rotated or spun maps (Supplementary Fig. S8). The scores are warnings, not calibrated probabilities.

2.8 Contrastive Learning

Contrastive learning maximizes cosine similarity between true text-image pairs and minimizes similarity between mismatched pairs [15, 17]. After training, unlabeled images can be labeled by ranking cosine similarity between the target image and any text corpus (Fig. 1b). Pretrained encoders can be used to label neuroimages with networks [1], cognitive concepts [21], or publication descriptions. Inversely, a set of images may be ranked in similarity, relative to target text.

2.9 Generative Models

2.9.1 Brain-to-Text: Generating Interpretations

~~A Qformer [26]~~ An MLP projector is used to generate descriptions of neuroimages (Fig. S1a). The ~~Qformer transforms from~~ projector transforms latent image representations to the token embedding matrix. The token embedding matrix is decoded by ~~a fine-tuned~~ NeuroQwen3-0.6B, a neuroscience-focused version of Qwen3-0.6B [25].

~~Training data included image-(title, abstracts). Text was summarized to general text, removing study-specific details~~ The stage 1 projector learns from training maps paired with summaries of their papers. Llama-3.3-70B-Instruct summarized each title and abstract with a fixed prompt, removing low-level study details (Supplementary Section S2.2). This allows the model to generate higher-level descriptions of activation maps. ~~A final fine-tuning dataset was generated using the text-to-brain model and synthetic descriptions of common networks,regions, and cognition.~~ The stage 2 projector learns from 1,338 curated `[NETWORK]`, `[REGION]`, and `[FUNCTION]` texts written by Qwen3.5-27B, each paired with the map that the anatomically centered projection head generates from it. A self-consistency filter keeps 1,106 of these pairs. This step allowed more control of the style of the LLM output, pushing outputs towards shorter, general descriptions. The target was concise brain-to-text captions, rather than plausible abstracts. Examples of brain-to-text generation are provided (Figures 2 and 3).

Each stage 2 training map is binarized at a percentile drawn anew for each batch, and title tokens weigh five times in the loss. The semantic input of each text is projected onto the canonical bank of its own level, as at inference. Stage 2 was trained with five seeds, and the seed with the lowest final validation loss was kept.

2.9.2 Text-to-Brain: Generating Activation Maps

Activation maps are generated by projecting text embeddings to the latent space of the neuroimage autoencoder (~~Fig S~~Supplementary Table S1) and using the neuroimage decoder (Fig. S1b). ~~Specter2~~ SPECTER2 [19] is used to embed arbitrary text: titles,

abstracts, descriptions, or user queries. A small neural network, the ~~projection~~ text-to-brain head, maps these frozen embeddings to the ~~shared latent space. The projection head allows the language model weights to be frozen while providing fine-tuning, similar to low-rank adaptation (LoRA)adapters [36]~~autoencoder latent (Supplementary Table S2). It is trained to predict each training paper's latent, with a loss that also scores the decoded map. Its settings and training epoch were chosen on the validation set (Supplementary Section S4). The neuroimage decoder then transforms the latent embeddings to MNI space. Generated maps are calibrated to the scale of the training maps by one factor per model, fitted on the validation set and applied identically to every map (clipped at 1), before they are encoded or scored. The factor is the median, over validation-set maps, of the training maps' peak (1) divided by the generated map's peak. No map is rescaled individually. These maps are coarse spatial priors. They show where effects are likely, without effect size, sign, or uncertainty.

A second, anatomically centered projection head maps text to the same latent. It learns from synthetic text-map pairs, built from atlas regions and training papers only, and from PubMed training pairs. It generates the maps on which the stage 2 projector trains.

2.10 Performance Metrics

The main performance metric used for test evaluation was the recall@k curve. This measure was used for both contrastive and generative models. Cosine similarity between all pairs of latent text and neuroimages is computed. Recall@k measures how often the true neuroimage-text pair was found in the top-k most similar pairs [16]. Given a reference latent neuroimage, recall@k measures how often the paired latent text is in the top-k most similar pairs. The inverse is performed using a reference latent text and computing recall@k against latent neuroimages. We step k from 1 to the max size of the test set and compute recall@k for each threshold. The area under this curve is reported, where 0.5 is chance performance and 1.0 is perfect performance. This chance level holds when each query has one target. A NeuroVault paper has several maps, so its chance level is estimated by shuffling texts across papers and applying the same choice of target (Supplementary Section S2.3). Retrieval is also reported as recall@1, recall@5, recall@10, mean reciprocal rank, and median rank. These are computed among all test papers and among hard negatives, the test papers of the query's own cognitive domain (Supplementary Table S8). Recall@k is used for the contrastive model, examining how well aligned the text and neuroimage representations are. For generative models, recall@k is computed from latent representations of generated outputs and true targets.

In addition to recall@k, we also provided contrastive cosine similarity, Dice score, Pearson correlation, and spin-tests for text-to-brain generation. For brain-to-text generation, contrastive cosine similarity, BERTScore, semantic cosine similarity, and a contrastive null test were performed. Matched distributions, or true (image, text) pairs, were compared against a randomly shuffled distribution of (image, text) pairs (~~).~~ Fig. 4). Every generated map is calibrated, as described above, before it is encoded or scored.

Text-to-brain maps were also scored outside NeuroVLM's representation space. Each map was compared in voxel space with the test paper's map as the autoencoder reconstructs it, by Dice and Pearson correlation. The simple baselines were linear (ridge) regression from the SPECTER2 embedding to the autoencoder latent, the mean training map, the mean map of the query's cognitive domain, atlas-based labeling of the regions named in the text, and the map of the nearest training paper. The published models were Neurosynth [3], NeuroQuery [11], and Text2Brain [27], used as released. Their training corpora each hold nearly half the test papers. Generated captions were compared with the caption of the nearest training paper, a retrieval baseline. They were also scored for study details, such as sample sizes, patient groups, or tasks. Each qualitative example is placed by its percentile among all items of its set, by cosine similarity in the shared space.

3 Results

3.1 Generative Models

Text-to-brain outputs are coarse spatial priors, without effect size, sign, or uncertainty. Brain-to-text outputs are broad network-level labels, not interpretations of a specific study or mechanism.

3.1.1 Networks

The network dataset [1] was used to demonstrate text-to-brain and brain-to-text generative models, because it provides some degree of ground truth, given that the collection of networks were labeled by experts (Fig. 2 for examples, Fig. 5d for quantitative metrics). The text-to-brain model transformed networks' definitions to the latent space using the encoder. The latent vectors were then transformed to neuroimages using the decoder. The brain-to-text model generates text related to a neuroimage input from the network dataset. This brain-to-text model uses ~~a Qformer~~ the stage 2 MLP projector to map from latent image space to the latent token embedding space ~~[26] ().~~ (Fig. S1). Seven of the eight network titles name the reference network exactly, and all eight name the right system. Generated maps lie at the 31st to 76th percentile and generated texts at the 19th to 99th percentile of all atlas networks. Supplementary Table S5 gives the complete texts.

3.1.2 NeuroVault

Generative neuroimage-to-text and text-to-neuroimage models were evaluated on the ~~neurovault~~ NeuroVault test set, which provides unthresholded brain maps not in the PubMed training set. Qualitative results are shown in Fig. 3, where each row ~~in this figure~~ corresponds to one group-level activation map and study from ~~neurovault~~NeuroVault. Titles and abstracts from ~~neurovault~~ NeuroVault were used to generate 3D brain maps~~; only true and predicted (generated) titles are shown.~~ . Each true title and each complete generated caption is shown. The rows were chosen from the top, middle, and bottom thirds of both similarity scores.

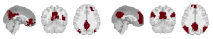
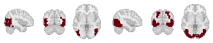
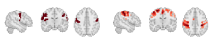
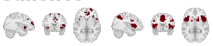
	Generated	Generated (brain-to-text)		
True	(text-to-brain)	Network	Anatomy	Cognition
Auditory		**Auditory Network.** The auditory network includes Heschl's gyrus, superior temporal gyrus, planum temporale, and inferior frontal cortex. It is commonly associated with the processing of sound, speech, and environmental sounds. The network is often described as a hierarchical system for auditory perception. Its coordinated activity is thought to support the integration of auditory information across the brain.	**Superior Temporal Gyrus.** The superior temporal gyrus is centered on the superior temporal sulcus. It is commonly associated with auditory processing, speech perception, and sound-related associations. Its anatomy supports representation of acoustic features such as frequency, timbre, and pitch.	**Early Auditory Processing.** Early auditory processing is the analysis of sound input at high frequencies. It involves primary auditory cortex in Heschl's gyrus, superior temporal gyrus, and planum temporale. This concept supports speech recognition, pitch detection, and low-frequency sound processing.
Default mode		**Default Mode Network.** The default mode network includes medial prefrontal cortex, posterior cingulate cortex, precuneus, angular gyrus, lateral parietal cortex, and medial temporal regions. It is commonly associated with internally oriented cognition, autobiographical memory, self-referential thought, and the integration of personal and social context.	**Precuneus.** The precuneus lies on the medial parietal cortex. It is commonly associated with autobiographical memory, self-related imagery, and mental context. This region is often described as supporting integration of personal and external information.	**Autobiographical Memory.** Autobiographical memory is commonly associated with the medial prefrontal cortex, posterior cingulate cortex, precuneus, angular gyrus, parahippocampal cortex, retrosplenial cortex, and lateral parietal cortex. This function supports remembering personal experiences, spatial context, and internally constructed events.
Visual		**Object Perception Network**	**Lateral Occipital Cortex**	**Visual Analysis**
Language		**Language Network**	**Middle Temporal Gyrus**	**Language Comprehension**
Somatomotor		**Somatomotor Network**	**Primary Somatosensory Cortex**	**Tactile Encoding**
Interoception		**Interoceptive Network**	**Anterior Insula**	**Interoceptive Processing**
Frontoparietal		**Frontoparietal Control Network**	**Inferior Frontal Gyrus**	**Working Memory**
Salience		**Salience Network**	**Anterior Cingulate Cortex**	**Executive Control**

Fig. 2 Networks: text-to-brain and brain-to-text generation. ~~The input to text-to-brain generation was a high-level summary of~~ Auditory and default mode networks show the ~~reference image~~complete generated texts, and the other six their three generated titles. ~~The input to brain-to-text generation were~~ True is the atlas ~~networks~~map, ~~see~~ Generated (text-to-brain) the ~~reference column for~~ map that the ~~group who produced~~ text-to-brain head generates from the ~~map. Text-to-brain generation used text~~ network's description~~of networks to generate a set of activation maps. Brain-to-text generation used atlas images to generate interpretations of~~ , and Generated (brain-to-text) the ~~brain map. Network~~network, anatomy, and ~~functional tokens were stacked into token embedding matrix, predicted~~ cognition texts that the stage 2 projector generates from the ~~Qformer~~true map. ~~This provides steering~~ Maps show their top 5% of ~~brain-to-text interpretation to corresponding aspects~~voxels in sagittal, coronal, and axial views. Supplementary Table S5 gives every text.

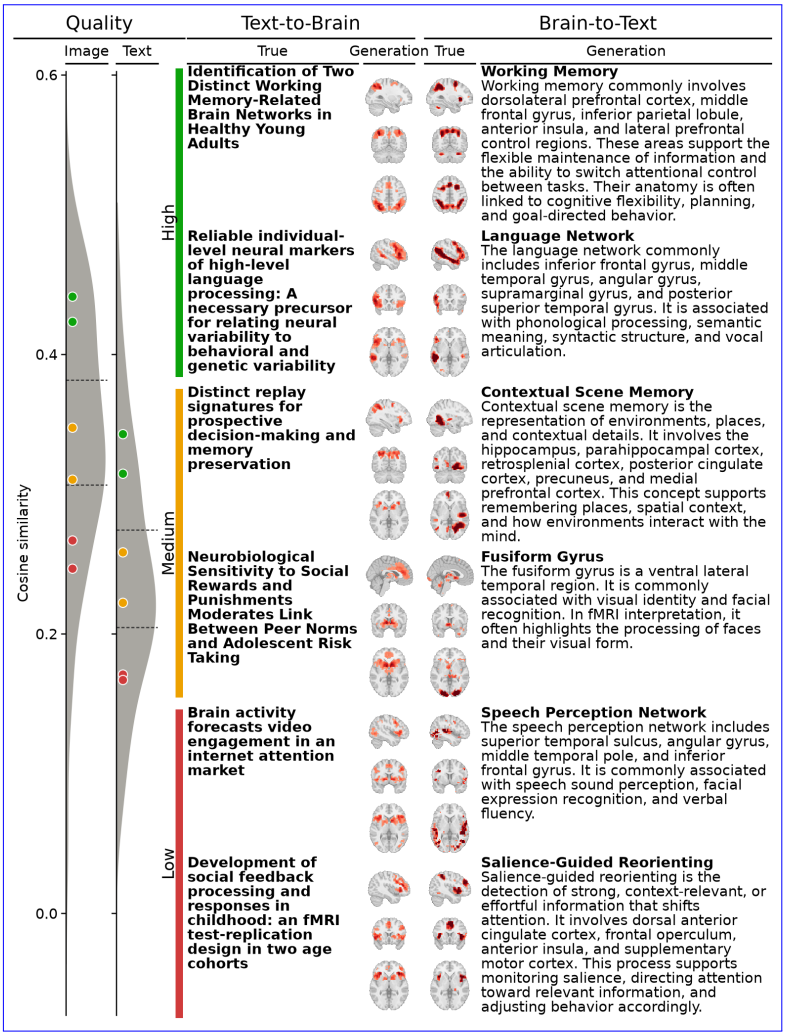


Fig. 3 NeuroVault: text-to-brain and brain-to-text generation. ~~Pairs~~ Rows are papers with clean captions nearest the center of ~~group-level activation maps and publication text~~ the third holding both shared-space cosines (~~titles and abstracts~~densities over 305 papers)~~were taken from NeuroVault~~, High and Medium with a one-domain title and clustered true map. ~~Text was used to generate brain maps~~ Low rows show a scattered true map (~~left~~top) and ~~true brain maps were used to generate related text~~ a model miss on a clean map (~~right~~bottom). ~~Quality~~ Maps show their top 5% of ~~generations is measured with cosine similarity between true and generated~~voxels.

3.1.3 PubMed

The PubMed test set (~~$n = 2987$~~ $n = 2{,}971$) was used to evaluate both generative directions (Fig. 4). Retrieval performance was quantified with a normalized recall@k curve. For each generated output, we computed its similarity to every candidate item in the test set using the NeuroVLM contrastive embedding space, then asked whether the correct paired item appeared within the top-k ranked candidates. For text-to-brain generation, generated brain maps were embedded with the image contrastive projection head and ranked against the source text embeddings. For brain-to-text generation, generated text was embedded with the text projection head and ranked against the source image embeddings. We report the area under the recall curve over normalized rank, $\frac{k}{N}$, so that random retrieval has expected AUC 0.5, while perfect retrieval has AUC 1.0. Perfect performance would mean that every generated output is most similar to its true paired input than to any mismatched item in the test set.

In addition to retrieval AUC, we evaluated the paired contrastive cosine similarity between each generated output and its true conditioning input. This metric directly measures alignment in the learned NeuroVLM multimodal space. Matched pairs were compared against a null distribution formed by mismatching generated outputs with inputs from other PubMed examples. Thus, higher matched cosine similarity indicates that the generated brain map or generated text preserves information that is specifically aligned with the correct paired modality, rather than only matching broad dataset-level structure.

For text-to-brain generation, we also quantified spatial accuracy of the generated brain maps. Dice score measured overlap between thresholded predicted and true activation maps. Pearson correlation measured voxelwise spatial correspondence between predicted and true maps, capturing graded similarity beyond binary overlap. Finally, spin-test p-values assessed whether the observed spatial overlap exceeded what would be expected under spatially autocorrelated cortical null maps. The spin-tests were limited to examples that had predominantly surface activation. This provides a more conservative test of anatomical specificity, since neighboring brain locations are not statistically independent.

For brain-to-text generation, we evaluated the generated descriptions using complementary text-based and contrastive metrics. BERTScore F1 measured contextual token-level overlap between generated and reference PubMed text, while semantic cosine similarity measured sentence-level semantic agreement using text embeddings from a generic language embedding model, `all-MiniLM-L6-v2`. Because reference PubMed abstracts can contain broad methodological and domain-general content, these lexical and semantic metrics may remain high even for mismatched pairs. We also included a contrastive null test: for each generated text, we compared its NeuroVLM contrastive similarity to the true source brain map against its similarity to mismatched brain maps from the test set. The resulting empirical p-value measures whether the generated text is more specifically aligned to its own source brain map than to other plausible PubMed brain maps.

Generated outputs find their true pair, with retrieval AUC 0.95 in both directions (Fig. 4 and Supplementary Table S4). Generated maps overlap their paper's map more than shuffled pairs do (Dice 0.53 against 0.41) and correlate more with it (Pearson

r 0.32 against 0.12). Of the 1,332 cortical maps, 58% beat their spatial nulls at $p < 0.05$. Generated captions are closer to their paper's text than to other papers' texts (MiniLM cosine 0.46 against 0.40). BERTScore separates them only weakly (0.553 against 0.548), because unrescaled BERTScore stays near 0.58 even for two unrelated test papers and rewards the style of the reference summaries more than their content. The matched caption still scores higher for 57% of maps. In the contrastive null test, 70% of captions reach $p < 0.05$, against 5% expected by chance. Compared with the Qformer, the MLP projector raises brain-to-text retrieval AUC from 0.82 to 0.90 at stage 1 and from 0.94 to 0.95 at stage 2 (Supplementary Fig. S9).

NeuroVLM gives the most accurate text-to-brain maps of the methods compared (Supplementary Fig. S6). It leads narrowly ahead of linear regression and ahead of Text2Brain, Neurosynth, NeuroQuery, mean and domain maps, and atlas labeling (Dice 0.53 against at most 0.49). Generated maps are also closer to each test paper's map than the map of the nearest training paper (Dice 0.53 against 0.40). By MiniLM cosine, generated captions are as close to the paper's text as the caption of the nearest training paper. But 57% of the retrieved captions report details of another study, against 2% of generated captions (Supplementary Fig. S10).

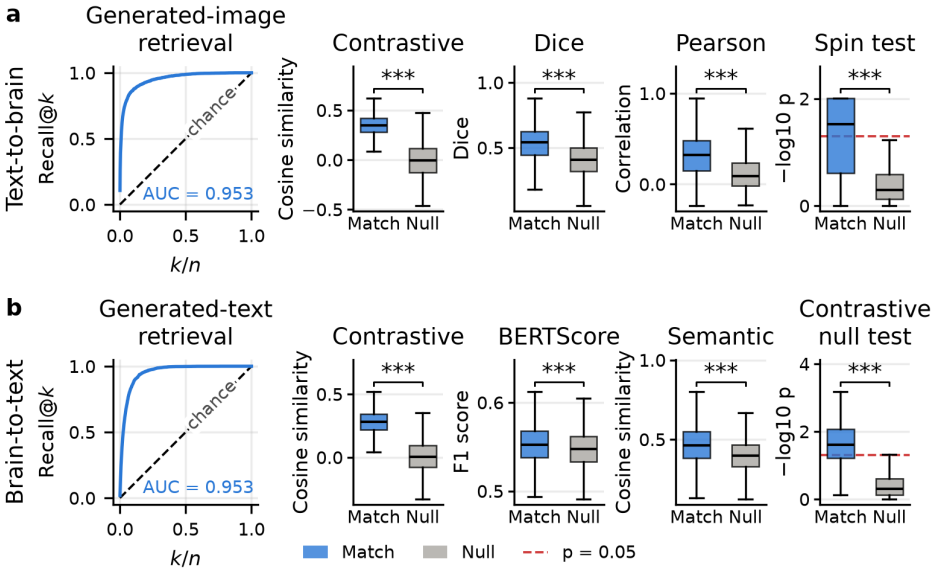


Fig. 4 ~~**Generative Performance.**~~ **Generative performance.** Text-to-brain (a) and brain-to-text (b) generation ~~performance~~ on the PubMed test set (~~$n = 2987$~~$n = 2{,}971$). ~~Top row: text-to-brain generation was evaluated by normalized recall~~Recall@k ~~k~~ k curves show retrieval of the true pair, ~~contrastive cosine similarity~~with k/n the fraction of all candidates. Boxes compare matched pairs (Match, ~~Dice overlap~~blue) with shuffled pairs (Null, ~~Pearson correlation, and spin-test significance~~gray). ~~Bottom row: brain-to-text generation was evaluated by normalized recall@k retrieval~~Dice uses each map's top 30% of voxels. The spin test (1,~~contrastive cosine similarity, BERTScore F1, semantic cosine similarity,~~ 332 cortical maps) and ~~a~~ the contrastive null test show each map's $-\log_{10} p$, with $p = 0.05$ dashed. ~~Matched pairs are compared against mismatched null pairs where applicable; recall AUC ranges from random chance~~ Stars mark Welch t-tests in the box panels (~~0.5~~Supplementary Table S4) ~~to perfect retrieval (1.0). Null distributions are~~ and one-sided binomial tests of the share below $p = 0.05$ in ~~gray, they represent randomly shuffled pairs~~the two p panels. ~~T-tests were conducted to compare all match versus null distributions~~One, ~~$^*p < 1e-100$ was found for all comparisons~~two, and three stars mark $p < 0.05$, 0.01, and 0.001.

3.2 Contrastive Model

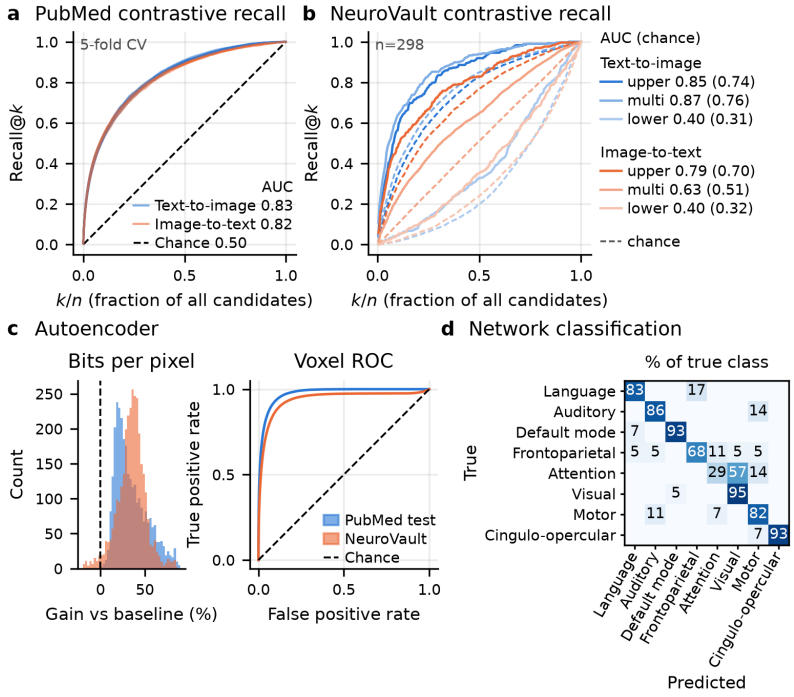


Fig. 5 ~~**Contrastive Performance.** Results are presented for the neuroimage autoencoder, contrastive models, and generative models.~~ **Contrastive performance. a) PubMed.** Recall@k ~~has expected random performance of area-under-the-curve (AUC=0.5), along the diagonal~~from five-fold cross-validation. ~~Perfect performance is AUC=1.0.~~ **b) NeuroVault.** Recall@k ~~measures how often the true pair~~ when each paper's target is ~~in the top-k most cosine similar pairs.~~ ~~**a) PubMed.** Contrastive recall curves~~ its best map (~~AUC=0.81~~upper bound)~~.~~ ~~**b) NeuroVault.** Contrastive recall~~ , every map (~~AUC=0.78~~multi-target)~~curve~~ , or its worst map (~~AUC=0.85~~lower bound), with p-maps inverted. ~~**c) Autoencoder.** Binary voxel performance (AUCROC=0.94)~~ With the best map, the AUC is 0.85 from text to brain and ~~bits-per-pixel gain~~ 0.79 from ~~a naive baseline~~brain to text. ~~**d) Network classification.** Multi-class predictions~~Dashed lines show the chance of each rule, from ~~argmax of cosine similarity between contrastive embeddings~~texts shuffled across papers. Chance lies above the diagonal for the upper bound and for multi-target retrieval from text to brain, because the best map or any of a paper's maps counts as a ~~confusion matrix~~hit, and below it for the lower bound. ~~The~~ Only multi-target retrieval from brain to text, with one abstract per map, follows the diagonal~~is correct classifications (dark blue)~~. ~~Counts~~ **c) Autoencoder.** Bits-per-pixel gain over a naive baseline and ~~proportion~~ voxel ROC, with AUROC 0.974 on the PubMed test set and 0.939 on NeuroVault. **d) Network classification.** Confusion matrix of ~~true-sample~~ the 119 labeled atlas maps (accuracy 0.82), with the percentage of each true class ~~are given~~ per cell.

3.2.1 Network Labeling

Cosine similarity between text-image embeddings was used to label ~~300 network images~~ the 119 atlas maps that carry one of eight network labels [1]. The argmax of cosine

similarity is taken as the predicted label. ~~The confusion matrix shows a vast majority of images~~ Of the 119 maps, 98 were labeled correctly ~~using this approach (~~(82%, Fig. 5d). Network images were embedded to a common latent space and then compared against latent text embeddings corresponding to networks, regions, and cognition. Example network images and associated text are shown in Fig. 2.

The network dataset [1] also contains unlabeled independent components (IC) for the Human Connectome Project and UK Biobank. Independent component analysis (ICA) of resting state networks has historically required manual classification of networks versus noise components [28]. Semi-supervised approaches also require manual classification for a classifier to learn from [29, 30]. These methods classify components as either noise or target components. The 40 ICA components from the network dataset are labeled automatically using the contrastive model (Fig. S4), with more detail, e.g., the names of various networks. Although noise is not directly modeled, noise components will have low similarity with a resting-state text corpus.

3.2.2 NeuroVault

The NeuroVault [22] test set was used to test the contrastive model. This shows that the model generalizes from coordinate tables during training to group-level activation maps at inference. ~~Recall@k measures how often the true neuroimage-text pair was found in the top-k most similar pairs~~ A NeuroVault paper has several maps but one text. With each paper's best-matching map as its target, the AUC is 0.85 from text to brain and 0.79 from brain to text (Fig. 5b~~, left)~~[16]~~. Similar to AUROC, the area under the curve is expected to be 0.5 for a randomly initialized, untrained model. After training with the PubMed set~~). This choice inflates the score and gives an upper bound. Each paper's worst-matching map gives a lower bound, and a multi-target score counts any of a paper's maps as a hit. Each rule has its own chance curve, from texts shuffled across papers. Under every rule, the AUC stays about 0.1 above chance. With every map as its own target, the AUC ~~reached 0.85.~~ is 0.62 from text to brain and 0.63 from brain to text, against 0.50. Supplementary Section S2.3 reports each map type separately (Supplementary Fig. S7 and Table S9).

3.2.3 PubMed

The PubMed test set was used to test the contrastive and generative models (Fig. 5a). These results are from 5-fold cross-validation, showing moderate AUC for the contrastive models (~~AUC=~~0.83 from text to brain and 0.82 ~~). Additional results show that the contrastive model performs similarly to neurocontext and niclip ()~~from brain to text). On the test set, the true pair is among the top 10 of 2,971 candidates for 9% of queries, against 0.3% by chance (Supplementary Table S8). Hard negatives, the test papers of the query's own cognitive domain, are only slightly harder than random papers of the same number (recall@10 0.24 against 0.28). On the 366 test papers that neither NiCLIP nor NeuroConText was trained on, NeuroVLM is slightly ahead of both (Fig. S3), with AUC ranging from 0.80 to ~~0.83. All test examples in this comparison were left out of training from all three models.~~ 0.84.

3.3 Autoencoder

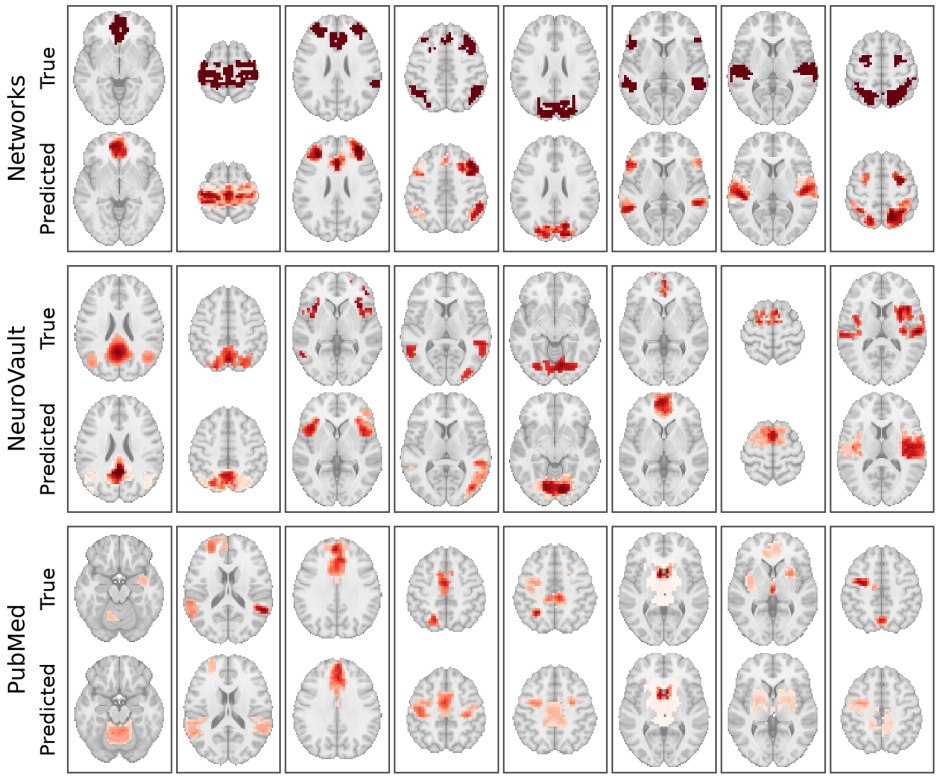


Fig. 6 Autoencoder test set reconstructions. Example ~~reconstructions for~~ maps from the networks, NeuroVault, and PubMed ~~datasets. The~~ test sets (top ~~in~~ of each box~~is the true image. The autoencoder encodes the images to latent space~~) and their reconstructions (bottom), ~~then decodes~~ each at the ~~latent space to recreate~~ axial slice through the ~~image~~reconstruction's main cluster. The ~~reconstructed predictions are shown below each true image. All images~~ NeuroVault and PubMed maps were ~~randomly selected from~~ drawn at random, and the ~~test sets~~eight networks are curated examples of the network dataset.

The first step in training is to learn an autoencoder that can reconstruct neuroimages. This was performed on the PubMed training set. Each MNI space activation map is encoded to a 384-dimensional latent vector then decoded back to MNI space. This latent vector space is the target for encoding text, resulting in text-to-brain decoding. The autoencoder was trained on binarized targets and predicts logits, e.g., each voxel was modeled as a binary classification problem. The model output is interpreted as the probability of each voxel being active. Performance is quantified by area under the receiver operating characteristic (AUROC ~~= 0.975~~0.974 on the PubMed test set and 0.939 on NeuroVault) and normalized negative log likelihood (bits/voxel) [31] that

measures improvement relative to a naive baseline (Fig. 5c). The test set included network maps, NeuroVault images, and PubMed (Fig. 6).

4 Discussion

One of the most central dogmas of neuroscience is that of functional localization: the idea that specific cognitive and behavioral functions can be localized to specific brain regions. This view has been updated to a richer, network-level view of behavior and cognition. The history of human brain imaging is such that vast amounts of natural language data have been generated in attempts to describe the populations from whom brain images were recorded, under what experimental conditions, with expert narratives describing the resulting brain maps. Together, these texts and images provide a rich set of data upon which we collectively build novel hypotheses about the neural basis of behavior, cognition, and disease.

Here, we introduce NeuroVLM, a generative AI framework that can serve as an expert collaborator in this process of narrative building and hypothesis generation. This generative approach builds upon past works that focused on text-to-brain models, constrained to a finite, predefined corpus [3, 11]. The text-to-brain model was expanded to BERT-based text encoders [27]. More recent efforts have trained contrastive models to rank similarity between text-brain pairs using LLM embeddings [16, 17].

NeuroVLM is an integrative framework (Fig. 1a) that is capable of generating text from neuroimages or neuroimages from text. This framework relies on encoding to, and decoding from, a common latent space. ~~This general framework is not dependent on the choice of encoders or decoders. Instead, we~~ The text encoder can be replaced by retraining only the small projection heads. Replacing SPECTER2 with a context-free word2vec encoder [18] lowers performance modestly (retrieval AUC 0.81 against 0.83, Supplementary Fig. S11). We provide a neuroimage autoencoder, a model to project text embeddings to the shared latent space, and an interpretable method to generate text from neuroimages. These models were trained on ~~30,826~~ 23,894 publications and extend to group-level statistical maps [22] and network atlases [1]. Use cases include generating ~~atlases or maps given a textcorpus, generating text interpretations~~ coarse spatial priors from text, generating network-level descriptions of 3D brain maps, labeling networks with coactive brain regions, finding publications most related to a statistical map query, or finding statistical maps most related to a text query.

Generation adds to retrieval. Generated maps are closer to held-out papers' maps than the maps of the most similar training papers. Generated captions match a paper's text as well as retrieved captions, but rarely carry details of other studies. Context also helps, by a modest margin. Decoded latent dimensions match the Neurosynth maps of their top terms (Supplementary Fig. S12). The contrastive cross-entropy loss matters most for retrieval (Supplementary Table S10).

A central limitation to our approach is that our training neuroimages are not native statistical maps, but proxy maps reconstructed from reported peak coordinates and smoothed with a fixed kernel. Coordinate tables are not intended to be a sufficient statistic for the underlying effects — they omit subthreshold structure, cluster shape,

effect-size information, variance, and many study-specific details such as contrast definitions, preprocessing choices, and sample characteristics. This choice is pragmatic for scale, but it bounds the quality of what the model can learn. Each map also pools all experiments, contrasts, and tables of a paper, and its positive and negative effects, because the released coordinates do not label them. It shows where a paper reports effects, not their sign. We trained on titles and abstracts, which exist for every paper. Table captions exist only for papers with open-access full text, about 40% of the corpus. They also rarely say where the coordinates lie. A classifier predicts a table's main region from its caption at chance level (macro-F1 0.06 against 0.05), but does better from the abstract.

Another limitation involves topics that are under-powered and under-studied. While common networks and widely used terminology are well-represented, rare constructs, emerging topics, or underrepresented populations can be mapped less reliably. For more niche, specific topics, additional fine-tuning is recommended.

In addition, it is important to emphasize that generative models often can introduce confident predictions that go beyond the evidence. Using NeuroVLM, brain maps will be generated from any text input, no matter how ludicrous. For example, ~~show~~ "show me the hot dog areas of the brain,~~"~~" will result in beautiful 3D brain images – largely regions related to gustation and reward, as it turns out (so perhaps not entirely ludicrous). Nevertheless, NeuroVLM should be viewed as an AI collaborator – a tool for organizing and navigating the literature, suggesting candidate networks, concepts, and related papers – rather than as an oracle that provides mechanistic explanations or ground truth maps.

Text-to-brain outputs are coarse spatial priors, without effect size, sign, or uncertainty. The values of a generated map are not probabilities. Brain-to-text outputs are broad network-level labels, not interpretations of a specific study or mechanism. Failures are not rare. In 16% of test captions, the similarity to the paper's text is negative. A typical failure is a fluent description of a different system. `[NETWORK]` titles are the least reliable of the three outputs, especially for single-study maps. Titles are steered toward canonical names, but the body text can still contain invented words or stray fragments. NeuroVLM also ignores negation. A sentence and its negation give nearly the same map ($r = 0.84$ to 1.00). Composition is roughly additive. Joining two concepts gives a map that covers both, and we find no evidence of composition beyond addition. The out-of-distribution scores flag unusual inputs, but they are warnings, not calibrated probabilities.

Despite these caveats, our work suggests that large-scale, paired text-neuroimage corpora are sufficient to learn a useful shared representation that supports practical tasks such as retrieval, labeling, and coarse generation. ~~Our strongest limitations involve access to neuroimaging data rather than model architecture.~~ Contributions to open-access repositories, such as NeuroVault [22] and OpenNeuro [32], will improve the future generations of multimodal neuroscience models; or more simply: higher quality, expertly labeled and annotated data can yield better results [33].

Looking ahead, we believe that it is critical to strengthen the ecosystem around standardized benchmarks and reproducible evaluation. Network labeling and recall-style retrieval provide accessible demonstrations, but they do not fully measure

neuroscientific correctness, robustness to analysis pipelines, or calibration under dataset shift. Progress will benefit from community benchmarks that specify train-test splits, map types, preprocessing, and query sets, and that evaluate multiple axes of performance: retrieval utility, interpretability, out-of-distribution generalization, and failure modes. Open model checkpoints, transparent preprocessing, and reproducible training scripts should be prioritized.

NeuroVLM frames the neuroimaging literature as a bidirectional link between how studies are described in language and where effects are reported in the brain. This enables a single framework to support retrieval, labeling, and generation from ~~tens-of-thousands of~~ tens of thousands of paired examples. These models can help organize a fragmented literature into a searchable, cross-modal interface: a way to move from maps to candidate concepts and related papers, and from text queries to spatial priors and exemplar results. By making these capabilities accessible with ~~modular components and~~ modest compute, this work reduces the barrier to using modern multimodal methods for everyday neuroscientific workflows. NeuroVLM runs on a consumer CPU, and the whole model retrains from scratch in about 1.8 hours on one GPU.

~~Our results underscore that the limiting factor for biological fidelity is often the structure of what the field reports and shares. When the community primarily provides~~ NeuroVLM learns from peak coordinates, ~~models inherit the constraints of peak-coordinate reconstructions; in contrast, sharing more data, such as~~ not full statistical maps. Coordinate-based meta-analysis loses information that image-based meta-analysis keeps, such as effect sizes and subthreshold effects [34]. Models trained on coordinates inherit these losses. Sharing full statistical maps, contrast definitions, and rich metadata ~~, creates a virtuous cycle and allows~~ would allow models to learn quantitative, uncertainty-aware representations that better reflect the evidence [33].

NeuroVLM is a step toward computational tools that augment, rather than replace, expert interpretation. Its most immediate value is as a literature navigation and hypothesis-generation aid that can surface relevant work, highlight commonalities across maps and descriptions, and provide interpretable suggestions. The broader opportunity is to couple these models with standardized benchmarks and richer open data so that future systems are reliable, reproducible, and grounded in the full statistical record of human neuroimaging research.

Data availability. All data used in this work are public. The datasets are in the NeuroVLM organization on Hugging Face (https://huggingface.co/neurovlm), at the revisions below.

- neurovlm/neuro_image_papers (revision bfda6c9): PubMed titles, abstracts, coordinate tables, and summaries.
- neurovlm/embedded_text (revision 7f861d8): PubMed maps, text embeddings, network maps, and NeuroVault maps.
- neurovlm/neurovault_maps (revision 4324288): NeuroVault maps, collections, and texts.
- neurovlm/pubmed_summary_qa (revision 2d53307): summaries of the PubMed abstracts.
- neurovlm/synthetic_pairs (revision 3005bb0): synthetic and curated concept texts.

- neurovlm/pubmed_ale_maps (revision 1766fad) and neurovlm/cognitive_atlas (revision 46194ce).

The language model is neurovlm/NeuroQwen3-0.6B (revision 23f50c9). Its adaptation corpus consists of PubMed titles and abstracts, which are public. The released model weights are in the same organization. The final model of this paper is retrained from these inputs by one command (see Code availability). The code repository lists the revision and checksum of every file it reads. The coordinates come from PubMed and Neurosynth Compose [20], the group-level maps from NeuroVault [22], and the network maps from the network correspondence toolbox [1]. Every figure and table is drawn by a script in the code repository from these data. Fig. S3 also uses the published NiCLIP and NeuroConText checkpoints.

Code availability. The code is at https://github.com/neurovlm/neurovlm under the Apache-2.0 license. It holds the library, the training pipeline, and a script for every figure and table, with a guide that names the script behind each figure. One command builds the pinned environment, fetches every input from Hugging Face at pinned, checksummed revisions, retrains the model, redraws the figures, and checks each metric against the paper. Two runs from scratch gave byte-identical weights on the same GPU model and library versions. Continuous integration trains every model for one batch on a CPU, which tests the whole pipeline.

Supplementary information.

S1 Generation

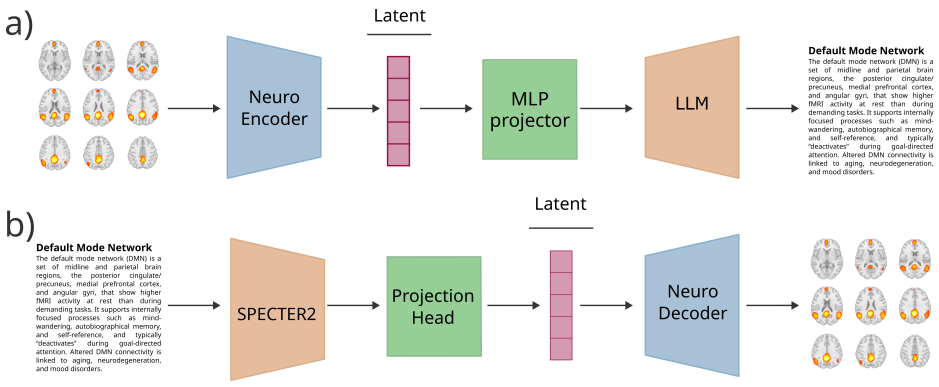


Fig. S1 Generating and interpreting neuroimages. a) Brain-to-text generation is used to predict text interpretations of input brain maps. An activation map is first encoded to the latent vector (pink). The ~~Qformer~~ MLP projector (green) then transforms the latent vector to the token embedding matrix of the LLM. ~~Qwen3-0~~NeuroQwen3-0.6B (orange) generates text descriptions. **b)** A text query is first encoded with ~~Specter2~~SPECTER2, a frozen, pre-trained text encoder. The projection head is a small, trainable network that transforms from ~~Specter2~~SPECTER2's embedding space to the ~~bi-modal, text-neuroimage embedding~~ latent space (pink). This latent vector is decoded (blue) to generate activation maps.

S2 Preprocessing

S2.1 Coordinate Tables

Coordinate tables (~~n=30826~~n = 30,826 papers, 857,298 peaks, median 17 per paper) were collected from PubMed and NeuroSynth Compose databases. All peaks of a paper, from every experiment, contrast, and table, were pooled into one map. The released coordinates carry no experiment, contrast, table, or sign labels. Coordinates were convolved with a 9mm Gaussian kernel. The kernel was normalized to values between zero and one. The zero-to-one range allows using binary cross-entropy loss when training the autoencoder. The MNI-space convolutional product, $\mathbf{X}$, was clamped to values between zero and one. The 512-dimensional DiFuMo atlas, $\mathbf{P}$, was used to transform the convolved products (Fig. S2). This is in contrast to using the 512-dimensional embedding directly, as in [16, 17]. Instead, the autoencoder is trained to learn a data-driven, low-dimensional embedding space.

$$\hat{\mathbf{X}} = \mathbf{X}\mathbf{P}\text{diag}(s)^{-1}\mathbf{P}^{\top}, \text{where } s_j = \sum_v \mathbf{P}_{vj}$$

The DiFuMo reconstructions were resampled to 4mm isotropic voxel space, matching [11]. Masking and flattening produced a 28,542 dimensional vector per image.

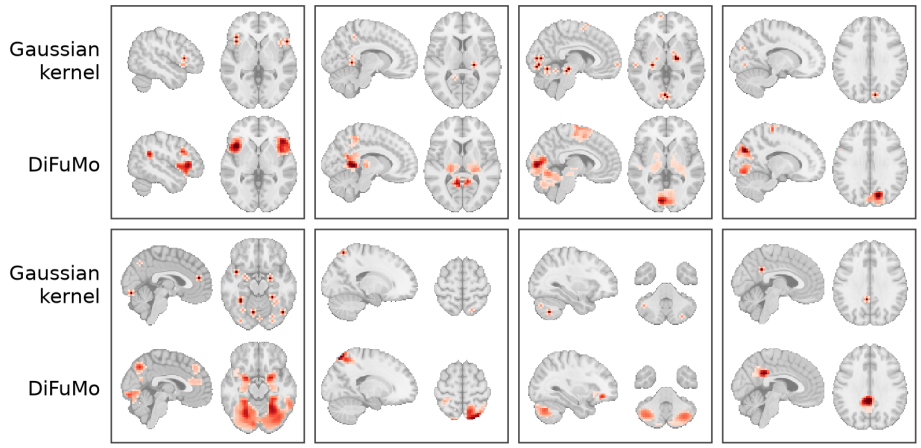


Fig. S2 Gaussian products transformed to DiFuMo. Points, from each study's coordinate ~~tables~~table, were ~~first~~ convolved with a Gaussian kernel (top). With a single study, the ALE kernel reduces to this Gaussian kernel. The image output from Gaussian kernel convolution was reconstructed with DiFuMo (bottom).

S2.2 Text

Text included titles and abstracts (~~n=30826~~n = 30,826) from PubMed publications. All text is paired to a DOI. Text was embedded using ~~Specter2 [19]~~SPECTER2 [19] , the allenai/specter2_aug2023refresh model with its ad hoc query adapter. This produces a 768-dimensional vector per publication. ~~Specter2~~ SPECTER2 was trained with contrastive loss, promoting ranking similarity between pairs. These embeddings almost collapse to a mean. This results in high cosine similarity for unrelated pairs, e.g. the expected cosine similarity distribution is shifted upwards by a constant. For text-to-brain decoding, the embeddings are demeaned with an empty text embedding, shifting the expected value of cosine similarity between random pairs to zero. Additionally, text embeddings were unit normalized, as contrastive loss is scale insensitive.

Llama-3.3-70B-Instruct (meta-llama/Llama-3.3-70B-Instruct) summarized each paper with the summary prompt below as its system message, removing low-level study details. The user message was the paper's title and abstract, joined by a separator token (`title[SEP]abstract`). Decoding sampled with temperature 0.3, top-p 0.9, and top-k 50, with at most 256 new tokens, so a rerun words each summary differently. The stage 1 projector learns from these summaries.

Qwen3.5-27B wrote short entries with the synthetic-text prompt below. The seeds were region names from nilearn's brain atlases, Cognitive Atlas concepts, and the network names of the network atlases. The user message was the seed alone, such as "Insula". Decoding used temperature 0.1 and top-p 0.9, with thinking mode off.

Sampling each seed several times gave re-wordings of one concept. The anatomically centered projection head learns from these entries.

The stage 2 projector learns from 1,338 curated texts about networks, regions, and functions, which Qwen3.5-27B wrote in the same way. The original generation code was not kept. Scripts in the code repository re-create the summaries and the synthetic texts. Their outputs cover the same topics as the released texts, in different words.

Summary prompt (Llama-3.3-70B-Instruct)

```
You are writing a short, topic-focused neuroscience mini-review.

INPUT:
You will receive:
    title + " [SEP] " + abstract from an fMRI publication.

GOAL:
Write a single coherent paragraph (4–6 sentences) that:
- Describes the scientific topic implied by the paper (e.g., meditation and the default mode
  network, exercise and working memory), not the paper's methods.
- Explains the key cognitive processes (e.g., working memory, self-referential thought,
  emotion regulation) and the main brain systems or networks involved.
- May mention typical effects or relationships in general terms (e.g., "has been associated
  with changes in connectivity" or "can influence executive control"), but should stay
      high-level.

ALLOWED:
- General statements about effects or relationships (e.g., "exercise has been associated
  with improved cognition", "meditation can alter DMN connectivity").
- Generic references to "studies" or "research" in the field.
- Qualitative descriptions of brain activity or connectivity (e.g., "increased activation
  in prefrontal regions", "changes in functional connectivity").

AVOID:
- Exact numerical details (sample sizes, p-values, coordinates, time points).
- Detailed task or protocol descriptions from the specific study.
- Directly quoting or closely paraphrasing long sentences from the abstract.
- Overly narrow focus on this single experiment; write as if summarizing the topic area.

STYLE:
- Neutral, scientific tone, like a short review or encyclopedia entry.
- One paragraph only. No bullet points or headings.
- Focus on giving a clear sense of "what this topic is about" in the context of the field.
```

Synthetic-text prompt (Qwen3.5-27B)

```
Write a short educational neuroscience entry for the provided fMRI concept.

Requirements:
- Explain the concept in general neuroscience terms.
- Always include relevant anatomical regions.
- Connect the anatomy to the cognitive or neural function.
- Use cautious, broad language; avoid overclaiming.
- Do not mention studies, patients, diseases, disorders, treatments, age groups, tasks,
      results, methods, or statistics.
- Do not use: critical for, responsible for, causal, proven, this study, activation, altered,
      reduced, increased.
- Do not invent highly specific subregions unless they are commonly associated with the
      concept.
- If the concept is broad, name the main commonly associated regions or networks.
- If the concept is anatomical, describe its broader network/function associations.
- Write 3-5 complete sentences.

Return valid JSON only:
```

```
{
  "title": "1-5 words",
  "description": "3-5 sentences"
}
```

One complete example of each kind of text follows.

- Summary of a training paper (stage 1 projector). The human parietal and opercular regions are involved in processing somatosensory information. Somatotopy refers to the organization of these regions according to the location of sensory inputs on the body. The parietal cortex, including areas such as the postcentral gyrus and intraparietal sulcus, contains multiple somatosensory areas with distinct somatotopic representations. Attentional modulation can enhance activity in these areas, particularly in higher-order regions.
- Synthetic text "Insula" (anatomically centered projection head). The insula is a deep cortical region situated within the lateral sulcus of the cerebral hemisphere. This area maintains connections with structures such as the anterior cingulate cortex and the amygdala. Processes occurring in this region are often linked to perceptions of internal bodily states and emotional awareness. It contributes to the integration of sensory information with subjective experience across various brain networks.
- Curated `[NETWORK]` text "Salience Network" (stage 2 projector). The salience network commonly includes anterior insula and dorsal anterior cingulate cortex. It is associated with detecting and prioritizing behaviorally relevant internal and external information. Its connections with limbic, autonomic, and frontoparietal systems support flexible shifts between internal monitoring and externally oriented control.

S2.3 NeuroVault

NeuroVault ~~images ($n = 4553$~~maps ($n = 3{,}183$ from 312 papers) span several group-level map types reflecting different analysis outputs, including inferential statistics maps: t-, z-, and F-maps (voxelwise test statistics for a contrast model) and p-maps (the corresponding voxelwise significance values). We also included beta maps (univariate GLM estimates and multivariate/decoding weight maps)~~as well as ROI/masks (binary/probabilistic inclusion maps) and parcellations (discrete region labels)~~. Masks, parcellations, anatomical maps, and variance maps were removed before evaluation. Seven papers (29 maps) duplicate a PubMed training paper and were also removed, leaving 3,~~which define spatial subsets or atlas assignments rather than continuous statistics~~154 maps from 305 papers. Of the 480 maps their uploaders labeled Other, 70 mean or variance images and 10 maps that could not be typed were also excluded, leaving 3,074 maps from 298 papers for retrieval.

P-maps were inverted to $1 - p$, so that high values mark strong effects. Signed maps keep their positive tail. Using both tails, or adding negative tails as extra maps, reduced evaluation performance. Supplementary Table S9 reports each map type separately. Activation maps were cluster-thresholded. For each map, we applied a high-intensity threshold at the 99th percentile and removed small clusters using a minimum cluster extent of 50 voxels. The surviving suprathreshold clusters were then re-vectorized with the same masker into a fixed-length feature vector, $d = 28542$, and

used in testing. Finally, we binarized the resulting vectors (presence/absence of surviving voxels) because the absolute scaling of values across map types (e.g., t-, z-, p-, F-, and beta maps) is not directly comparable; binarization yields a uniform support that emphasizes spatial localization over intensity.

NeuroVault text was limited to titles and abstracts, since per-image contrast definitions were often noisy or missing. Since multiple images were paired to each title-abstract pair, the contrastive model was used to select the most similar image. This inflates scores and provides an upper performance bound. ~~All $n = 4553$ images~~ A lower bound keeps each paper's worst-matching map, and a multi-target score counts any of a paper's maps as a hit. A per-map score makes every map its own target. The chance level of each rule comes from the same rule applied after shuffling texts across papers (mean of 20 shufflings). It is not 0.5 for the upper and lower bounds, because the target is chosen with the same similarities that are ranked (Supplementary Fig. S7). All 3,154 maps were used when evaluating the autoencoder, since the autoencoder does not require text-image pairs.

S3 Autoencoder

S3.1 Architecture

Table S1 Autoencoder architecture (layer-wise).

Layer	Module	Input → Output	Activation
Encoder 1	Linear	28542 → 1024	ReLU
Encoder 2	Linear	1024 → 512	ReLU
Encoder 3	Linear	512 → 384	None
Decoder 1	Linear	384 → 512	ReLU
Decoder 2	Linear	512 → 1024	ReLU
Decoder 3	Linear	1024 → 28542	None

Shapes omit batch dimension.

After resampling to 4 mm^3 voxel space, masking, and flattening, each neuroimage is a 28,542 dimensional vector. The exact layers, sizes, and activation functions are provided in Tab. S1. The encoder maps from 28,542 to 384, then the decoder maps from 384 back to 28,542. ReLU activations are used. The model can be thought of as a single masked convolution per layer, with a kernel size equal to the input size. This simplifies handling sparsity from outside-of-brain voxels. Alternatively, convolutional and transposed convolutional layers with a smaller kernel could be explored in future work.

S3.2 Binary Cross-Entropy Loss

The autoencoder was trained with binary cross-entropy (BCE)-with-logits loss. Let $\mathbf{x}^{(i)} \in [0,1]^D$ denote a masked and flattened activation map with $d = 28542$. The

decoder produces logits $\mathbf{z}^{(i)} \in \mathbb{R}^d$, that are transformed to probabilities using the sigmoid function, $\sigma(u)$.

$$\hat{\mathbf{x}}^{(i)} = \sigma\left(\mathbf{z}^{(i)}\right), \qquad \sigma(u) = \frac{1}{1+\exp(-u)} \tag{1}$$

We minimize the binary cross-entropy between $\mathbf{x}^{(i)}$ and $\hat{\mathbf{x}}^{(i)}$ using the numerically stable BCE-with-logits form:

$$\mathcal{L}_{\text{BCELogits}}\left(\mathbf{x}^{(i)}, \mathbf{z}^{(i)}\right) = -\frac{1}{D}\sum_{j=1}^{D}\left[x_j^{(i)} \log \sigma(z_j^{(i)}) + \left(1 - x_j^{(i)}\right) \log\left(1 - \sigma(z_j^{(i)})\right)\right] \tag{2}$$

$$= \frac{1}{D}\sum_{j=1}^{D}\left(\max(z_j^{(i)}, 0) - z_j^{(i)} x_j^{(i)} + \log\left(1 + \exp(-|z_j^{(i)}|)\right)\right). \tag{3}$$

The overall reconstruction loss is the minibatch average over B examples:

$$\mathcal{L}_{\text{recon}} = \frac{1}{B}\sum_{i=1}^{B} \mathcal{L}_{\text{BCELogits}}\left(\mathbf{x}^{(i)}, \mathbf{z}^{(i)}\right). \tag{4}$$

S3.3 Training

The autoencoder was trained for ~~100~~ 101 epochs using AdamW with a learning rate of 5×10^{-5} and a batch size of ~~256.~~ 256, and the epoch with the lowest validation loss was kept. Reconstruction was optimized with binary cross-entropy loss with logits as described above.

S4 Projection Head

S4.1 Architecture

Table S2 Projection head architecture (layer-wise).

Layer	Module	Input → Output	Activation
Text 1	Linear	768 → 512	ReLU
Text 2	Linear	512 → 384	None
Neuro 1	Linear	384 → 384	ReLU
Neuro 2	Linear	384 → 384	None

Shapes omit batch dimension.

There are three types of projection heads. The first two ~~are used to~~ transform 768 dimensional ~~Specter2~~ SPECTER2 embeddings to 384 dimensions. These two projection heads were the same size but trained on different targets. ~~One was trained to map~~

~~latent~~ The text-to-brain head maps text to the autoencoder's latent space, providing text-to-brain decoding. The other text projection head was learned alongside the neuro projection head, optimized with the contrastive objective, InfoNCE loss. The neuro projection head maps from the autoencoder latent space to the bi-modal latent space. Tab. S2 contains the exact layers sizes. A larger, anatomically centered projection head (3.8 million parameters) maps concept texts to the autoencoder latent.

S4.2 InfoNCE Loss

Contrastive projection heads were trained with Info Noise Contrastive Estimation (InfoNCE) loss. Let $\mathbf{I} \in \mathbb{R}^{B\times d}$ and $\mathbf{T} \in \mathbb{R}^{B\times d}$ represent a batch of image and text embeddings, respectively, where B is the batch size and d is the embedding dimension. We used $B = 2048$ and ℓ_2-normalize each embedding for $i \in \{1, \ldots, B\}$:

$$\tilde{\mathbf{i}}_i = \frac{\mathbf{i}_i}{\|\mathbf{i}_i\|_2}, \qquad \tilde{\mathbf{t}}_i = \frac{\mathbf{t}_i}{\|\mathbf{t}_i\|_2}, \tag{5}$$

We then compute the pairwise cosine similarity matrix:

$$\mathbf{S} = \frac{1}{\tau}\,\tilde{\mathbf{I}}\,\tilde{\mathbf{T}}^{\top} \in \mathbb{R}^{B\times B}, \tag{6}$$

The temperature ~~parameter was learnable and initialized to~~ was fixed at $\tau = 0.07$ [15]. The correct text-image pair for each index i is the diagonal entry S_{ii}, all off-diagonal entries are mismatched pairs, i.e., text-image pairs from mismatched studies. Thus, target labels are the identity matrix. We optimize a symmetrized cross-entropy loss over image-to-text ($\mathcal{L}_{\text{i2t}}$) and text-to-image ($\mathcal{L}_{\text{t2i}}$) retrieval. The final loss is the mean of the two.

$$\mathcal{L}_{\text{i2t}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(\mathbf{S}_{ii})}{\sum_{k=1}^{B}\exp(\mathbf{S}_{ik})} \tag{7}$$

$$\mathcal{L}_{\text{t2i}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(\mathbf{S}_{ii})}{\sum_{k=1}^{B}\exp(\mathbf{S}_{ki})} \tag{8}$$

$$\mathcal{L}_{\text{InfoNCE}} = \frac{1}{2}\left(\mathcal{L}_{\text{i2t}} + \mathcal{L}_{\text{t2i}}\right). \tag{9}$$

S4.3 Training

The ~~projection head for~~ text-to-brain ~~generation was trained for 200 epochs with a batch size of 1024 using AdamW with a learning rate of 5×10^{-5}. Optimization of this text-to-brain model used a BCE-with-logits objective.~~head predicts the autoencoder latent of each training paper. Its loss is the mean squared error of the latent plus two correlation terms between the decoded map and the decoded true latent, one on logits and one on probabilities. During training, Gaussian noise is added to the input embeddings, and the output bias starts at the mean training latent. The head was trained with AdamW (learning rate 10^{-3}, weight decay 0.01, batch size 512) for

up to 80 epochs, and the epoch with the best validation Dice was kept (epoch 30). Linear (ridge) regression from SPECTER2 to the latent is the simplest form of this mapping and is compared in Supplementary Fig. S6. For joint text–image contrastive alignment, we trained paired projection heads using a symmetric InfoNCE contrastive objective with temperature, $\tau = 0.07$. Models were optimized with AdamW for ~~300~~ 301 epochs using a batch size of 2048 and learning rate of 1×10^{-5}. The anatomically centered head learns from 6,309 synthetic text-map pairs, built from atlas regions and training papers, and from 512 random PubMed training pairs per step with the text-to-brain loss. It was trained for up to 2,000 epochs, and the epoch with the best PubMed validation recall was kept (epoch 200). Supplementary Table S10 compares other losses for these heads.

S5 ~~Query Transformer~~Brain-to-Text Projector

The ~~Qformer~~ projector maps raw (r) and semantic (s) latent image embeddings to the token embedding space of the language model, here ~~, the neuroscience-adapted Qwen3-0~~NeuroQwen3-0.6B. The semantic embeddings are raw image embeddings projected onto the text-semantic space using the contrastive projection heads. The language model predicts text, given these latent representations:

$$P(\text{text}|r, s)$$

The ~~Qformer~~ projector transforms the latent embeddings to the token embedding matrix (Z) of the LLM. ~~During training,~~ Stage 1 trains on s~~is used. At inference~~. Stage 2 trains on s_{proj}~~may alternatively be used~~, the projection of s onto the canonical bank of the text's own level, as at inference.

$$Z = \text{~~Qformer~~Projector}(r, s) \tag{10}$$

The projector is a residual multilayer perceptron (Supplementary Table S3). Layer normalization is applied to each input, and the two are concatenated. A linear layer maps them to a width of 704, followed by two pre-normalized residual blocks. A final linear layer produces 32 tokens of 1024 dimensions. Each token adds a learned query embedding and passes through layer normalization. An alignment head maps the mean token back to the semantic space for the alignment term of stage 1. A query transformer (Qformer) [26] of nearly the same size, with 32 learned queries that attend over the two inputs through six transformer layers, was less accurate (Supplementary Fig. S9).

Semantic embeddings may optionally be projected onto a canonical semantic space. This improves robustness to noise and preprocessing variation by replacing a predicted semantic embedding with a soft mixture of clean, interpretable canonical prototypes. Let s denote the predicted semantic embedding and let C denote a bank of k canonical embeddings in the same d-dimensional contrastive space:

$$r, s \in \mathbb{R}^d, \qquad C \in \mathbb{R}^{k \times d}.$$

Both s and the rows of C are ℓ_2-normalized. Projecting s onto C is the cosine similarity between s and each canonical prototype. These similarities are passed through

softmax with a temperature parameter (τ) that determines smoothing across concept embeddings. With lower temperature, the projection behaves more like nearest-neighbor assignment. With higher temperature, it averages across more canonical concepts. This improves the decoding of maps that are similar enough to, yet outside of, the training distribution.

$$w = \text{softmax}\left(\frac{sC^\top}{\tau}\right) \in \mathbb{R}^k \tag{11}$$

$$\tilde{s}_{\text{proj}} = wC \in \mathbb{R}^d \tag{12}$$

$$s_{\text{proj}} = \frac{\tilde{s}_{\text{proj}}}{\|\tilde{s}_{\text{proj}}\|_2} \tag{13}$$

The canonical embedding matrix can be split into subtypes: regions, networks, and function. This allows steering the LLM towards these targets. For example appending the `[NETWORK]` token to the bottom of Z steers language generation towards network-level descriptions while `[FUNCTION]` token steers towards higher-level, cognitive interpretations. Appending `[REGION]` generates text related to the most dominant region in the image. Future work could increase these steering tokens to various sub-domains, given curated datasets.

Captions are decoded greedily or by beam search with a repetition penalty, with the vocabulary restricted to Latin-script tokens. After a `[NETWORK]` or `[REGION]` token, tokens that cannot continue a canonical name of that bank are masked until the line ends. The language model's own probabilities choose among the remaining tokens, so the title is still generated, steered toward the canonical names rather than looked up. Without the mask, 23% of test `[NETWORK]` titles and 9% of `[REGION]` titles matched no canonical or atlas name. With it, none do. The body text and `[FUNCTION]` titles are not masked.

Stage 1 was trained with AdamW (learning rate 3×10^{-4} with cosine decay, batch size 32) and early stopping, and the epoch with the lowest validation loss was kept (epoch 4). Its loss is next-token cross-entropy through the frozen language model plus an alignment term. Stage 2 starts from stage 1 and trains for 30 epochs with AdamW (learning rate 10^{-4} with cosine decay, batch size 18) on 972 concept texts, with 134 for validation. Each training map is binarized at a percentile drawn from 90 to 97 for each batch, with voxel dropout in half the maps, and title tokens weigh five times in the loss. Stage 2 was trained with seeds 1 to 5, and the seed with the lowest final validation loss (seed 3) was kept.

Component	Operation	Output	Params
Image input	brain latent	$B \times 384$	–
Semantic input	projected/canonical semantic latent	$B \times 384$	frozen
Input ~~projections~~ normalization	~~two LN-MLPs, 384 →512 →512~~ LN of each input, concatenated	~~$B \times 2 \times 512$~~ $B \times 768$	~~921,088~~ 1,536
~~Query tokens~~ Input layer	~~32 learned queries~~ Linear, 768→704	~~$B \times 32 \times 512$~~ $B \times 704$	~~16,384~~ 541,376
~~Transformer decoder~~ Residual blocks	~~6 layers, 8 heads, FFN width 2048~~ 2 pre-LN MLP blocks, 704→1408→704	~~$B \times 32 \times 512$~~ $B \times 704$	~~25,224,192~~ 3,971,968
~~LM projection~~ Output layer	~~LN-MLP, 512 →1024 →1024~~ LN, Linear, 704→32 × 1024	$B \times 32 \times 1024$	~~1,575,936~~ 23,070,080
Query tokens	32 learned embeddings, LN	$B \times 32 \times 1024$	34,816
Alignment head	mean pool, LN-MLP, 1024→512→384	$B \times 384$	723,840
~~QFormer~~ Projector total	–	–	28,~~461,440~~ 343,616

Table S3 ~~**Query transformer architecture.**~~ **Brain-to-text projector architecture.** B denotes batch size, LN denotes layer normalization, MLP denotes multilayer perceptron, ~~FFN denotes feed-forward network,~~ and LM denotes language model. The packaged model also includes a frozen 295,680-parameter image projection head and frozen canonical semantic projection matrix.

S6 Generative Performance

In the submitted version, the BERTScore null mean (0.52) came from a cached score of an earlier generation run, and every null in Supplementary Table S4 is computed from the same captions as the matched scores.

Table S4 Match-versus-null evaluation of brain-to-text and text-to-brain generation on the PubMed test set (~~$n = 2987$~~ $n = 2{,}971$). Null scores were computed from mismatched image-text pairs. Welch's t-tests compare matched and null distributions~~; p-values are reported in scientific notation~~.

Direction	Metric	Match mean	Null mean	t	p
Brain-to-text	Contrastive cosine	~~0.27~~ 0.28	~~0.02~~ 0.01	~~86.54~~ 95.46	~~1.61×10^{-1025}~~ 7×10^{-1158}
Brain-to-text	BERTScore F1	~~0.56~~ 0.553	~~0.52~~ 0.548	~~68.03~~ 8.88	~~8.44×10^{-737}~~ 9×10^{-19}
Brain-to-text	Semantic cosine	~~0.47~~ 0.46	0.40	~~22.94~~ 22.73	~~1.51×10^{-111}~~ 1×10^{-109}
Text-to-brain	Contrastive cosine	~~0.28~~ 0.34	~~0.11~~ 0.00	~~52.00~~ 95.41	~~3.22×10^{-482}~~ 4×10^{-1116}
Text-to-brain	Dice	~~0.47~~ 0.53	~~0.40~~ 0.41	~~24.07~~ 35.18	~~3.25×10^{-122}~~ 2×10^{-246}
Text-to-brain	Pearson	~~0.29~~ 0.32	~~0.17~~ 0.12	~~26.66~~ 38.61	~~7.46×10^{-148}~~ 6×10^{-290}

S7 Model Comparison

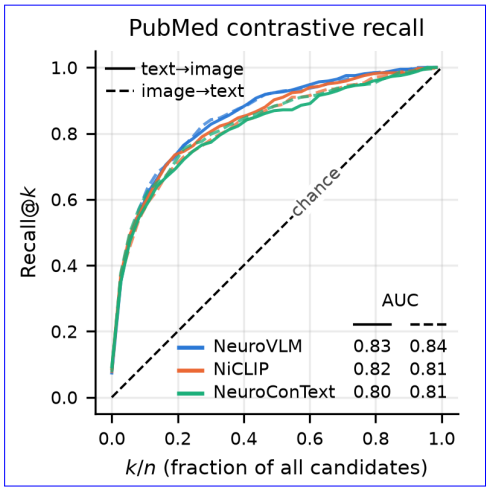


Fig. S3 Contrastive model comparison. Recall@k ~~test set curves~~ for NeuroVLM (blue), NiCLIP (orange), and NeuroConText (green) ~~. Models performed similarly across~~ on the ~~methods~~366 PubMed test papers that none of the three models was trained on. ~~Text includes titles~~ Solid lines are text to image and ~~abstracts only~~dashed lines image to text. NeuroVLM uses a language encoder that is 63 times smaller than NiCLIP and NeuroConText, 110 million versus 7 billion parameters.

S8 Independent Component Labeling



Fig. S4 Independent Component Labeling. Automated labeling of resting-state components from UK Biobank (top) and the Human Connectome Project ~~and UK Biobank~~(bottom). ~~Each row corresponds to one network. The contrastive columns are~~ Bars give the ~~top-5~~ five most similar ~~pairs~~ labels from the contrastive models, and taking `argmax` gives the predicted label~~for that image~~. The generated column is ~~brain-to-text generation using~~ the ~~Qformer model~~ `[NETWORK]` caption of the stage 2 MLP projector, its title in bold. Components lie at the 6th to 99th percentile of all 40 components by brain-to-text cosine.

S9 Network Examples

Fig. 2 shortens the generated texts so that its maps stay legible. A generated text is best judged in full, so Supplementary Table S5 gives the complete input description and the complete generated texts of every network in Fig. 2. The texts describe networks, regions, and functions in general terms, as the stage 2 projector was trained to do.

Table S5: Complete texts of Fig. 2. The input description is the text the text-to-brain head turns into a map. The stage 2 projector writes the generated titles and texts for the atlas map.

Text (generated title)	Complete text
Visual (YeoLab VisualA)	
Input description	The visual network supports perception and interpretation of visual information. It is centered on primary visual cortex, extrastriate occipital cortex, and lateral occipital cortex. It supports visual feature processing, object perception, motion perception, spatial layout, and visual attention.
Network **Object Perception Network**	The object perception network includes extrastriate cortex, particularly the right inferior temporal cortex, and includes lateral occipital regions. It is commonly associated with the perception of object form, location, motion, and depth. The network is often described as supporting the coordinated analysis of visual input.
Anatomy **Lateral Occipital Cortex**	The lateral occipital cortex is a ventral occipital region. It is commonly associated with object form and visual identity. This region is often described as supporting perception of complex visual structures.
Cognition **Visual Analysis**	Visual analysis is the processing of visual form, object structure, and visual identity. It is associated with extrastriate occipitotemporal cortex, intrastriate temporoparietal cortex, lateral temporal cortex, and fusiform gyrus. This concept supports recognizing complex visual patterns, object form, spatial layout, and visual object identity.
Language (Glasser Language)	
Input description	The language network supports comprehension and production of spoken and written language. It is mainly left-lateralized, involving inferior frontal cortex, posterior superior temporal cortex, middle temporal cortex, and angular gyrus. It supports semantic access, syntax, phonology, speech comprehension, reading, and verbal expression.
Network **Language Network**	The language network commonly includes inferior frontal gyrus, middle temporal pole, superior temporal gyrus, and lateral temporal cortex. It is associated with phonological processing, syntactic analysis, and meaning construction. The network is often described as supporting the comprehension and production of spoken or written language.
Anatomy **Middle Temporal Gyrus**	The middle temporal gyrus lies along the lateral surface of the superior temporal gyrus. It is commonly associated with auditory processing, language comprehension, and semantic content. This region is often described as supporting the analysis of spoken words.

Table S5 (continued)

Text (generated title)	Complete text
Cognition **Language Comprehension**	Language comprehension commonly involves inferior frontal gyrus, middle temporal gyrus, and superior temporal gyrus. These regions support processing of spoken and written language, including phonology, syntax, semantics, and meaning retrieval.
Auditory (Shirer Auditory)	
Input description	The auditory network supports perception and interpretation of sound. It is centered on Heschl's gyrus, primary auditory cortex, planum temporale, and superior temporal gyrus. It supports pitch, loudness, rhythm, timbre, speech-sound processing, and auditory scene analysis.
Network **Auditory Network**	The auditory network includes Heschl's gyrus, superior temporal gyrus, planum temporale, and inferior frontal cortex. It is commonly associated with the processing of sound, speech, and environmental sounds. The network is often described as a hierarchical system for auditory perception. Its coordinated activity is thought to support the integration of auditory information across the brain.
Anatomy **Superior Temporal Gyrus**	The superior temporal gyrus is centered on the superior temporal sulcus. It is commonly associated with auditory processing, speech perception, and sound-related associations. Its anatomy supports representation of acoustic features such as frequency, timbre, and pitch.
Cognition **Early Auditory Processing**	Early auditory processing is the analysis of sound input at high frequencies. It involves primary auditory cortex in Heschl's gyrus, superior temporal gyrus, and planum temporale. This concept supports speech recognition, pitch detection, and low-frequency sound processing.
Somatomotor (WashU LateralSM)	
Input description	The somatomotor network supports movement and body sensation. It spans precentral gyrus, postcentral gyrus, supplementary motor area, and paracentral lobule. It supports voluntary movement, touch, proprioception, motor planning, and sensorimotor integration.
Network **Somatomotor Network**	The somatomotor network is commonly associated with somatomotor representation in the primary motor cortex. It is broadly tied to bodily motion, tactile sensation, and motor control. The network is often described as a central hub for somatosensory-to-motor integration.
Anatomy **Primary Somatosensory Cortex**	Primary somatosensory cortex lies in the posterior parietal lobe. It is commonly associated with tactile, proprioceptive, and body-related sensory representations. It supports the integration of sensory information across the body.
Cognition **Tactile Encoding**	Tactile encoding is the representation of touch in the body. It involves primary somatosensory cortex in the parietal operculum, secondary somatosensorium in the insular cortex, and posterior insula. This concept supports tactile discrimination, body awareness, and the integration of sensory feedback with cognitive processing.

Table S5 (continued)

Text (generated title)	Complete text
Interoception (Laird Interoception)	
Input description	The interoception network supports awareness of internal bodily state. It is centered on the insula and anterior cingulate cortex, with contributions from midcingulate cortex. It supports visceral sensation, autonomic monitoring, bodily awareness, pain appraisal, and affective feeling states.
Network **Interoceptive Network**	The interoceptive network includes anterior insula, posterior insula, anterior cingulate cortex, thalamus, and somatosensory cortex. It is commonly associated with salience of internal states, bodily awareness, and integration of bodily and emotional information.
Anatomy **Anterior Insula**	The anterior insula is a region of the brain commonly associated with salience, emotion, and bodily awareness. It is often linked with interoception, affective salience, and the integration of internal and external states. Its anatomy supports the interplay between subjective experience and cortical processing.
Cognition **Interoceptive Processing**	Interoceptive processing is the evaluation of internal bodily signals. It is commonly associated with anterior insula, posterior insula, anterior cingulate cortex, and corticospinal tracts. This function supports awareness of bodily state, salience of internal concerns, and integration of interoceptive and affective information.
Default Mode Network (Shen Default mode)	
Input description	The default mode network supports internally oriented cognition. It includes medial prefrontal cortex, posterior cingulate cortex, precuneus, angular gyrus, and medial temporal lobe. It supports autobiographical memory, self-referential thought, future simulation, social inference, and narrative reasoning.
Network **Default Mode Network**	The default mode network includes medial prefrontal cortex, posterior cingulate cortex, precuneus, angular gyrus, lateral parietal cortex, and medial temporal regions. It is commonly associated with internally oriented cognition, autobiographical memory, self-referential thought, and the integration of personal and social context.
Anatomy **Precuneus**	The precuneus lies on the medial parietal cortex. It is commonly associated with autobiographical memory, self-related imagery, and mental context. This region is often described as supporting integration of personal and external information.
Cognition **Autobiographical Memory**	Autobiographical memory is commonly associated with the medial prefrontal cortex, posterior cingulate cortex, precuneus, angular gyrus, parahippocampal cortex, retrosplenial cortex, and lateral parietal cortex. This function supports remembering personal experiences, spatial context, and internally constructed events.

Table S5 (continued)

Text (generated title)	Complete text
Frontoparietal Control Network (Du FPN-A)	
Input description	The frontoparietal control network supports flexible cognitive control. It links lateral prefrontal cortex, inferior frontal junction, intraparietal sulcus, and inferior parietal lobule. It supports task goals, working memory, planning, rule use, attentional control, and adaptive decision-making.
Network **Frontoparietal Control Network**	The frontoparietal control network includes dorsolateral prefrontal cortex, angular gyrus, intraparietal cortex, inferior parietal lobule, and posterior parietal cortex. It is commonly associated with executive control, working memory, and goal-directed behavior.
Anatomy **Inferior Frontal Gyrus**	This region is broadly tied to the integration of information. It is often associated with the handling of language and the manipulation of conceptual knowledge. Its function is generally described as supporting high-level cognition. It is thought to play a central role in executive control.
Cognition **Working Memory**	Working memory commonly involves dorsolateral prefrontal cortex, inferior parietal lobule, intraparietal sulcus, and lateral frontal cortex. These regions support the maintenance and manipulation of information in working memory. Their anatomy aligns with the frontoparietal network.
Salience Network (Du SAL/PMN)	
Input description	The salience network supports detection of behaviorally relevant events. It is centered on anterior insula, frontal operculum, dorsal anterior cingulate cortex, and midcingulate cortex. It supports salience detection, interoceptive appraisal, attentional orienting, conflict monitoring, and response selection.
Network **Salience Network**	The salience network includes the anterior insula, anterior cingulate cortex, and posterior cingulate gyrus. It is commonly associated with monitoring internal and external events, detecting relevant information, and shifting foci of inquiry.
Anatomy **Anterior Cingulate Cortex**	The anterior cingulate cortex is a medial region of the frontal lobe. It is commonly linked with control, monitoring, and self-regulation. This region is often described as supporting the evaluation of salient information.
Cognition **Executive Control**	Executive control is the regulation of thought, behavior, and performance. It involves dorsolateral prefrontal cortex, anterior cingulate cortex, frontoparietal regions, basal ganglia, and dorsal anterior insula. This concept supports adaptation, flexibility, goal-directed behavior, and conflict monitoring.

S10 Data Flow and Training Details

Several components learn from the same training papers, and a held-out paper could reach a later stage through the synthetic or language-model data. Supplementary Fig. S5 therefore shows which data train each component and which sets are used

only for evaluation. Supplementary Table S6 gives the training and validation sizes of each component, and Supplementary Table S7 gives the settings needed to retrain it. The training papers are reused across stages, and the stage 1 projector uses the 22,547 training papers that have a summary. No validation or test paper trains any component.

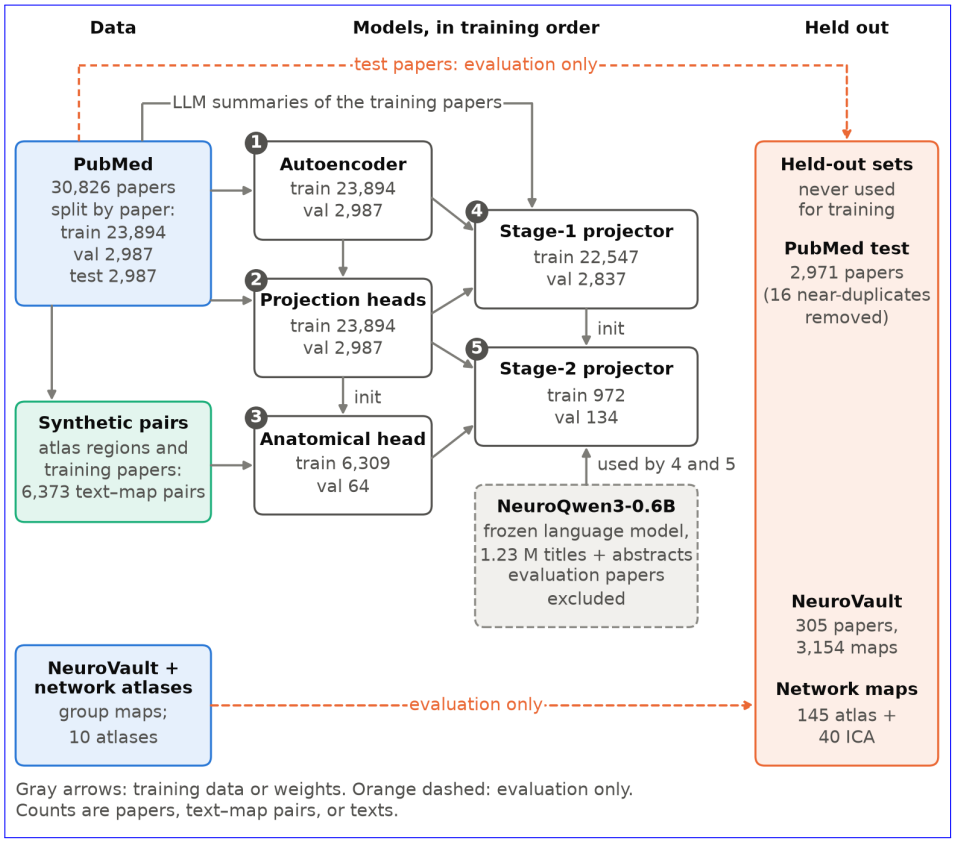


Fig. S5 Data flow of the final model. Boxes 1–5 are the trained components in training order, with their training and validation sizes. Gray arrows carry training data or weights, and orange dashed arrows mark sets used only for evaluation.

Table S6 Training data per component. Training and validation sizes of each trained component of the final model.

Component	Data	Train	Validation	Unit
Autoencoder	PubMed coordinate maps	23,894	2,987	papers
Text-to-brain and contrastive heads	SPECTER2 text and map latents	23,894	2,987	papers
Anatomically centered projection head	synthetic concept text-map pairs from training papers	6,309	64	pairs
Stage 1 projector (MLP)	map latents and LLM summaries	22,547	2,837	papers
Stage 2 projector (MLP)	`[NETWORK]`, `[REGION]`, `[FUNCTION]` texts (1,106 of 1,338 kept)	972	134	texts
NeuroQwen3-0.6B	PubMed titles and abstracts	1.22 M	12,317	articles

Table S7 Training details. Trainable parameters, training data, epochs, learning rate, batch size, seed, and training time of each component of the final model. All use AdamW, with epochs selected on the validation set. Times are for one RTX 5090 GPU, except for NeuroQwen3-0.6B (one H100).

Component	Params	Data, train / validation	Epochs (selection)	Learning rate (decay), batch	Seed	Time
Autoencoder	59.9 M	PubMed maps 23,894 / 2,987	101 (lowest loss)	5e-5, 256	0	10 min
Contrastive heads (text, image)	0.59 M, 0.30 M	Text, map latents 23,894 / 2,987	301 (last)	1e-5, 2,048	123	6 s
Text-to-brain head	0.59 M	Text, map latents 23,894 / 2,961	30 of at most 80 (best Dice)	1e-3, 512	0	22 s
Anatomically centered head	3.8 M	Synthetic pairs 6,309 / 64, and PubMed pairs	200 of 2,000 (best recall)	2e-4, 2,048	123	103 s
Stage 1 projector (MLP)	28.3 M	Maps, summaries 22,547 / 2,837	4 of at most 30 (lowest loss)	3e-4 (cosine), 32	17	17 min
Stage 2 projector (MLP)	28.3 M	Concept texts 972 / 134	30 (last)	1e-4 (cosine), 18	3 (best of 1–5)	5 min per seed
NeuroQwen3-0.6B	596 M	PubMed abstracts 1.22 M / 12,317 (385 M tokens)	1	5e-6 (cosine), 16 × 1,024 tokens, bf16	12345	not recorded

S11 Text-to-Brain Baselines and Retrieval

Scores computed in NeuroVLM's own latent space could favor NeuroVLM's own maps. We therefore also scored text-to-brain maps in voxel space, against simple baselines and published models. Each method turns a test paper's title and abstract into a map, which is scored against that paper's map as the autoencoder reconstructs it (Supplementary Fig. S6). Ridge regression maps the SPECTER2 embedding to the autoencoder latent, with its penalty chosen on the validation set. The domain mean map is the mean training map of the query's nearest of ten cognitive domains, found by k-means on the training latents. Atlas-based labeling is the union of the Harvard-Oxford regions named in the text. Neurosynth, NeuroQuery, and Text2Brain were used as released. Their corpora include 1,386 (Neurosynth) or 1,295 (NeuroQuery and Text2Brain) of the 2,971 test papers. NeuroVLM gives the most accurate maps by both Dice and correlation. It narrowly leads ridge regression, which learns the same mapping without a hidden layer, and it leads Text2Brain (Dice 0.53 against 0.49) and the map of the nearest training paper. The area under the recall curve summarizes all ranks, so Supplementary Table S8 also reports conventional retrieval measures among all test papers and among hard negatives. Hard negatives are only slightly harder than random papers of the same number (recall@10 0.24 against 0.28).

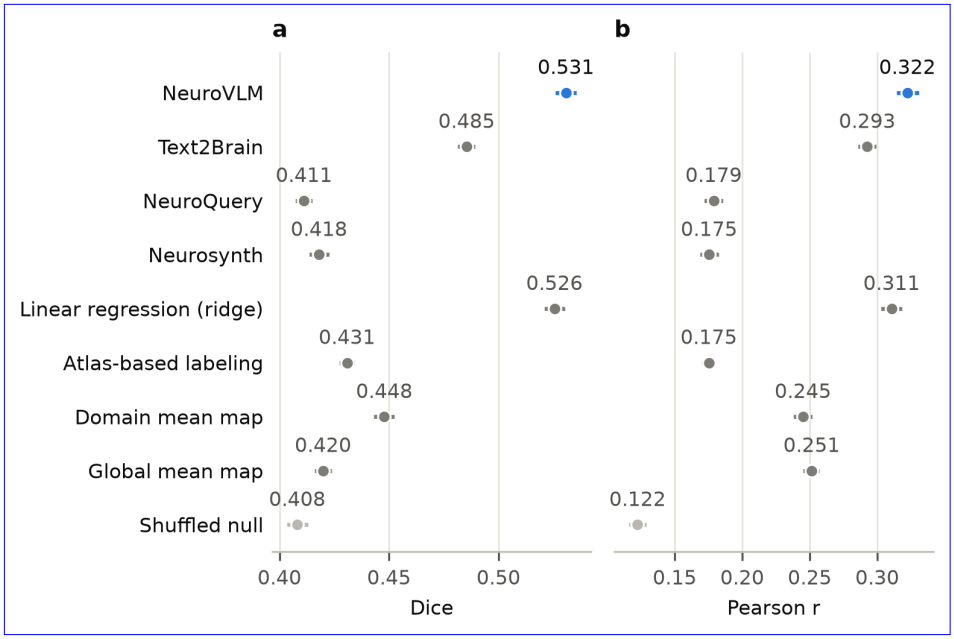


Fig. S6 **Text-to-brain baselines.** Text-to-brain maps from NeuroVLM and other methods, scored against each test paper's reconstructed map (n = 2,971). Panels show (a) Dice and (b) voxelwise Pearson r, with means and 95% bootstrap intervals. Dice uses each map's top 30% of voxels. Neurosynth, NeuroQuery, and Text2Brain were used as published. Their corpora include 1,386 (Neurosynth) or 1,295 (NeuroQuery and Text2Brain) test papers, which can only favor them. Text2Brain's training split is unreleased, so instead of retraining we scored the 1,157 papers in neither corpus, and NeuroVLM still beats all three.

Table S8 Contrastive retrieval on the PubMed test set. Retrieval among all 2,971 test papers, among hard negatives, the test papers of the query's own cognitive domain, and among the same number of random papers. MRR is the mean reciprocal rank.

	Recall@1	Recall@5	Recall@10	MRR	Median rank
All candidates (2,971 test papers)					
Text → brain	0.018	0.053	0.088	0.045	230
Brain → text	0.013	0.052	0.088	0.041	217
Chance	<0.001	0.002	0.003	0.003	1,486
Hard negatives (papers of the query's own domain, mean 585)					
Text → brain	0.054	0.160	0.242	0.118	44
Brain → text	0.059	0.176	0.263	0.126	40
Chance	0.003	0.017	0.034	0.019	250
Random negatives (random papers, as many as the hard negatives)					
Text → brain	0.069	0.196	0.283	0.142	35
Brain → text	0.071	0.201	0.289	0.144	35
Chance	0.003	0.017	0.034	0.019	250

S12 NeuroVault Bounds and Map Types

Each NeuroVault paper has several maps, and choosing the map most similar to the text inflates the score. Supplementary Fig. S7 therefore compares rules that choose the target in other ways. The best map gives an upper bound, the worst map a lower bound, and the multi-target rule counts any of a paper's maps as a hit. Under every rule, retrieval AUC stays about 0.1 above its own chance level. NeuroVault also mixes map types whose values mean different things. P-maps were inverted and signed maps keep their positive tail (Supplementary Section S2.3). Supplementary Table S9 reports each map type separately. Every declared map type with more than ten maps scores above chance in both directions. Components and gradients labeled Other, whose sign is arbitrary, score at chance.

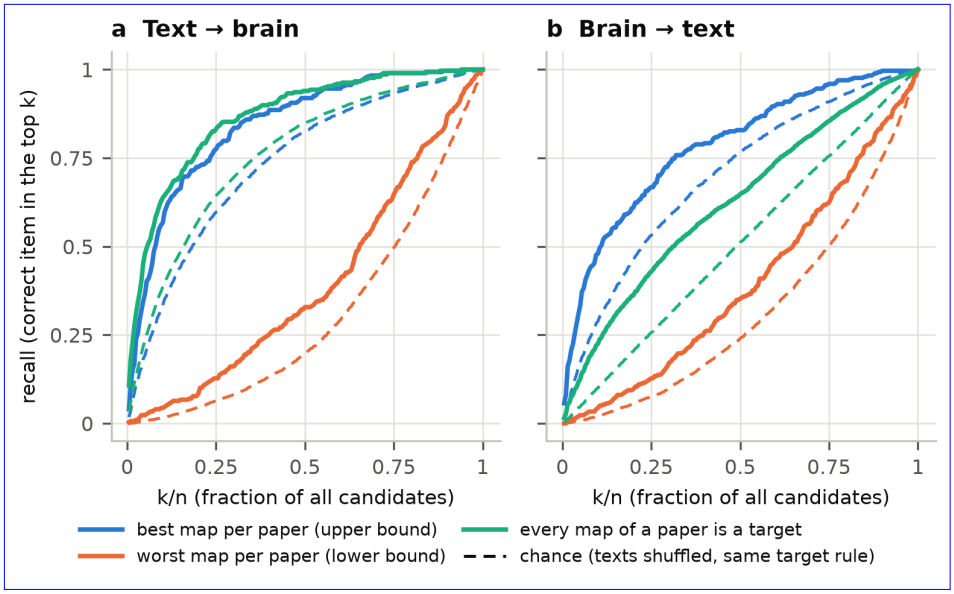


Fig. S7 **NeuroVault retrieval.** Recall for (a) text to brain and (b) brain to text when each paper's target is its best map, its worst map, or every map, with p-maps inverted. The x-axis is k/n (fraction of all candidates). Dashed lines show chance, the same rule applied to texts shuffled across papers. The best map, or counting any of a paper's maps as a hit in (a), lifts chance above the diagonal, and the worst map lowers it.

Table S9 **NeuroVault results by map type.** AUC with every map as its own target (chance 0.50). Other marks maps their uploaders labeled Other, typed from file names, descriptions, and values. Masks, parcellations, anatomical and variance maps, mean images, and untyped maps were excluded.

Map type	Maps	Papers	Handling	AUC Text → brain	AUC Brain → text
T	1,761	189	positive tail	0.64	0.64
Z	811	75	positive tail	0.57	0.58
beta, univariate	43	5	positive tail	0.61	0.77
F	40	17	positive tail	0.64	0.55
beta, multivariate	14	5	positive tail	0.84	0.83
p	3	2	inverted (1 − p)	0.64	0.60
1 − p	2	1	used as is	0.54	0.44
Other, t, z, or F	204	23	positive tail	0.63	0.70
Other, effect estimate	103	15	positive tail	0.69	0.62
Other, component or gradient	93	8	positive tail	0.49	0.51

S13 Out-of-Distribution Inputs

NeuroVLM returns a caption for any map and a map for any text, even when the input is unlike its training data. We therefore tested warning scores on inputs built to lie outside the training distribution. Maps were rotated by 90 degrees or at random, spun on the cortical surface, shuffled across voxels, replaced by white noise, or scaled by 100 or 0.01. Off-topic texts came from Wikipedia (WikiText) and cooking recipes. Supplementary Fig. S8 shows the warning scores of each input. A map is flagged when its Pearson correlation with its autoencoder reconstruction is lower than in 97.5% of validation maps, or when the Mahalanobis distance of its latent from the training latents is larger than in 97.5% of them. Correlation, unlike the plain cosine, ignores the constant offset that a dense map shares with its smooth reconstruction, which hid white noise. A text is flagged when its cosine similarity to the nearest training abstract is lower than in 95% of validation abstracts. The scores flag 4% of test maps, 5% of test abstracts, every shuffled, white-noise, and rescaled map, every off-topic text, and 61 to 80% of rotated or spun maps. The scores are warnings, not calibrated probabilities.

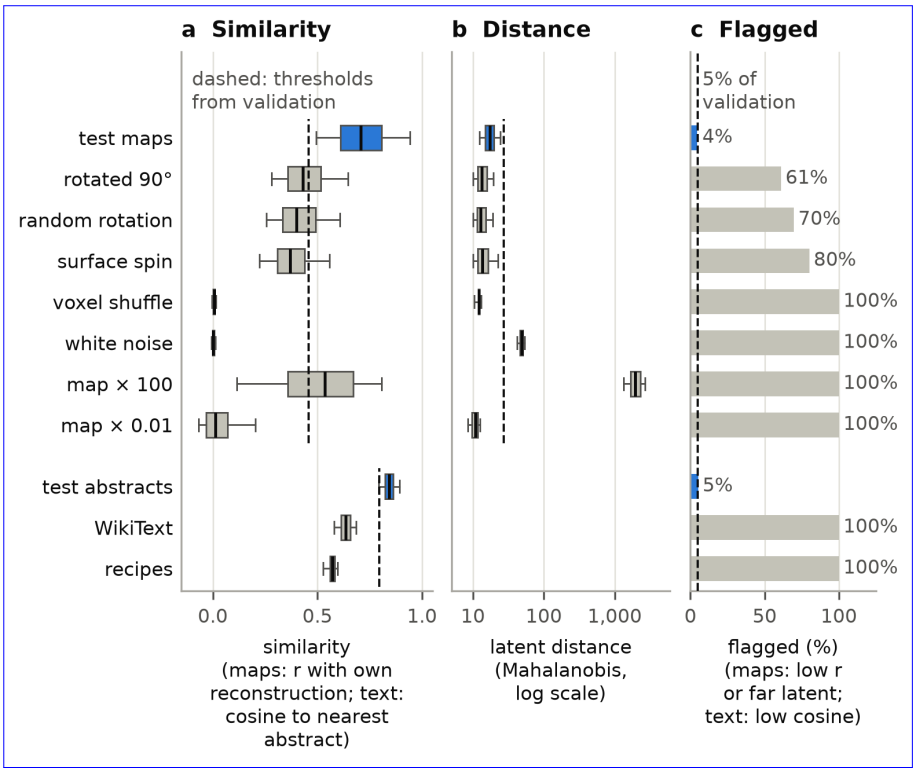


Fig. S8 Out-of-distribution warnings. Panels show (a) each input's similarity to the training data, (b) the distance of a map's latent from the training latents, and (c) the share of inputs flagged. Dashed lines mark thresholds set on the validation split. Blue marks held-out test inputs.

S14 Projector Comparison

The original model used a Qformer, which attended over only two input tokens, the map latent and its semantic vector. With two tokens, its attention has little to select, so we compared it with a simpler MLP projector of the same size. Supplementary Fig. S9 compares the original model, with the Qformer, and the final model, with the MLP projector, on the PubMed test set. The MLP projector raises brain-to-text retrieval AUC from 0.82 to 0.90 at stage 1 and from 0.94 to 0.95 at stage 2. It replaces the Qformer in both stages.

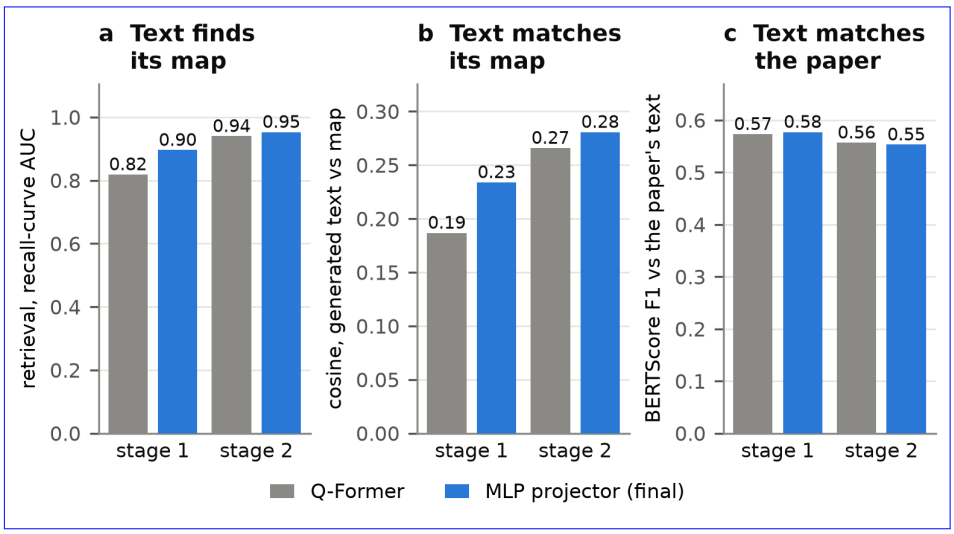


Fig. S9 Qformer and MLP projector. Gray bars show the original model with the Qformer and blue bars the final model with the MLP projector, on PubMed test maps. Panels show (a) retrieval AUC, (b) caption-to-map cosine, and (c) BERTScore F1 against the paper's text.

S15 Caption Granularity

A decoder that collapses to a fixed set of captions would be no better than retrieving the caption of the nearest training paper. The canonical projection could also push captions toward its prototypes. We therefore measured how specific the captions are. A caption names a study detail if it names a sample size, patient group, or task. Supplementary Fig. S10 compares retrieved captions with captions generated at each stage. Over 99% of stage 2 test captions are unique, so the decoder does not collapse. The share that names a study detail falls from 36% at stage 1 to 2% at stage 2, against 57% of retrieved captions. Generated captions are as close to the paper's text as retrieved captions. The canonical projection does pull captions toward its prototypes, and about one stage 2 caption in ten nearly copies a curated training text.

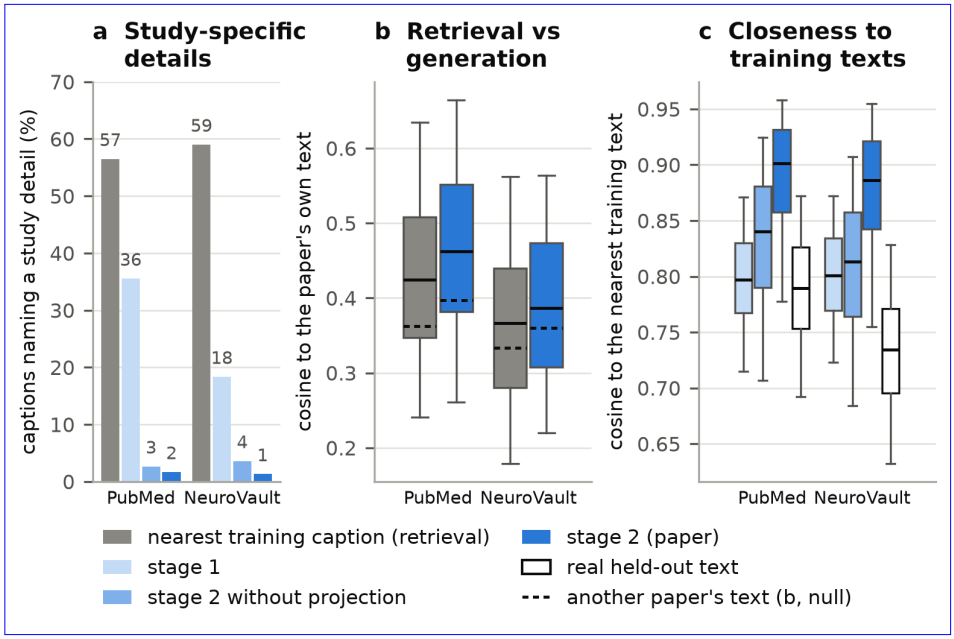


Fig. S10 **Granularity of generated captions.** Captions of PubMed test maps and NeuroVault papers, retrieved or generated at each stage. Panels show (a) the share naming a study detail and the MiniLM cosine (b) with the paper's own text and (c) with the nearest training text.

S16 Context-Free Encoder

SPECTER2 reads each word in the context of the whole title and abstract. To measure what this context adds, we compared it with a context-free baseline. We replaced SPECTER2 with the mean word2vec embedding [18] of each title and abstract and retrained only the projection heads (Supplementary Fig. S11). The word2vec vocabulary covers 87% of the words. SPECTER2 leads on both tasks by a modest margin. Retrieval AUC is 0.83 against 0.81, and text-to-brain Dice and voxel correlation are also higher with SPECTER2.

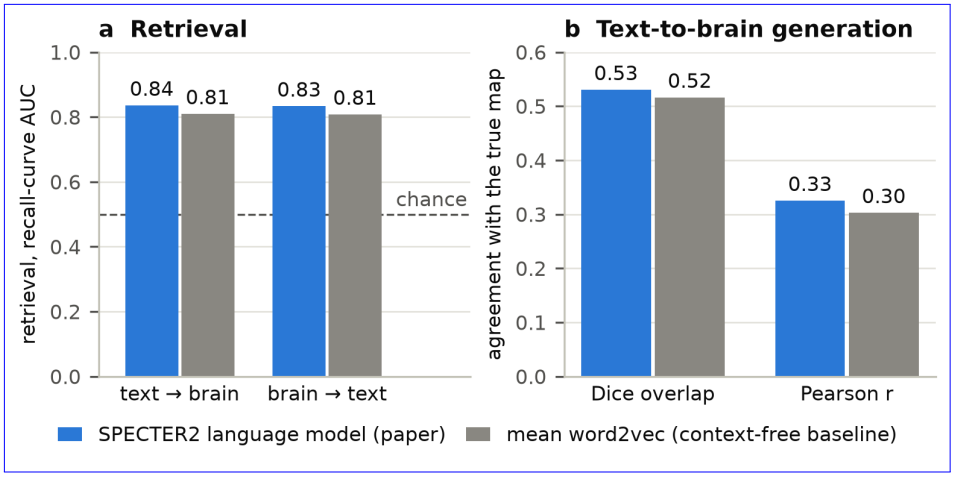


Fig. S11 Context-free encoder. PubMed test-set performance with SPECTER2 or with the mean word2vec embedding of each title and abstract, after retraining only the projection heads. Panels show (a) retrieval AUC in both directions and (b) text-to-brain Dice and voxel correlation.

S17 Latent Dimensions and Loss Ablations

Encoder weights are hard to read directly. We therefore interpreted the latent dimensions that texts and maps share. Each dimension was decoded to a map and labeled with the Cognitive Atlas terms most associated with it. We also retrained the heads with other losses, to test which modeling choices matter. For the clearest latent dimensions, the decoded map matches the Neurosynth map of the dimension's top term, such as listening in superior temporal cortex ($r = 0.65$, Supplementary Fig. S12). Text and maps agree on what each dimension means. The contrastive cross-entropy loss matters most for retrieval (Supplementary Table S10). Without it, retrieval AUC drops from 0.83 to 0.62, while the temperature and a one-directional loss barely matter. The final text-to-brain head also beats heads trained with other losses.

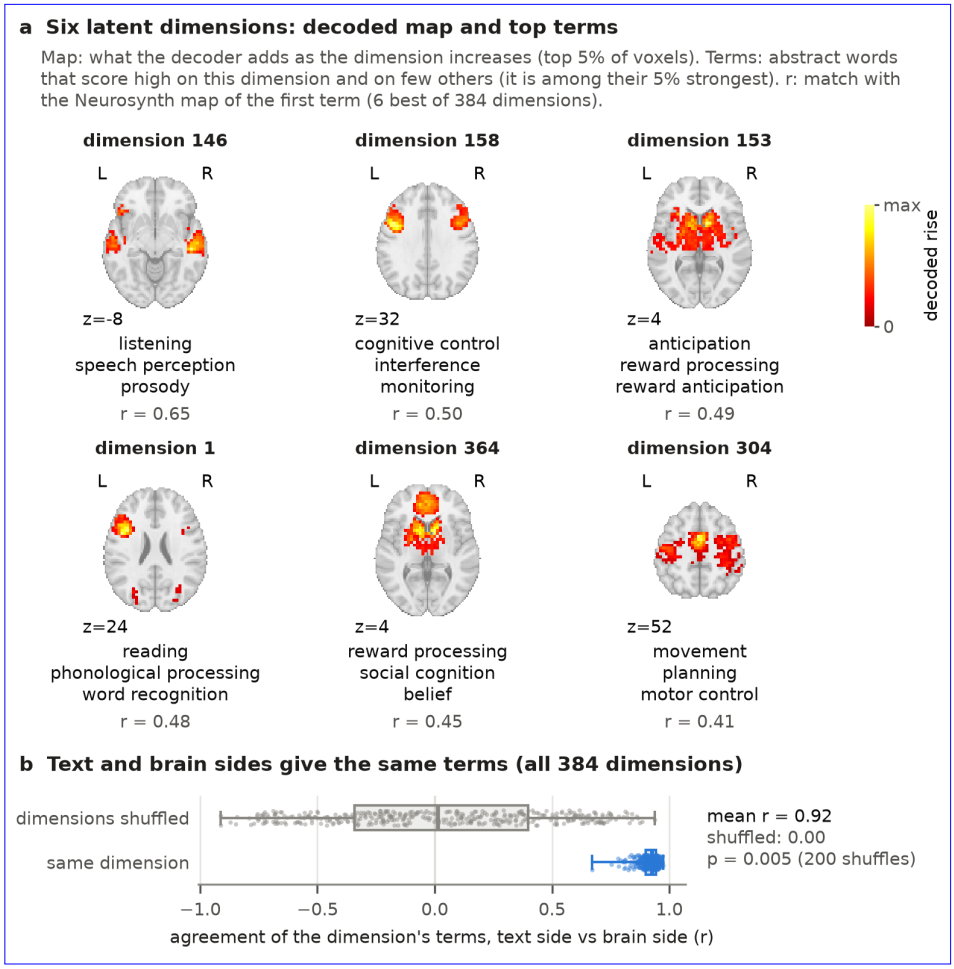


Fig. S12 **Latent dimensions.** (a) Decoded maps of six latent dimensions, each labeled with its three most associated Cognitive Atlas terms and its correlation r with the top term's Neurosynth map. (b) Agreement of each dimension's term associations between the text and brain sides, against shuffled dimensions.

Table S10 Loss ablations. Heads retrained on the final model's latents with the loss shown and scored on the PubMed test set. Values are means of three seeds.

Loss	Test-set score
Contrastive heads, retrieval AUC	
Symmetric InfoNCE cross-entropy, $\tau = 0.07$ (final)	0.833
One direction only, text → brain / brain → text	0.834 / 0.827
$\tau = 0.03$ / $\tau = 0.20$ / learnable τ	0.829 / 0.827 / 0.833
No cross-entropy, cosine of matched pairs only	0.618
Text-to-brain head, Dice	
Latent MSE and map correlation, final head with its own training settings	0.530
Latent MSE only	0.512
Voxel-wise binary cross-entropy through the decoder	0.491
Latent cosine only	0.385

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